OmniCode: Chapter 18: Track 3 - Applications, Bridges & Studio

C99 Bare-Metal, Lua Coroutines, Boundary Demos & HUD

CHAPTER IDENTIFIER
CH 18

NOTEBOOKLM WORD COUNT
22,863 words

CATALOGED CHAPTER FILES
34 files (6,769 lines)

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EPISTEMIC COVENANT BEDROCK & IMMUTABLE LANDMARKS

"And he gave unto Moses... two tables of testimony, tables of stone, written with the finger of God." — Exodus 31:18

"Prepare thy work without, and make it fit for thyself in the field; and afterwards build thine house." — Proverbs 24:27

"Remove not the ancient landmark, which thy fathers have set." — Proverbs 22:28

GOOGLE NOTEBOOKLM GROUNDING FOCUS & LEARNING PROMPT

- **Chapter Focus:** *"How does Lua coroutine engine run tape rolls as fibers?"*
- **Architectural Rule:** All definitions conform to 4-block structure & spatial ruler.

CHAPTER MANIFEST & INDEX (34 FILES)

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2	**<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/01_boundaries_and_asides.omni**	OMNI	168	1,456	10.4
3	**<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/02_logic_and_quad_state.omni**	OMNI	60	302	2.3
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12	**<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/while_statement.omni**	OMNI	67	351	2.8
13	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/README.md**	MD	79	379	2.8
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19	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/package.json**	JSON	54	92	1.1
20	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/syntaxes/omnicode.tmLanguage.json**	JSON	321	529	8.7
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22	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/app.js**	JS	509	1,537	20.4
23	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/cloud.html**	HTML	201	718	10.8
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34	**<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/terminal.js**	JS	305	885	11.9

<TRACK 3> : {Applications & OmniStudio Interfaces}/TRACK.md MD 10 LINES 52 WORDS 0.4 KB

```
# <TRACK 3> : {Applications & OmniStudio Interfaces}

## Scope & Invariant

User-facing tooling, visual studio interfaces, IDE integration, and canonical demonstration tapes.

- `apps/gui/`: OmniStudio HUD web interface & desktop application.
- `apps/editors/`: Visual Studio Code extension, grammar syntaxes, and language tooling.
- `apps/examples/`: Canonical verification reels and language showcase tapes.
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/01_boundaries_and_asides.omni

OMNI | 168 LINES | 1456 WORDS | 10.4 KB

```
# ================================================================================
# BLOCK 1: METADATA BLOCK
# ( Ranked Morphological Boundaries & Preserved Intent Asides )
# ================================================================================
# ARCHITECTURAL TOPOLOGY:
#   Foundational Showcase: The 6 Ranked Morphological Boundary Types.
#   - Rank 1: < ... > Invariant Axiom      (Immutable Law & Ontological Truth)
#   - Rank 2: [ ... ] Denotation           (Concrete Definition & Mutable State)
#   - Rank 3: { ... } Connotation          (Context, Environment & Payload)
#   - Rank 4: [ ... } Inside-Out Emergence (Entity Projects into Context)
#   - Rank 5: { ... ] Outside-In Imposition(Context Shapes Entity)
#   - Rank 6: ( ... ) Intent Aside         (First-Class Pronunciation & Teleology)
# ================================================================================

= [Module]: < Foundations { Boundaries } > & [ Example { Boundaries and Asides } ] ( Demonstrates the six ranked morphological boundaries and first-class intent asides ) {
    [ Version ]:   { "1.0.0" } ( The Version is in the context of 1.0.0 ),
    [ Created ]:   { "2026-10-11" } ( Created in the context of October 11, 2026 ),
    [ Updated ]:   { "2026-10-12" } ( Last updated in the context of October 12, 2026 ),
    [ Author ]:    { "CreativeWorkzStudio LLC / ProfessorSeanEX" } ( Author in the context of CreativeWorkzStudio LLC and ProfessorSeanEX ),
    [ System ]:    { OmniCode { Core Foundations } } ( System is in the context of OmniCode Core Foundations ),
    [ Witness ]:   < Scripture { Proverbs 8:29 KJV } > { "When he gave to the sea his decree, that the waters should not pass his commandment: when he appointed the foundations of the earth." } ( Invariant Scripture Witness Proverbs 8:29 in the context: When he appointed the foundations of the earth ),
    [ Nature ]:    < FoundationalExample > & { BaseBlock } ( Nature is a Foundational Example in the context of Base Block ),
    [ Semantics ]: { BoundaryPhysics :: MorphologicalClosure :: IntentPreservation } ( Semantics in the context of Boundary Physics, Morphological Closure, and Intent Preservation )
}

# Freestanding Foundational Axiom Declaration
# Rank 1: < ... > Invariant Axiom (Immutable Law)
< Axiom Truth > = < +1 > ( Ground truth axiom: immutably sealed in positive unity ) {
    [ Law ]:       < +1 > ( The immutable truth law is sealed in affirmative Proceed ),
    [ Geometry ]:  ENCLOSE [ Truth ] WITHIN < Invariant Space > ( Truth is bounded by immutable axiom walls )
}

# ================================================================================
# BLOCK 2: OPENING BLOCK
# ================================================================================
Opening {
    # Rank 1: Invariant Axiom (Inherited law)
    [ Ground Law ]: < Axiom Truth > ( Rank 1: Immutable axiom law ),

    # Rank 2: [ ... ] Denotation (Mutable Definition / State Container)
    [ State Alpha ]: [ +1 ] ( Rank 2: Concrete discrete definition with mutable state initialized to Proceed ),

    # Rank 3: { ... } Connotation (Ambient Universe / Environmental Context)
    { Universe Omega }: { Ground Law } ( Rank 3: Surrounding environmental context holding ground truth ),

    # Rank 4: [ ... } Inside-Out Emergence (Entity Projects Outward into Universe)
    [ Emergent Node }: [ State Alpha } ( Rank 4: Internal definition emerges outward into ambient context ),

    # Rank 5: { ... ] Outside-In Imposition (Environmental Context Shapes Internal Entity)
    { Imposed Node ]: { Universe Omega ] ( Rank 5: Surrounding context actively imposes constraints on internal entity ),

    # Rank 6: ( ... ) Intent Aside (Teleological Purpose & Cognitive Voice)
    ( Preserved Intent ): ( Rank 6: First-class teleological intent preserved in AST for human-AI shared reasoning ),

    # Hardware & Memory Preconditions
    [ Alignment ]: [ Memory Aligned 64 ] ( Memory aligned to 64-bit boundary ),
    [ Preconditions ]: [ (+1) plus (-1) <=> 0 ] & [ Status = +1:PROCEED ] ( ALU equilibrium holds and the machine status is armed in Proceed )
}

# ================================================================================
# BLOCK 3: BODY BLOCK
# ================================================================================
Body [
    # ------------------------------------------------------------------------
    # ROLL 1: RANK 1 INVARIANT VERIFICATION
    # Verify immutable ground truth identity
```

```
    # ------------------------------------------------------------------------
    [ ROLL 1 ] : [ Verify Axiom ] : < Axiom Truth > <=> VERIFY [ +1 ] = [ Truth` ]
        ( Rank 1: Verify immutable ground truth axiom against positive unity ),

    # ------------------------------------------------------------------------
    # ROLL 2: RANK 2 DENOTATIVE STATE MUTATION
    # Advance mutable state within defined boundary
    # ------------------------------------------------------------------------
    [ ROLL 2 ] : [ Mutate State ] : [ State Alpha ] --> ADVANCE [ +1 ] = [ State Alpha` ]
        ( Rank 2: Advance discrete denotative state yielding new state State-Alpha-prime ),

    # ------------------------------------------------------------------------
    # ROLL 3: RANK 3 CONNOTATIVE CONTEXT EXPANSION
    # Expand ambient contextual universe with verified state
    # ------------------------------------------------------------------------
    [ ROLL 3 ] : [ Expand Context ] : { Universe Omega } <-- ABSORB { State Alpha` } = { Universe Omega` }
        ( Rank 3: Ambient context absorbs new denotative state expanding environmental scope ),

    # ------------------------------------------------------------------------
    # ROLL 4: RANK 4 INSIDE-OUT ASYMMETRIC EMERGENCE
    # Project internal discrete state outward into ambient context
    # ------------------------------------------------------------------------
    [ ROLL 4 ] : [ Project Emergence ] : [ State Alpha` } --> EMERGE { Universe Omega` } = [ Emergent Node` }
        ( Rank 4: Discrete state emerges through asymmetrical boundary into ambient universe ),

    # ------------------------------------------------------------------------
    # ROLL 5: RANK 5 OUTSIDE-IN ASYMMETRIC IMPOSITION
    # Impose ambient contextual constraints onto internal discrete entity
    # ------------------------------------------------------------------------
    [ ROLL 5 ] : [ Impose Context ] : { Universe Omega` ] --> IMPOSE [ State Alpha` ] = { Imposed Node` ]
        ( Rank 5: Ambient universe imposes environmental boundaries onto internal discrete state ),

    # ------------------------------------------------------------------------
    # ROLL 6: RANK 6 FIRST-CLASS INTENT ASIDE PRESERVATION
    # Preserve teleological intent as a first-class AST node for human-AI cognition
    # ------------------------------------------------------------------------
    [ ROLL 6 ] : [ Preserve Intent ] : ( Preserved Intent ) --> RECORD [ AST ] = ( Teleology Verified` )
        ( Rank 6: Intent aside preserved as live AST node for human-AI cognitive continuity )
]

# ============================================================================
# BLOCK 4: CLOSING BLOCK
# ============================================================================
Closing ( Sealing Gate: Reconcile, validate, tripwire, debug, log, and affirm permanent boundary execution ) {
    # ------------------------------------------------------------------------
    # 1. VERIFICATION OF OPERATIONAL ROLLS
    # ------------------------------------------------------------------------
    [ Verify ]: [
        [ ROLL 1 ] = PROCEED ::
        [ ROLL 2 ] = PROCEED ::
        [ ROLL 3 ] = PROCEED ::
        [ ROLL 4 ] = PROCEED ::
        [ ROLL 5 ] = PROCEED ::
        [ ROLL 6 ] = PROCEED
    ] ( Verify that all six boundary operational rolls achieved affirmative consensus in PROCEED ),

    # ------------------------------------------------------------------------
    # 2. VALIDATION OF INVARIANT INTEGRITY & CLOSURE
    # ------------------------------------------------------------------------
    [ Validate ]: { Truth` = +1 } & { State Alpha` = +1 } & { Emergent Node` } & { Imposed Node` } & ( Teleology
Verified` )
        ( Validate that invariant truth holds, state remains affirmative, and all morphological closures and asides are
sealed ),

    # ------------------------------------------------------------------------
    # 3. ZERO-TOLERANCE TRIPWIRING
    # ------------------------------------------------------------------------
    [ Tripwire ]: { Status` = +1:PROCEED } --> [ ARMED ] | { Status` != +1:PROCEED } --> [ TRIPPED ] --> [ Commit Status
= -1:ROLLBACK ]
        ( Tripwire remains armed in Proceed or trips on anomaly to veto execution ),

    # ------------------------------------------------------------------------
    # 4. DIAGNOSTIC DEBUG TELEMETRY
    # ------------------------------------------------------------------------
    [ Debug ]: { [ Truth` ] :: [ State Alpha` ] :: [ Emergent Node` ] :: [ Imposed Node` ] :: ( Teleology Verified` ) ::
[ Status` ] }
        ( Expose diagnostic telemetry across all verified morphological boundaries and machine status ),

```

```
    # ------------------------------------------------------------------------
    # 5. EPISODIC INVENTORY LOGGING
    # ------------------------------------------------------------------------
    [ Log ]: [ Journal ]: INSCRIBE { [ Boundaries Sealed ] :: [ Rank 1 Through 6 ] :: [ Truth` ] }
        ( Inscribe the verified boundary physics demonstration into the permanent episodic system journal ),

    # ------------------------------------------------------------------------
    # 6. PERMANENT COVENANT EXECUTION & FOURFOLD COMMIT STATUS LOOP
    # ------------------------------------------------------------------------
    [ Commit Status ]: [
        # +1: PROCEED (+t: Advance forward along the causal tape)
        { [ Verify ] = PROCEED } & { [ Tripwire ] = ARMED }
            --> [ +1:PROCEED ] |

        #  0: AWAIT (t=t: Hold in equilibrium, pending resolution)
        { [ Status` ] = PENDING }
            --> [ 0:AWAIT ] :: REPEAT |

        # NULL: HALT (t=empty: Cessation of clock, absolute stop)
        { [ Status` ] = NULL }
            --> [ HALT:NULL ] |

        # -1: ROLLBACK (-t: Reverse causal tape to origin t=0)
        { [ Tripwire ] = TRIPPED } | { [ Verify ] != PROCEED }
            --> [ -1:ROLLBACK ]
    ] ( Fourfold state machine: Proceed advances forward, Await holds equilibrium, Halt ceases execution in Null,
Rollback reverts to origin )
}
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/02_logic_and_quad_state.omni

OMNI | 60 LINES | 302 WORDS | 2.3 KB

```
# ================================================================================
# BLOCK 1: METADATA BLOCK
# ( Balanced Ternary Logic & Micro/Macro Quad-State Truth Space )
# ================================================================================
Module < Logic.QuadStateEngine > ( Showcase Balanced Ternary and Micro/Macro Duality ) {
    version: < "1.0.0" >,
    system: < OmniCode.Core >,
    semantics: < BalancedTernary :: MicroMacroDuality :: CaveatPhysics >
}

AxiomAffirmation < yes >    ( Micro: yes == Macro: PROCEED == +1 )
AxiomRefutation  < no >     ( Micro: no  == Macro: ROLLBACK == -1 )
AxiomBalance     < ok >     ( Micro: ok  == Macro: AWAIT    ==  0 )

# ================================================================================
# BLOCK 2: OPENING BLOCK
# ( Preconditions & State Initialization )
# ================================================================================
Opening {
    requires: [ TruthSpace.T4 ],
    preconditions: [ AxiomAffirmation = PROCEED ]
}

flag_affirm [ yes ]   ( Initial affirmative state )
flag_negate [ -- yes ] ( Negation flips +1 to -1: no )

# ================================================================================
# BLOCK 3: BODY BLOCK
# ( Truth Space Algebra: Conjunction, Disjunction, Negation, and Caveat Gate )
# ================================================================================
Body [
    # 1. Micro/Macro Equivalence Checks
    affirm_check [ flag_affirm = PROCEED ] ( Affirmation matches PROCEED )
    refute_check [ flag_negate = ROLLBACK ] ( Negated affirmation matches ROLLBACK )

    # 2. Conjunction (:: AND) and Disjunction (| OR)
    and_test [ yes :: yes = PROCEED ]
    or_test  [ yes | no = PROCEED ]

    # 3. Adversative Caveat Gate (! BUT)
    # The action proceeds ONLY because the caveat holds (+1)
    gated_action [ flag_affirm ! [ affirm_check = yes ] ] ( Caveat permits forward commit )
]

# ================================================================================
# BLOCK 4: CLOSING BLOCK
# ( Verification & Macro State Commit )
# ================================================================================
Closing {
    verify: [
        affirm_check = yes ::
        refute_check = yes ::
        and_test = yes ::
        or_test = yes ::
        gated_action = PROCEED
    ],
    commit_status [ PROCEED ] ( Commit verified quad-state execution )
}

```

<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/03_cardinal_directions.omni

OMNI | 62 LINES | 328 WORDS | 2.5 KB

```
# ================================================================================
# BLOCK 1: METADATA BLOCK
# ( 4 Cardinal Directions Across Causality and Abstraction Axes )
# ================================================================================
Module < Geometry.CardinalDirections > ( Showcase 4 cardinal directional vectors and shorthands ) {
    version: < "1.0.0" >,
    system: < OmniCode.Core >,
    semantics: < VectorDynamics :: CausalityAndScale >
}

BaseDomain < 100 >   ( Ground level primitive )
TargetDomain < 200 > ( Elevated abstract domain )

# ================================================================================
# BLOCK 2: OPENING BLOCK
# ( Vector Preconditions & Initial States )
# ================================================================================
Opening {
    requires: [ CoordinatePlane.2D ],
    preconditions: [ BaseDomain = 100 ]
}

step_x [ 10 ] ( State moving across X temporal axis )
step_z [ 10 ] ( State moving across Z scale axis )

# ================================================================================
# BLOCK 3: BODY BLOCK
# ( Cardinal Vector Executions: Right, Left, Up, Down )
# ================================================================================
Body [
    # 1. Right (--> and ->): Forward Causation / Act Upon
    forward_canonical [ step_x --> 20 ] ( Forward step advancing state to 20 )
    forward_shorthand [ step_x -> 30 ]  ( Shorthand forward step to 30 )

    # 2. Left (<-- and <-): Retroaction / Feedback / Rollback
    feedback_canonical [ step_x <-- 10 ] ( Retroactive feedback step )
    feedback_shorthand [ step_x <- 10 ]  ( Shorthand feedback step )

    # 3. Up (--^ and -^): Synthesis / Abstraction / Ascend
    ascend_canonical [ step_z --^ TargetDomain ] ( Synthesis into target domain )
    ascend_shorthand [ step_z -^ TargetDomain ]  ( Shorthand synthesis )

    # 4. Down (--v and -v): Decomposition / Analysis / Descend
    descend_canonical [ TargetDomain --v BaseDomain ] ( Decomposition into base domain )
    descend_shorthand [ TargetDomain -v BaseDomain ]  ( Shorthand decomposition )
]

# ================================================================================
# BLOCK 4: CLOSING BLOCK
# ( Vector Integrity Verification & Commit )
# ================================================================================
Closing {
    verify: [
        forward_canonical = 20 ::
        forward_shorthand = 30 ::
        ascend_canonical = 200 ::
        descend_canonical = 100
    ],
    commit_status [ PROCEED ] ( Commit verified vector trajectory )
}

```

<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/04_turing_track_tape.omni

OMNI | 55 LINES | 348 WORDS | 2.4 KB

```
# ================================================================================
# BLOCK 1: METADATA BLOCK (THE TRACK)
# ( Track Manifest, Semantic Intent, and Inferred Ontological Types )
# ================================================================================
Module < ArithmeticReel.Tape1 > ( Add 1 ball to the group A which contains 4 balls ) {
    version: < "1.0.0" >,
    system: < OmniCode.Core >,
    semantics: < TrackPhysics :: MultiTape :: AIInference >
}

# The ? (Infer Operator) marks concepts whose structural types are inferred by the AI brain:
Ball < Entity? > ( Inferred discrete unit: resolved from natural sentence )
Group < Set? >   ( Inferred collection container: resolved from natural sentence )

# ================================================================================
# BLOCK 2: OPENING BLOCK (THE HEAD)
# ( The Focal Setup: Reading Initial Conditions and Contextual State )
# ================================================================================
Opening {
    import: [ Core.Arithmetic :: Core.Logic ],
    requires: [ Group : Set? :: Ball : Entity? ],
    preconditions: [ +1 ]
}

# The Head sets Group A with initial count of 4 balls
A [ 4 : Ball? ] { Group? } ( Group A initialized with 4 inferred balls )
new_ball [ 1 : Ball? ]     ( 1 additional ball to be introduced )

# ================================================================================
# BLOCK 3: BODY BLOCK (THE TAPE & ROLLS)
# ( The Tape Execution: SVO Rolls advancing the state forward using 'plus' )
# ================================================================================
Body [
    # ROLL 1 (SVO Action): Subject A acts upon target, advancing by 'plus' new_ball
    roll_1 [ A --> A plus new_ball ] ( Act upon A: causal transition from 4 to 5 balls )

    # ROLL 2 (SVO Verification): Subject A is 5 under the caveat that new_ball is 1
    roll_2 [ A = 5 ! [ new_ball = 1 ] ] ( State is 5, provided caveat of 1 added ball holds )
]

# ================================================================================
# BLOCK 4: CLOSING BLOCK (REEL COMMIT & PROOF)
# ( Tape Stop: Verification Invariants and State Commit )
# ================================================================================
Closing {
    verify: [
        roll_1 = 5 ::
        roll_2 = PROCEED ::
        A = 5 ::
        ADD = InvariantMacro
    ],
    commit_status [ PROCEED ] ( Commit the verified tape state )
}

```

<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/05_control_flow_and_loops.omni

OMNI | 146 LINES | 792 WORDS | 6.4 KB

```
# ================================================================================
# BLOCK 1: METADATA BLOCK
# ( Control Flow Manifest: Quad-State Accumulator Loops & Conditional Branching )
# ================================================================================
Module < ControlFlow.QuadStateShowcase > ( Canonical showcase of Simple Loops, Nested Loops, While statements, and If/Then/Else branches ) {
    version: < "1.0.0" >,
    author: < "ProfessorSeanEX & Antigravity" >,
    system: < OmniCode.Core >,
    semantics: < BalancedTernary :: QuadStateControlFlow :: NamedSiliconRegisters >
}

# Axiomatic Boundary Limits & Invariant Constants
LOOP_BOUND < 10 >          ( Invariant boundary ceiling for linear simple loop )
GRID_ROWS  < 3 >           ( Invariant outer dimension: 3 rows )
GRID_COLS  < 4 >           ( Invariant inner dimension: 4 columns )
THRESHOLD  < 100 >         ( Invariant threshold boundary for while condition )

# ================================================================================
# BLOCK 2: OPENING BLOCK
# ( Hardware Context Setup, Register Allocations & Preconditions )
# ================================================================================
Opening {
    import: [ Core.Logic :: Core.Arithmetic ],
    requires: [ TruthSpace.T4 :: MemoryAligned ],
    preconditions: [ +1 ]
}

# Physical Register Initializations for Body Execution
loop_acc         [ 0 ] ( Silicon Accumulator IT container )
loop_turn        [ 0 ] ( Temporal Cadence TURN container )
sensor_reading   [ 25 ] ( Dynamic reading for while statement )
condition_affirm [ yes ] ( Affirmative condition flag: +1 )
condition_refute [ no ]  ( Refuted condition flag: -1 )

# ================================================================================
# BLOCK 3: BODY BLOCK
# ( The 5 Canonical Control Flow Constructs )
# ================================================================================
Body [
    # --------------------------------------------------------------------------
    # 1. SIMPLE LOOP: Linear Quad-State Accumulator cycling from 0 up to BOUND
    # --------------------------------------------------------------------------
    simple_loop [
        STATE (
            < BOUND = LOOP_BOUND >,
            [ IT = 0, TURN = 0 ],
            { SPOOL }
        ) = [
            PROCEED  -WHEN-> { (IT plus 1 --> IT) & (TURN plus 1 --> TURN) },
            AWAIT    -WHEN-> { BUSY },
            ROLLBACK -WHEN-> { FAULT & (HOME --> IT) },
            HALT     -WHEN-> { TURN = BOUND }
        ]
    ] ( Simple linear loop: ticks up IT and TURN until watchdog BOUND is met )

    # --------------------------------------------------------------------------
    # 2. NESTED LOOP: 2D Matrix Grid Iteration (Outer Row x Inner Column)
    # --------------------------------------------------------------------------
    nested_loop [
        STATE (
            < ROW_BOUND = GRID_ROWS, COL_BOUND = GRID_COLS >,
            [ ROW_TURN = 0, ROW_SUM = 0 ],
            { SPOOL }
        ) = [
            PROCEED -WHEN-> {
                # Inner Column State Loop: accumulates COL_BOUND cells per row
                STATE (
                    < BOUND = COL_BOUND >,
                    [ COL_TURN = 0, IT = ROW_SUM ],
                    { SPOOL }
                ) = [
                    PROCEED  -WHEN-> { (IT plus 1 --> IT) & (COL_TURN plus 1 --> COL_TURN) },
```

```
                    AWAIT    -WHEN-> { BUSY },
                    ROLLBACK -WHEN-> { FAULT & (HOME --> IT) },
                    HALT     -WHEN-> { COL_TURN = BOUND }
                ] &
                (IT --> ROW_SUM) &
                (ROW_TURN plus 1 --> ROW_TURN)
            },
            AWAIT    -WHEN-> { BUSY },
            ROLLBACK -WHEN-> { FAULT },
            HALT     -WHEN-> { ROW_TURN = ROW_BOUND }
        ]
    ] ( Nested 2D loop: outer row cadence driving inner column accumulator )

    # ------------------------------------------------------------------------
    # 3. WHILE STATEMENT: Predicated Quad-State Loop
    # Advances while sensor_reading < THRESHOLD; halts when condition ceases
    # ------------------------------------------------------------------------
    while_statement [
        STATE (
            < BOUND = THRESHOLD >,
            [ IT = sensor_reading, TURN = 0 ],
            { FROM }
        ) = [
            PROCEED  -WHEN-> { (sensor_reading < THRESHOLD) & (IT plus 15 --> IT) & (TURN plus 1 --> TURN) },
            AWAIT    -WHEN-> { SENSOR_BUSY },
            ROLLBACK -WHEN-> { SENSOR_ERROR & (HOME --> IT) },
            HALT     -WHEN-> { (IT >= THRESHOLD) | (TURN = 100) }
        ]
    ] ( While loop: continues PROCEED while predicate holds, HALT escapes boundary )

    # ------------------------------------------------------------------------
    # 4. IF / THEN STATEMENT: Unilateral Conditional Gate
    # Evaluates condition: if affirmative (+1), actuates then branch; else awaits
    # ------------------------------------------------------------------------
    if_then_statement [
        STATE (
            [ IT = 0 ],
            { INTO }
        ) = [
            PROCEED -WHEN-> { condition_affirm & (42 --> IT) },
            AWAIT   -WHEN-> { -- condition_affirm }
        ]
    ] ( If/Then gate: proceeds with forward action only under verified affirmative condition )

    # ------------------------------------------------------------------------
    # 5. IF / ELSE STATEMENT: Bilateral Ternary Branch
    # Evaluates condition: +1 forks to then branch, -1 forks to else branch
    # ------------------------------------------------------------------------
    if_else_statement [
        STATE (
            [ IT = 0 ],
            { INTO }
        ) = [
            PROCEED  -WHEN-> { (condition_refute = yes) & (100 --> IT) },
            ROLLBACK -WHEN-> { (condition_refute = no)  & (200 --> IT) },
            AWAIT    -WHEN-> { condition_refute = ok }
        ]
    ] ( If/Else branch: balanced ternary fork selecting between affirmative and negative vectors )
]

# ============================================================================
# BLOCK 4: CLOSING BLOCK
# ( Postconditions, Invariant Integrity Proofs & Macro State Commit )
# ============================================================================
Closing {
    verify: [
        LOOP_BOUND = 10 ::
        GRID_ROWS = 3 ::
        GRID_COLS = 4 ::
        THRESHOLD = 100
    ],
    commit_status [ PROCEED ] ( Verified quad-state control flow suite sealed )
}
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/canonical_notation.omni

OMNI | 75 LINES | 476 WORDS | 3.2 KB

```
# ================================================================================
# BLOCK 1: METADATA BLOCK
# ( File Identity, Version, Author, Intent Aside, and Ontological Axioms )
# ================================================================================
Module < CanonicalNotationShowcase > ( Foundations of Being, Causality, and Abstraction ) {
    version: < "1.0.0" >,
    author: < "ProfessorSeanEX" >,
    system: < OmniCode.Core >,
    semantics: < BalancedTernary :: QuadState :: BoundaryPhysics >
}

# Axiomatic Categories (Invariant Sets)
A < +1 > ( Invariant category A: axiomatic ground truth )
B < +1 > ( Invariant category B: independent set domain )


# ================================================================================
# BLOCK 2: OPENING BLOCK
# ( Environment Setup, Preconditions, Scoped Universes, and Context Dictation )
# ================================================================================
Opening {
    requires: [ MemoryAligned :: TypeChecking ],
    initial_truth: [ +1 ]
}

# Mutable Denotations initialized in Opening
a [ +1 ] ( Particular instance 'a' initialized with mutable proceed state )
b [ +1 ] ( Particular instance 'b' initialized with mutable proceed state )

# Contextual Connotations & Asymmetric Causality
a_in_A [ a ] { A }  ( Instance 'a' validated within contextual universe A )
a_top_down { A ]    ( Outside-In: Environment A actively forces definition of a )
b_bottom_up [ B }   ( Inside-Out: Definition of b emerges from context B )


# ================================================================================
# BLOCK 3: BODY BLOCK
# ( Core Transformations, Being Relations, Causality Vectors, and Connectives )
# ================================================================================
Body [
    # 1. Ontological Relations of Being (Y-Axis)
    instance_typing [ a : A ]    ( 'a' stands in relation to Category A )
    identity_check [ a = a ]     ( 'a' is identical to 'a' )
    distinction_check [ a != b ] ( 'a' is not 'b' )
    similarity_check [ a ~= b ]  ( 'a' is liken to 'b' in structural form )

    # 2. Temporal & Causal Vectors (X-Axis)
    forward_action [ a --> b ]   ( 'a' acts upon 'b' forward in time )
    feedback_action [ b <-- a ]  ( 'b' reflects back upon 'a' as retroactive feedback )

    # 3. Hierarchical Abstraction Vectors (Z-Axis)
    synthesis [ a ^-- A ]        ( Low-level primitive 'a' builds up to high-level concept A )
    decomposition [ A v-- a ]    ( High-level category A decomposes down into primitive 'a' )

    # 4. Connectives & Adversative Caveat Pipeline
    negation [ -- a ]            ( Inverts state of 'a' across ternary zero balance )
    conjunction [ a : A :: b : B ]
    pipeline [ a --> b ! [ a : A :: b : B ] ] ( 'a' acts on 'b', BUT caveat must be satisfied )
]


# ================================================================================
# BLOCK 4: CLOSING BLOCK
# ( Postconditions, Invariant Integrity Proofs, Teardown, and Commit / Return )
# ================================================================================
Closing {
    verify: [
        identity_check = +1 ::
        forward_action = +1 ::
        pipeline = +1
    ],
    on_rollback: [ <-- unwind_state ],
```

```
    commit_status: [ +1 ] ( Commit state: verified execution )
}
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/if_else_statement.omni

OMNI | 64 LINES | 336 WORDS | 2.6 KB

```
# ================================================================================
# BLOCK 1: METADATA BLOCK
# ( File Manifest: Canonical Quad-State If/Else Bilateral Ternary Branch )
# ================================================================================
Module < ControlFlow.IfElseStatement > ( Demonstrates bilateral ternary branching in Quad-State logic ) {
    version: < "1.0.0" >,
    author: < "ProfessorSeanEX & Antigravity" >,
    system: < OmniCode.Core >,
    semantics: < BalancedTernary :: TernaryBranching :: NamedSiliconRegisters >
}

# Invariant Branch Targets
THEN_RESULT < 100 > ( Value emitted when condition is affirmative: +1 )
ELSE_RESULT < 200 > ( Value emitted when condition is refutative: -1 )

# ================================================================================
# BLOCK 2: OPENING BLOCK
# ( Context Preconditions & Branch Flag Setup )
# ================================================================================
Opening {
    import: [ Core.Logic :: Core.Arithmetic ],
    requires: [ TruthSpace.T4 ],
    preconditions: [ +1 ]
}

# The active condition to evaluate (set to 'no' to trigger else branch)
branch_condition [ no ] ( Initial condition evaluating to -1 / ROLLBACK )
branch_acc       [ 0 ]  ( Accumulator register receiving selected branch output )

# ================================================================================
# BLOCK 3: BODY BLOCK
# ( Execution: If/Else Bilateral Branch )
# ================================================================================
Body [
    # THE IF / ELSE STATEMENT:
    # `if (branch_condition == yes) { IT = THEN_RESULT; } else { IT = ELSE_RESULT; }`
    # - PROCEED (+1) : True branch (actuates THEN_RESULT into IT)
    # - ROLLBACK (-1): False branch (actuates ELSE_RESULT into IT)
    # - AWAIT (0)    : Neutral / uncommitted state (holds position)
    if_else_branch [
        STATE (
            [ IT = branch_acc ],
            { INTO }
        ) = [
            PROCEED  -WHEN-> { (branch_condition = yes) & (THEN_RESULT --> IT) },
            ROLLBACK -WHEN-> { (branch_condition = no)  & (ELSE_RESULT --> IT) },
            AWAIT    -WHEN-> { branch_condition = ok }
        ]
    ] ( If/Else branch: selects between affirmative then and refutative else vectors )
]

# ================================================================================
# BLOCK 4: CLOSING BLOCK
# ( Postconditions & Commit Status )
# ================================================================================
Closing {
    verify: [
        THEN_RESULT = 100 ::
        ELSE_RESULT = 200 ::
        branch_condition = no
    ],
    commit_status [ PROCEED ] ( If/Else ternary branch verified and sealed )
}
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/if_then_statement.omni

OMNI | 60 LINES | 275 WORDS | 2.3 KB

```
# ================================================================================
# BLOCK 1: METADATA BLOCK
# ( File Manifest: Canonical Quad-State If/Then Unilateral Conditional Gate )
# ================================================================================
Module < ControlFlow.IfThenStatement > ( Demonstrates unilateral conditional gating in Quad-State logic ) {
    version: < "1.0.0" >,
    author: < "ProfessorSeanEX & Antigravity" >,
    system: < OmniCode.Core >,
    semantics: < BalancedTernary :: ConditionalGating :: NamedSiliconRegisters >
}

# Invariant Target Payload
PAYLOAD_VALUE < 42 > ( Value to be transduced upon affirmative condition )

# ================================================================================
# BLOCK 2: OPENING BLOCK
# ( Context Preconditions & Flag Setup )
# ================================================================================
Opening {
    import: [ Core.Logic :: Core.Arithmetic ],
    requires: [ TruthSpace.T4 ],
    preconditions: [ +1 ]
}

# Initial Register & Condition States
permission_flag [ yes ] ( Affirmative precondition flag: +1 )
accumulator     [ 0 ]   ( Destination IT register )

# ================================================================================
# BLOCK 3: BODY BLOCK
# ( Execution: If/Then Conditional Gate )
# ================================================================================
Body [
    # THE IF / THEN STATEMENT:
    # `if (permission_flag) { accumulator = PAYLOAD_VALUE; }`
    # - PROCEED: When condition is affirmative (+1), actuates vector into IT
    # - AWAIT  : When condition is non-affirmative, holds state without mutating
    if_then_dispatch [
        STATE (
            [ IT = accumulator ],
            { INTO }
        ) = [
            PROCEED -WHEN-> { permission_flag & (PAYLOAD_VALUE --> IT) },
            AWAIT   -WHEN-> { -- permission_flag }
        ]
    ] ( If/Then gate: proceeds with forward action only when permission holds )
]

# ================================================================================
# BLOCK 4: CLOSING BLOCK
# ( Postconditions & Commit Status )
# ================================================================================
Closing {
    verify: [
        PAYLOAD_VALUE = 42 ::
        permission_flag = yes
    ],
    commit_status [ PROCEED ] ( If/Then conditional gate verified and sealed )
}
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/nested_loop.omni OMNI 79 LINES 375 WORDS 2.9 KB

```
# ================================================================================
# BLOCK 1: METADATA BLOCK
# ( File Manifest: Canonical Quad-State Nested 2D Grid Loop )
# ================================================================================
Module < ControlFlow.NestedLoop > ( Demonstrates 2D matrix row and column grid iteration ) {
    version: < "1.0.0" >,
    author: < "ProfessorSeanEX & Antigravity" >,
    system: < OmniCode.Core >,
    semantics: < BalancedTernary :: NestedStateLoop :: NamedSiliconRegisters >
}

# Invariant Grid Dimensions
NUM_ROWS < 3 > ( Outer loop dimension: 3 rows )
NUM_COLS < 4 > ( Inner loop dimension: 4 columns )

# ================================================================================
# BLOCK 2: OPENING BLOCK
# ( Context Preconditions & Coordinate Initialization )
# ================================================================================
Opening {
    import: [ Core.Logic :: Core.Arithmetic ],
    requires: [ TruthSpace.T4 ],
    preconditions: [ +1 ]
}

# Initial Register Setup
grid_sum  [ 0 ] ( Running total accumulated across all cells )
grid_rows [ 0 ] ( Current outer row iteration count )

# ================================================================================
# BLOCK 3: BODY BLOCK
# ( Execution: Nested Row/Column Traversal )
# ================================================================================
Body [
    # THE NESTED LOOP:
    # Outer STATE manages ROW_TURN up to ROW_BOUND.
    # On each outer PROCEED, the inner STATE manages COL_TURN up to COL_BOUND,
    # accumulating into IT and retiring into grid_sum.
    matrix_traversal [
        STATE (
            < ROW_BOUND = NUM_ROWS, COL_BOUND = NUM_COLS >,
            [ ROW_TURN = 0, ROW_ACC = 0 ],
            { SPOOL }
        ) = [
            PROCEED -WHEN-> {
                # Inner Column State Loop:
                STATE (
                    < BOUND = COL_BOUND >,
                    [ COL_TURN = 0, IT = ROW_ACC ],
                    { SPOOL }
                ) = [
                    PROCEED  -WHEN-> { (IT plus 1 --> IT) & (COL_TURN plus 1 --> COL_TURN) },
                    AWAIT    -WHEN-> { BUSY },
                    ROLLBACK -WHEN-> { FAULT & (HOME --> IT) },
                    HALT     -WHEN-> { COL_TURN = BOUND }
                ] &
                (IT --> ROW_ACC) &
                (ROW_TURN plus 1 --> ROW_TURN)
            },
            AWAIT    -WHEN-> { BUSY },
            ROLLBACK -WHEN-> { FAULT },
            HALT     -WHEN-> { ROW_TURN = ROW_BOUND }
        ]
    ] ( Nested 2D loop traverses all rows and columns to completion )
]

# ================================================================================
# BLOCK 4: CLOSING BLOCK
# ( Postconditions & Commit Status )
# ================================================================================
Closing {
    verify: [
        NUM_ROWS = 3 ::
        NUM_COLS = 4 ::
```

```
        grid_sum = 0
    ],
    commit_status [ PROCEED ] ( Nested loop verified and sealed )
}
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/simple_loop.omni OMNI 66 LINES 316 WORDS 2.4 KB

```
# ================================================================================
# BLOCK 1: METADATA BLOCK
# ( File Manifest: Canonical Quad-State Simple Linear Loop )
# ================================================================================
Module < ControlFlow.SimpleLoop > ( Demonstrates single-dimensional linear accumulator loop ) {
    version: < "1.0.0" >,
    author: < "ProfessorSeanEX & Antigravity" >,
    system: < OmniCode.Core >,
    semantics: < BalancedTernary :: QuadStateLoop :: NamedSiliconRegisters >
}

# Invariant Watchdog Bound
CEILING < 10 > ( Invariant maximum bound for accumulator loop )

# ================================================================================
# BLOCK 2: OPENING BLOCK
# ( Context Preconditions & Register Initialization )
# ================================================================================
Opening {
    import: [ Core.Logic :: Core.Arithmetic ],
    requires: [ TruthSpace.T4 ],
    preconditions: [ +1 ]
}

# Initial Register Setup
initial_it   [ 0 ] ( Initial primary accumulator IT value )
initial_turn [ 0 ] ( Initial temporal cadence TURN value )

# ================================================================================
# BLOCK 3: BODY BLOCK
# ( Execution: Linear Accumulator Loop )
# ================================================================================
Body [
    # THE SIMPLE LOOP:
    # Cycles through T4 truth space:
    #   +1 (PROCEED)  : Ticks up IT and TURN
    #    0 (AWAIT)    : Yields / waits if bus is busy
    #   -1 (ROLLBACK) : Rewinds to baseline HOME on fault
    #   NULL (HALT)   : Breaks loop when TURN reaches BOUND
    loop_result [
        STATE (
            < BOUND = CEILING >,
            [ IT = initial_it, TURN = initial_turn ],
            { SPOOL }
        ) = [
            PROCEED  -WHEN-> { (IT plus 1 --> IT) & (TURN plus 1 --> TURN) },
            AWAIT    -WHEN-> { BUSY },
            ROLLBACK -WHEN-> { FAULT & (HOME --> IT) },
            HALT     -WHEN-> { TURN = BOUND }
        ]
    ] ( Linear accumulator loop runs to completion when TURN equals BOUND )
]

# ================================================================================
# BLOCK 4: CLOSING BLOCK
# ( Postconditions & Commit Status )
# ================================================================================
Closing {
    verify: [
        CEILING = 10 ::
        initial_it = 0 ::
        initial_turn = 0
    ],
    commit_status [ PROCEED ] ( Simple loop verified and sealed )
}
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/examples/while_statement.omni

OMNI | 67 LINES | 351 WORDS | 2.8 KB

```
# ================================================================================
# BLOCK 1: METADATA BLOCK
# ( File Manifest: Canonical Quad-State Predicated While Loop )
# ================================================================================
Module < ControlFlow.WhileStatement > ( Demonstrates predicate-governed loop advancing while condition holds ) {
    version: < "1.0.0" >,
    author: < "ProfessorSeanEX & Antigravity" >,
    system: < OmniCode.Core >,
    semantics: < BalancedTernary :: PredicatedWhileLoop :: NamedSiliconRegisters >
}

# Invariant Threshold Boundary
MAX_THRESHOLD < 100 > ( Invariant upper ceiling triggering while loop termination )
MAX_CYCLES    < 500 > ( Watchdog maximum turn count )

# ================================================================================
# BLOCK 2: OPENING BLOCK
# ( Context Preconditions & Initial Sensor Values )
# ================================================================================
Opening {
    import: [ Core.Logic :: Core.Arithmetic ],
    requires: [ TruthSpace.T4 ],
    preconditions: [ +1 ]
}

# Initial Stream Values
sensor_reading [ 10 ] ( Initial inbound value to be sampled )
sensor_status  [ ok ] ( Initial bus status: equilibrium 0 )

# ================================================================================
# BLOCK 3: BODY BLOCK
# ( Execution: Predicated While Loop )
# ================================================================================
Body [
    # THE WHILE STATEMENT:
    # `while (sensor_reading < MAX_THRESHOLD) { Process_Stream(); }`
    # - PROCEED: Advances while the predicate is affirmative (+1)
    # - AWAIT  : Yields/pauses while the hardware sensor is busy sampling (0)
    # - ROLLBACK: Restores baseline from HOME if parity fails (-1)
    # - HALT   : Escapes the loop when predicate becomes false or watchdog trips (NULL)
    while_sensor_stream [
        STATE (
            < BOUND = MAX_CYCLES >,
            [ IT = sensor_reading, TURN = 0 ],
            { FROM }
        ) = [
            PROCEED  -WHEN-> { (sensor_reading < MAX_THRESHOLD) & (IT plus 10 --> IT) & (TURN plus 1 --> TURN) },
            AWAIT    -WHEN-> { SENSOR_BUSY },
            ROLLBACK -WHEN-> { SENSOR_ERROR & (HOME --> IT) },
            HALT     -WHEN-> { (IT >= MAX_THRESHOLD) | (TURN = BOUND) }
        ]
    ] ( While statement loops until sensor exceeds threshold or bound is reached )
]

# ================================================================================
# BLOCK 4: CLOSING BLOCK
# ( Postconditions & Commit Status )
# ================================================================================
Closing {
    verify: [
        MAX_THRESHOLD = 100 ::
        MAX_CYCLES = 500 ::
        sensor_reading = 10
    ],
    commit_status [ PROCEED ] ( While loop verified and sealed )
}
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/README.md | MD | 79 LINES | 379 WORDS | 2.8 KB

```
# OmniCode VS Code & IDE Extension

> **Official Language Support Package for Visual Studio Code, Cursor, and Google Antigravity IDE.**

Provides morphological boundary colorization, 3D coordinate operator grammar, balanced ternary literals, and universal 4-Block file templates.

---

## 1. Extension Topology & Installation

### File Topology

```text
editors/vscode/
├── package.json                   # Extension manifest (v0.1.2, .omni & .scroll associations)
├── language-configuration.json    # Brackets, auto-closing pairs, comment syntax
├── snippets/
│   └── omnicode.json             # 4-Block auto-population snippets (omni-file & omni-scroll)
└── syntaxes/
    └── omnicode.tmLanguage.json  # TextMate syntax grammar scopes & keyword rules
```

Installation

```bash
# Package the VSIX extension
npx -y @vscode/vsce package --allow-missing-repository

# Install into Antigravity IDE / VS Code
code --install-extension editors/vscode/omnicode-language-0.1.2.vsix
```

---

2. Core Capabilities & Grammar

Morphological Boundary Highlighting

Visually distinguishes all six ontological boundary types to eliminate bracket blindness:

* `[ ... ]` : **Denotation** (Fixed definitions, physical registers)
* `< ... >` : **Invariant** (Immutable axioms, process types)
* `{ ... }` : **Connotation** (Ambient context, scoped universes)
* `{ ... ]` : **Top-Down** (Context dictates definition / Absorber)
* `[ ... }` : **Bottom-Up** (Definition emerges from context / Generator)
* `( ... )` : **Aside** (Machine-readable intent and AI reasoning traces)

3D Coordinate Operator Colorization

* **Y-Axis (Being/Identity):** `=`, `!=`, `~=` / `≈`
* **X-Axis (Dynamics/Causality):** `-->` (Act upon forward), `<--` (Retroactive feedback/rollback)
* **Z-Axis (Scale/Hierarchy):** `^--` (Builds to / Layer up), `v--` (Breaks down from / Layer down)
* **Connectives:** `!` (Adversative caveat), `::` (And), `:` (In relation to), `|` (Or), `--` (Not)

Balanced Ternary Truth Literals

* `+1`, `YES`, `PROCEED` (Monotonic commit)
* `0`, `OK`, `AWAIT` (Asynchronous yield / equilibrium)
* `-1`, `NO`, `ROLLBACK` (State unwinding)
* `NULL`, `VOID`, `HALT` (Terminal trap)

Universal 4-Block Auto-Population Snippets

* `omni-file` / `omni-4block`: Complete compliant `.omni` module template with interactive tab stops.
* `omni-scroll` / `scroll`: Complete compliant `.scroll` episodic trace and learning template.
* `block-1`: Scaffolds Block 1: Metadata Block.
* `block-2`: Scaffolds Block 2: Opening Block.
* `block-3`: Scaffolds Block 3: Body Block.
* `block-4`: Scaffolds Block 4: Closing Block.

```

```
## 3. Verified Extension State & Invariants

* **Language ID:** `omnicode`
* **File Associations:** `*.omni`, `*.scroll`
* **Grammar Version:** `0.1.2`
* **Editor Integration:** Tested on VS Code 1.75+ and Antigravity IDE.
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/extension.js

JS | 93 LINES | 306 WORDS | 3.9 KB

```
// ================================================================================
// OmniCode VS Code Extension: Hover & Docstring Subsystem
// ================================================================================
// Connects the editor to the native OmniCode language engine (CLI & HTTP).
// Delivers 3-Tier composite hovercards bridging high-level .tape process ribbons
// down to low-level .omni invariant contracts and physical silicon execution states.
// ================================================================================

const vscode = require('vscode');
const { execFile } = require('child_process');
const http = require('http');

/**
 * Activates the OmniCode VS Code Extension.
 * Registers the HoverProvider for .omni, .scroll, and .tape files.
 * @param {vscode.ExtensionContext} context
 */
function activate(context) {
    const hoverProvider = vscode.languages.registerHoverProvider('omnicode', {
        provideHover(document, position, token) {
            const range = document.getWordRangeAtPosition(position);
            if (!range) {
                return null;
            }

            const symbol = document.getText(range);
            const line = position.line + 1;
            const col = position.character + 1;
            const filePath = document.fileName;

            return new Promise((resolve) => {
                // Method 1: Try local omnicode CLI binary directly
                execFile(
                    'omnicode',
                    ['hover', filePath, String(line), String(col), '--json'],
                    { timeout: 3000 },
                    (err, stdout) => {
                        if (!err && stdout) {
                            try {
                                const data = JSON.parse(stdout.trim());
                                if (data && data.markdown) {
                                    const md = new vscode.MarkdownString(data.markdown);
                                    md.isTrusted = true;
                                    md.supportHtml = true;
                                    return resolve(new vscode.Hover(md, range));
                                }
                            } catch (_e) {
                                // Fall through to HTTP server fallback
                            }
                        }

                        // Method 2: Fallback to running OmniCode Studio / GUI server (default port 8080)
                        const url = `http://127.0.0.1:8080/api/hover?file=${encodeURIComponent(filePath)}&line=${line}&col=${col}`;
                        const req = http.get(url, (res) => {
                            let body = '';
                            res.on('data', (chunk) => (body += chunk));
                            res.on('end', () => {
                                try {
                                    const data = JSON.parse(body);
                                    if (data && data.markdown) {
                                        const md = new vscode.MarkdownString(data.markdown);
                                        md.isTrusted = true;
                                        md.supportHtml = true;
                                        return resolve(new vscode.Hover(md, range));
                                    }
                                } catch (_err) {
                                    // Ignored
                                }
                                resolve(null);
                            });
                        });

```

```
                        req.on('error', () => resolve(null));
                        req.setTimeout(1000, () => {
                            req.destroy();
                            resolve(null);
                        });
                    }
                );
            });
        }
    });

    context.subscriptions.push(hoverProvider);
}

function deactivate() {}

module.exports = {
    activate,
    deactivate,
};
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/language-configuration.json

JSON | 30 LINES | 76 WORDS | 0.5 KB

```json
{
  "comments": {
    "lineComment": "#"
  },
  "brackets": [
    ["[", "]"],
    ["<", ">"],
    ["{", "}"],
    ["{", "]"],
    ["[", "}"],
    ["(", ")"]
  ],
  "autoClosingPairs": [
    { "open": "[", "close": "]" },
    { "open": "<", "close": ">" },
    { "open": "{", "close": "}" },
    { "open": "(", "close": ")" },
    { "open": "\"", "close": "\"" },
    { "open": "'", "close": "'" }
  ],
  "surroundingPairs": [
    ["[", "]"],
    ["<", ">"],
    ["{", "}"],
    ["(", ")"],
    ["\"", "\""],
    ["'", "'"]
  ]
}
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/omnicode-language-0.1.0.vsix

DOC/BINARY | 1 LINES | 5 WORDS | 2.9 KB

[Binary Artifact: omnicode-language-0.1.0.vsix (2.9 KB)]

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/omnicode-language-0.1.1.vsix

DOC/BINARY | 1 LINES | 5 WORDS | 4.0 KB

[Binary Artifact: omnicode-language-0.1.1.vsix (4.0 KB)]

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/omnicode-language-0.1.2.vsix

DOC/BINARY | 1 LINES | 5 WORDS | 6.4 KB

[Binary Artifact: omnicode-language-0.1.2.vsix (6.4 KB)]

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/package.json

JSON | 54 LINES | 92 WORDS | 1.1 KB

```json
{
  "name": "omnicode-language",
  "displayName": "OmniCode Language Support",
  "description": "Syntax highlighting, boundary morphology, and 4-block architecture support for the OmniCode language",
  "version": "0.1.3",
  "publisher": "omnicode",
  "main": "./extension.js",
  "activationEvents": [
    "onLanguage:omnicode"
  ],
  "engines": {
    "vscode": "^1.75.0"
  },
  "categories": [
    "Programming Languages",
    "Snippets"
  ],
  "contributes": {
    "languages": [
      {
        "id": "omnicode",
        "aliases": [
          "OmniCode",
          "omnicode",
          "omni",
          "OmniCode Scroll",
          "scroll",
          "OmniCode Tape",
          "tape"
        ],
        "extensions": [
          ".omni",
          ".scroll",
          ".tape"
        ],
        "configuration": "./language-configuration.json"
      }
    ],
    "grammars": [
      {
        "language": "omnicode",
        "scopeName": "source.omnicode",
        "path": "./syntaxes/omnicode.tmLanguage.json"
      }
    ],
    "snippets": [
      {
        "language": "omnicode",
        "path": "./snippets/omnicode.json"
      }
    ]
  }
}
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/syntaxes/omnicode.tmLanguage.json

JSON | 321 LINES | 529 WORDS | 8.7 KB

```
{
  "$schema": "https://raw.githubusercontent.com/martinring/tmlanguage/master/tmlanguage.json",
  "name": "OmniCode",
  "scopeName": "source.omnicode",
  "patterns": [
    { "include": "#comments" },
    { "include": "#boundary_casts" },
    { "include": "#primes" },
    { "include": "#affixed_identifiers" },
    { "include": "#asides" },
    { "include": "#strings" },
    { "include": "#ternary_literals" },
    { "include": "#vectors" },
    { "include": "#being_relations" },
    { "include": "#connectives" },
    { "include": "#boundaries" },
    { "include": "#numbers" },
    { "include": "#keywords" },
    { "include": "#registers" },
    { "include": "#categories" },
    { "include": "#instances" }
  ],
  "repository": {
    "comments": {
      "patterns": [
        {
          "name": "comment.line.number-sign.omnicode",
          "match": "#.*$"
        }
      ]
    },
    "boundary_casts": {
      "patterns": [
        {
          "name": "keyword.operator.boundary_cast.omnicode",
          "match": "\\[\\+\\]|\\[\\-\\]|<\\+>|<\\->|\\{\\+\\}|\\{\\-\\}"
        }
      ]
    },
    "primes": {
      "patterns": [
        {
          "name": "variable.other.prime.omnicode",
          "match": "\\b[a-zA-Z_][a-zA-Z0-9_]*`+"
        }
      ]
    },
    "affixed_identifiers": {
      "patterns": [
        {
          "name": "variable.other.affixed.abstract.omnicode",
          "match": "\\+[a-zA-Z_][a-zA-Z0-9_]*|[a-zA-Z_][a-zA-Z0-9_]*\\+"
        },
        {
          "name": "variable.other.affixed.concrete.omnicode",
          "match": "\\-[a-zA-Z_][a-zA-Z0-9_]*|[a-zA-Z_][a-zA-Z0-9_]*\\-"
        }
      ]
    },
    "asides": {
      "patterns": [
        {
          "name": "meta.aside.omnicode",
          "begin": "\\(",
          "beginCaptures": {
            "0": { "name": "punctuation.definition.aside.begin.omnicode" }
          },
          "end": "\\)",
          "endCaptures": {
            "0": { "name": "punctuation.definition.aside.end.omnicode" }
          },
          "patterns": [
            {
```

```
            "name": "string.unquoted.aside.intent.omnicode",
            "match": "[^()]+"
          }
        ]
      }
    ]
  },
  "strings": {
    "patterns": [
      {
        "name": "string.quoted.double.omnicode",
        "begin": "\"",
        "end": "\"",
        "patterns": [
          {
            "name": "constant.character.escape.omnicode",
            "match": "\\\\."
          }
        ]
      },
      {
        "name": "string.quoted.single.omnicode",
        "begin": "'",
        "end": "'|$",
        "patterns": [
          {
            "name": "constant.character.escape.omnicode",
            "match": "\\\\."
          }
        ]
      }
    ]
  },
  "ternary_literals": {
    "patterns": [
      {
        "name": "constant.language.ternary.proceed.omnicode",
        "match": "\\+1\\b|\\b(YES|PROCEED)\\b"
      },
      {
        "name": "constant.language.ternary.rollback.omnicode",
        "match": "-1\\b|\\b(NO|ROLLBACK)\\b"
      },
      {
        "name": "constant.language.ternary.await.omnicode",
        "match": "\\b0\\b|\\b(OK|AWAIT)\\b"
      },
      {
        "name": "constant.language.ternary.halt.omnicode",
        "match": "\\b(NULL|VOID|HALT)\\b"
      }
    ]
  },
  "vectors": {
    "patterns": [
      {
        "name": "keyword.operator.vector.scale.diagonal.major.omnicode",
        "match": "/-\\^"
      },
      {
        "name": "keyword.operator.vector.scale.diagonal.minor.omnicode",
        "match": "\\\\-v"
      },
      {
        "name": "keyword.operator.vector.scale.diagonal.great.omnicode",
        "match": "\\\\-\\^"
      },
      {
        "name": "keyword.operator.vector.scale.diagonal.small.omnicode",
        "match": "/-v"
      },
      {
        "name": "keyword.operator.vector.quadrant.omnicode",
        "match": "/\\^|\\\\v|\\^\\\\|v/"
      },
      {
        "name": "keyword.operator.vector.causality.forward.omnicode",
        "match": "-->|->"
```

```
        },
        {
          "name": "keyword.operator.vector.causality.backward.omnicode",
          "match": "<--|<-"
        },
        {
          "name": "keyword.operator.vector.scale.up.omnicode",
          "match": "\\^--|--\\^|-\\^|↑"
        },
        {
          "name": "keyword.operator.vector.scale.down.omnicode",
          "match": "v--|--v|-v|↓"
        }
      ]
    },
    "being_relations": {
      "patterns": [
        {
          "name": "keyword.operator.relation.invariant_equivalence.omnicode",
          "match": "<=>"
        },
        {
          "name": "keyword.operator.relation.identity.omnicode",
          "match": "="
        },
        {
          "name": "keyword.operator.relation.distinction.omnicode",
          "match": "!="
        },
        {
          "name": "keyword.operator.relation.homomorphism.omnicode",
          "match": "~=|≈"
        },
        {
          "name": "keyword.operator.relation.comparison.omnicode",
          "match": "<=|>=|<|>"
        }
      ]
    },
    "connectives": {
      "patterns": [
        {
          "name": "keyword.operator.connective.caveat.omnicode",
          "match": "!"
        },
        {
          "name": "keyword.operator.connective.and.omnicode",
          "match": "::"
        },
        {
          "name": "keyword.operator.connective.relation.omnicode",
          "match": ":"
        },
        {
          "name": "keyword.operator.connective.or.omnicode",
          "match": "\\|"
        },
        {
          "name": "keyword.operator.connective.not.omnicode",
          "match": "--"
        },
        {
          "name": "keyword.operator.connective.bound_pair.omnicode",
          "match": "&"
        },
        {
          "name": "keyword.operator.connective.percent.omnicode",
          "match": "%"
        },
        {
          "name": "keyword.operator.address.mmio.omnicode",
          "match": "::@|@"
        },
        {
          "name": "keyword.operator.range.omnicode",
          "match": "\\.\\."
        }
      ]
```

```
    },
    "boundaries": {
      "patterns": [
        {
          "name": "punctuation.definition.boundary.denotation.omnicode",
          "match": "\\[|\\]"
        },
        {
          "name": "punctuation.definition.boundary.invariant.omnicode",
          "match": "<|>"
        },
        {
          "name": "punctuation.definition.boundary.connotation.omnicode",
          "match": "\\{|\\}"
        }
      ]
    },
    "numbers": {
      "patterns": [
        {
          "name": "constant.numeric.hex.omnicode",
          "match": "\\b0x[0-9a-fA-F]+\\b"
        },
        {
          "name": "constant.numeric.omnicode",
          "match": "\\b\\d+(\\.\\d+)?\\b"
        }
      ]
    },
    "registers": {
      "patterns": [
        {
          "name": "variable.language.register.omnicode",
          "match": "\\b(HEAD|HOME|SPOOL|IT|SELF|CALLER|OTHER|STATE|TURN|BOUND|FROM|INTO|DISCARD)\\b"
        }
      ]
    },
    "keywords": {
      "patterns": [
        {
          "name": "keyword.control.block.omnicode",
          "match": "\\b(Module|Scroll|Opening|Body|Closing)\\b"
        },
        {
          "name": "keyword.other.declaration.omnicode",
          "match":
"\\b(import|requires|preconditions|verify|commit_status|user_prompt|synthesized_reel|machine_utterance|repair_entry|failpoint_rec
ord|version|system|author|safety|semantics|timestamp|session_id|execution_duration_micros|cognitive_mode|Witness|Nature)\\b"
        },
        {
          "name": "keyword.control.loop.verb.omnicode",
          "match":
"\\b(LOOP|WHEN|GOVERN|GUARD|TRAP|INSCRIBE|TRANSDUCE|PROJECT|SEQUENCE|ENCAPSULATE|LOCATE|BIND|ROUTE|SET_ASIDE|TRANSITION|CALCULATE
|MEASURE|MODULATE|REPEAT|PLAY|PAUSE|REWIND|STOP)\\b"
        },
        {
          "name": "keyword.control.tape.omnicode",
          "match": "\\b(Tape|Track)\\b"
        },
        {
          "name": "keyword.other.clause.tape.omnicode",
          "match": "\\b(WHO|WHAT|WHEN|WHERE|WHY|HOW)\\b"
        },
        {
          "name": "entity.name.function.roll.omnicode",
          "match": "\\broll_\\d+\\b|\\bROLL\\b"
        },
        {
          "name": "keyword.operator.arithmetic.word.omnicode",
          "match": "\\b(plus|minus|times|divide|by|add|sub|mult|div)\\b"
        },
        {
          "name": "keyword.operator.infer.omnicode",
          "match": "\\?"
        }
      ]
    },
    "categories": {
```

```
      "patterns": [
        {
          "name": "entity.name.type.category.omnicode",
          "match": "\\b[A-Z][a-zA-Z0-9_]*\\b"
        }
      ]
    },
    "instances": {
      "patterns": [
        {
          "name": "variable.other.instance.omnicode",
          "match": "\\b[a-z_][a-zA-Z0-9_]*\\b"
        }
      ]
    }
  }
}
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/editors/vscode/snippets/omnicode.json

JSON | 163 LINES | 611 WORDS | 6.5 KB

```
{
  "OmniCode 4-Block Template": {
    "prefix": ["omni-file", "omni-4block", "omni"],
    "description": "Auto-populate standard OmniCode 4-Block file architecture",
    "body": [
      "# ================================================================================",
      "# BLOCK 1: METADATA BLOCK",
      "# ( File Identity, Version, Author, Intent Aside, and Invariants )",
      "# ================================================================================",
      "Module < ${1:ModuleName} > ( ${2:Module intent rationale} ) {",
      "    version: < \"${3:0.1.0}\" >,",
      "    author: < \"${4:ProfessorSeanEX}\" >,",
      "    system: < OmniCode.Core >,",
      "    safety: < MemorySafe :: Deterministic >",
      "}",
      "",
      "# Invariant Axioms & Types",
      "Type_${1:ModuleName} < +1 > ( Invariant definition )",
      "",
      "",
      "# ================================================================================",
      "# BLOCK 2: OPENING BLOCK",
      "# ( Environment Setup, Preconditions, Scoped Context, and Declarations )",
      "# ================================================================================",
      "Opening {",
      "    requires: [ ${5:MemoryAligned.64} ],",
      "    preconditions: [ +1 ]",
      "}",
      "",
      "# Initial state instances",
      "${6:state} [ +1 ] ( Initialized state instance )",
      "",
      "",
      "# ================================================================================",
      "# BLOCK 3: BODY BLOCK",
      "# ( Core Transformations, Causality Vectors, Logic, and Caveat Pipelines )",
      "# ================================================================================",
      "Body [",
      "    # Forward action: state acts upon transformation",
      "    ${6:state} --> ${7:next_state} [ +1 ]",
      "    ${7:next_state} ! [ ${6:state} = +1 ] ( Caveat verification )",
      "]",
      "",
      "",
      "# ================================================================================",
      "# BLOCK 4: CLOSING BLOCK",
      "# ( Postconditions, Invariant Proofs, Teardown, and Commit Status )",
      "# ================================================================================",
      "Closing {",
      "    verify: [ ${7:next_state} = +1 ],",
      "    commit_status: [ +1 ] ( Commit verified state )",
      "}",
      ""
    ]
  },
  "Metadata Block": {
    "prefix": "block-1",
    "description": "Insert Block 1: Metadata Block",
    "body": [
      "# ================================================================================",
      "# BLOCK 1: METADATA BLOCK",
      "# ================================================================================",
      "Module < ${1:Name} > ( ${2:Intent} ) {",
      "    version: < \"0.1.0\" >",
      "}",
      ""
    ]
  },
  "Opening Block": {
    "prefix": "block-2",
    "description": "Insert Block 2: Opening Block",
    "body": [
      "# ================================================================================",
```

```
      "# BLOCK 2: OPENING BLOCK",
      "# ================================================================================",
      "Opening {",
      "    requires: [ ${1:Prerequisites} ]",
      "}",
      ""
    ]
  },
  "Body Block": {
    "prefix": "block-3",
    "description": "Insert Block 3: Body Block",
    "body": [
      "# ================================================================================",
      "# BLOCK 3: BODY BLOCK",
      "# ================================================================================",
      "Body [",
      "    ${1:source} --> ${2:target}",
      "]",
      ""
    ]
  },
  "Closing Block": {
    "prefix": "block-4",
    "description": "Insert Block 4: Closing Block",
    "body": [
      "# ================================================================================",
      "# BLOCK 4: CLOSING BLOCK",
      "# ================================================================================",
      "Closing {",
      "    verify: [ ${1:condition} = +1 ],",
      "    commit_status: [ +1 ]",
      "}",
      ""
    ]
  },
  "OmniCode Episodic Scroll Template": {
    "prefix": ["omni-scroll", "scroll"],
    "description": "Auto-populate standard OmniCode Episodic Cognitive Scroll (.scroll)",
    "body": [
      "# ================================================================================",
      "# BLOCK 1: METADATA BLOCK (THE COGNITIVE SCROLL)",
      "# ( Execution Trace ID, Prompt Intent, Inferred Types, and Neural Snapshot )",
      "# ================================================================================",
      "Scroll < EpistemicTrace.${1:SessionId} > ( ${2:User Prompt Intent} ) {",
      "    version: < \"1.0.0\" >,",
      "    system: < OmniCode.AI.Model >,",
      "    timestamp: < \"${CURRENT_SECONDS_UNIX}\" >,",
      "    session_id: < \"${1:SessionId}\" >,",
      "    cognitive_mode: < NeuralSymbolicTransduction >",
      "}",
      "",
      "# ================================================================================",
      "# BLOCK 2: OPENING BLOCK (EXPERIENCE INITIALIZATION)",
      "# ( Hardware Invariants, Preconditions, and Initial Context )",
      "# ================================================================================",
      "Opening {",
      "    import: [ Core.Arithmetic :: Core.Logic :: Core.Vectors ],",
      "    preconditions: [ +1 ]",
      "}",
      "",
      "# Original Cognitive Discourse Prompt:",
      "user_prompt < \"${2:User Prompt Intent}\" >",
      "",
      "# ================================================================================",
      "# BLOCK 3: BODY BLOCK (CHRONICLE OF EXECUTION & RESTORATION)",
      "# ( Replay Tape: Synthesized Program, Sensory Telemetry & Immune Interventions )",
      "# ================================================================================ ",
      "Body [",
      "    # Immune System Interventions (Restorator):",
      "    repair_entry [ L1: CleanExecution --> \"Zero repairs required.\" ]",
      "",
      "    # Executed Turing Reel:",
      "    synthesized_reel [",
      "        ${3:state} [ +1 ]",
      "    ]",
      "]",
      "",
      "# ================================================================================",
```

```
      "# BLOCK 4: CLOSING BLOCK (EPISODIC COMMIT & LEARNING PROOF)",
      "# ( Verification Proof, Final Quad-State, and Conversational Synthesis )",
      "# ============================================================================== ",
      "Closing {",
      "    commit_status [ <PROCEED> ],",
      "    machine_utterance ( \"${4:Execution succeeded to +1:PROCEED.}\" )",
      "}",
      ""
    ]
  }
}
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/app.js JS 509 LINES 1537 WORDS 20.4 KB

```
// ================================================================================
// OMNICODE STUDIO & COGNITIVE TERMINAL APPLICATION LOGIC
// Grounded in Balanced Ternary Quad-State & Biblical Moral Ontology
// ================================================================================

document.addEventListener('DOMContentLoaded', () => {
    initNavigation();
    initPromptStudio();
    initTerminalGateway();
    initLexiconExplorer();
    initEpisodicScrolls();
    initHoverInspector();
});

// --------------------------------------------------------------------------------
// 1. NAVIGATION TAB SWITCHING
// --------------------------------------------------------------------------------
function initNavigation() {
    const tabs = document.querySelectorAll('.nav-tab');
    const panels = document.querySelectorAll('.view-panel');

    tabs.forEach(tab => {
        tab.addEventListener('click', () => {
            const targetViewId = tab.getAttribute('data-view');

            tabs.forEach(t => t.classList.remove('active'));
            panels.forEach(p => p.classList.remove('active'));

            tab.classList.add('active');
            const targetPanel = document.getElementById(targetViewId);
            if (targetPanel) {
                targetPanel.classList.add('active');
            }

            // Trigger data load if needed
            if (targetViewId === 'view-lexicon') {
                loadLexiconData();
            } else if (targetViewId === 'view-scrolls') {
                loadScrollsData();
            }
        });
    });
}

// --------------------------------------------------------------------------------
// 2. AI PROMPT STUDIO CONTROLLER
// --------------------------------------------------------------------------------
function initPromptStudio() {
    const promptInput = document.getElementById('prompt-input');
    const submitBtn = document.getElementById('btn-submit-prompt');
    const promptChips = document.querySelectorAll('.prompt-chip');
    const quadstateBadge = document.getElementById('studio-quadstate-badge');
    const stateLabel = document.getElementById('state-label');
    const responseContent = document.getElementById('response-content');
    const synthesizedCode = document.getElementById('synthesized-code');
    const badgeSession = document.getElementById('badge-session-id');
    const linkScroll = document.getElementById('link-scroll');
    const linkLog = document.getElementById('link-log');
    const copyCodeBtn = document.getElementById('btn-copy-code');

    // Quick Prompt Chips
    promptChips.forEach(chip => {
        chip.addEventListener('click', () => {
            const promptText = chip.getAttribute('data-prompt');
            promptInput.value = promptText;
            promptInput.focus();
        });
    });

    // Copy Code Button
    copyCodeBtn.addEventListener('click', () => {
        const codeText = synthesizedCode.textContent;
        navigator.clipboard.writeText(codeText).then(() => {
            const originalText = copyCodeBtn.textContent;
```

```
            copyCodeBtn.textContent = 'Copied!';
            setTimeout(() => { copyCodeBtn.textContent = originalText; }, 1800);
        });
    });

    // Submit Prompt Execution
    submitBtn.addEventListener('click', async () => {
        const prompt = promptInput.value.trim();
        if (!prompt) return;

        // Set Loading State
        submitBtn.disabled = true;
        submitBtn.querySelector('.btn-text').textContent = 'Deliberating...';
        stateLabel.textContent = 'DELIBERATING (0:AWAIT)';
        quadstateBadge.className = 'quadstate-indicator';

        try {
            const res = await fetch('/api/prompt', {
                method: 'POST',
                headers: { 'Content-Type': 'application/json' },
                body: JSON.stringify({ prompt })
            });

            if (!res.ok) {
                const errData = await res.json().catch(() => ({ error: 'Unknown server error' }));
                throw new Error(errData.error || `HTTP ${res.status}`);
            }

            const data = await res.json();
            renderPromptResult(data);
        } catch (err) {
            console.error('Prompt evaluation error:', err);
            responseContent.innerHTML = `<p style="color: var(--state-rollback);">✗ Error evaluating prompt:
${err.message}</p>`;
            stateLabel.textContent = 'ERROR (NULL:HALT)';
            quadstateBadge.className = 'quadstate-indicator halt';
        } finally {
            submitBtn.disabled = false;
            submitBtn.querySelector('.btn-text').textContent = 'Send to AI Cortex';
        }
    });

    function renderPromptResult(data) {
        // 1. Update QuadState Badge
        const state = (data.state || 'PROCEED').toUpperCase();
        quadstateBadge.className = 'quadstate-indicator';

        if (state === 'PROCEED') {
            quadstateBadge.classList.add('proceed');
            stateLabel.textContent = '+1: PROCEED';
        } else if (state === 'ROLLBACK') {
            quadstateBadge.classList.add('rollback');
            stateLabel.textContent = '-1: ROLLBACK (CAVEAT VETO)';
        } else if (state === 'AWAIT') {
            quadstateBadge.classList.add('await');
            stateLabel.textContent = '0: AWAIT (DISCERNMENT)';
        } else {
            quadstateBadge.classList.add('halt');
            stateLabel.textContent = 'NULL: HALT (TERMINAL TRAP)';
        }

        // 2. Session and Response Content
        badgeSession.textContent = `Session: ${data.session_id || 'active'}`;
        responseContent.textContent = data.response || 'No natural discourse generated.';

        // 3. Synthesized Code Block
        synthesizedCode.textContent = data.omnicode || '// Zero code generated.';

        // 4. File Links
        linkScroll.textContent = `📜 Scroll: ${data.scroll || 'Recorded'}`;
        linkLog.textContent = `📋 Log: ${data.log || 'Recorded'}`;
    }
}

// ---------------------------------------------------------------------------
// 3. CLI TERMINAL GATEWAY CONTROLLER
// ---------------------------------------------------------------------------
function initTerminalGateway() {
```

```
    const termInput = document.getElementById('terminal-cmd-input');
    const runBtn = document.getElementById('btn-run-cmd');
    const termStream = document.getElementById('terminal-stream');
    const clearBtn = document.getElementById('btn-clear-terminal');
    const shortcutBtns = document.querySelectorAll('.term-btn');
    const termWindow = document.getElementById('terminal-window');

    const desktopBtn = document.getElementById('btn-desktop-launch');
    if (desktopBtn) {
        desktopBtn.addEventListener('click', () => {
            executeTerminalCommand('omnicode desktop');
        });
    }

    clearBtn.addEventListener('click', () => {
        termStream.innerHTML = '';
    });

    shortcutBtns.forEach(btn => {
        btn.addEventListener('click', () => {
            const cmd = btn.getAttribute('data-cmd');
            termInput.value = cmd;
            executeTerminalCommand(cmd);
        });
    });

    runBtn.addEventListener('click', () => {
        const cmd = termInput.value.trim();
        if (!cmd) return;
        executeTerminalCommand(cmd);
        termInput.value = '';
    });

    termInput.addEventListener('keydown', (e) => {
        if (e.key === 'Enter') {
            const cmd = termInput.value.trim();
            if (!cmd) return;
            executeTerminalCommand(cmd);
            termInput.value = '';
        }
    });

    async function executeTerminalCommand(command) {
        appendTerminalLine(`omnicode> ${command}`, 'term-line-cmd');

        try {
            const res = await fetch('/api/terminal', {
                method: 'POST',
                headers: { 'Content-Type': 'application/json' },
                body: JSON.stringify({ command })
            });

            const data = await res.json();
            if (data.stdout) {
                appendTerminalLine(data.stdout, 'term-line-stdout');
            }
            if (data.stderr) {
                appendTerminalLine(data.stderr, 'term-line-stderr');
            }
        } catch (err) {
            appendTerminalLine(`Error executing command: ${err.message}`, 'term-line-stderr');
        }

        termWindow.scrollTop = termWindow.scrollHeight;
    }

    function appendTerminalLine(text, cssClass) {
        const div = document.createElement('div');
        div.className = cssClass;
        div.textContent = text;
        termStream.appendChild(div);
    }
}

// ---------------------------------------------------------------------------
// 4. COVENANT MANIFEST & BIBLICAL LEXICON CONTROLLER
// ---------------------------------------------------------------------------
let allLexiconEntries = [];
```

```
230
231 function initLexiconExplorer() {
232     const searchInput = document.getElementById('lexicon-search-input');
233     const filterBtns = document.querySelectorAll('.filter-btn');
234
235     searchInput.addEventListener('input', () => {
236         filterAndRenderLexicon();
237     });
238
239     filterBtns.forEach(btn => {
240         btn.addEventListener('click', () => {
241             filterBtns.forEach(b => b.classList.remove('active'));
242             btn.classList.add('active');
243             filterAndRenderLexicon();
244         });
245     });
246 }
247
248 async function loadLexiconData() {
249     const entriesGrid = document.getElementById('lexicon-entries-grid');
250     if (allLexiconEntries.length > 0) return; // Already cached
251
252     entriesGrid.innerHTML = '<p class="placeholder-text">Loading biblical knowledge base from offline memory...</p>';
253
254     try {
255         const res = await fetch('/api/lexicon');
256         const data = await res.json();
257         allLexiconEntries = data.entries || [];
258         filterAndRenderLexicon();
259     } catch (err) {
260         console.error('Failed to load lexicon:', err);
261         entriesGrid.innerHTML = `<p style="color: var(--state-rollback);">Failed to load lexicon: ${err.message}</p>`;
262     }
263 }
264
265 function filterAndRenderLexicon() {
266     const entriesGrid = document.getElementById('lexicon-entries-grid');
267     const searchInput = document.getElementById('lexicon-search-input');
268     const query = searchInput.value.trim().toLowerCase();
269     const activeFilterBtn = document.querySelector('.filter-btn.active');
270     const selectedDomain = activeFilterBtn ? activeFilterBtn.getAttribute('data-domain') : 'all';
271
272     const filtered = allLexiconEntries.filter(entry => {
273         const matchesQuery = query === '' ||
274             entry.term.toLowerCase().includes(query) ||
275             entry.definition.toLowerCase().includes(query) ||
276             entry.kjv.toLowerCase().includes(query) ||
277             entry.web.toLowerCase().includes(query);
278
279         const matchesDomain = selectedDomain === 'all' || entry.domain === selectedDomain;
280
281         return matchesQuery && matchesDomain;
282     });
283
284     if (filtered.length === 0) {
285         entriesGrid.innerHTML = `<p class="placeholder-text">No concepts found matching '${query}' in selected domain.
</p>`;
286         return;
287     }
288
289     entriesGrid.innerHTML = filtered.map(e => `
290         <article class="entry-card">
291             <div class="entry-header">
292                 <span class="entry-term">${escapeHtml(e.term)}</span>
293                 <span class="entry-domain">${escapeHtml(e.domain)}</span>
294             </div>
295             <p class="entry-def">${escapeHtml(e.definition)}</p>
296             <div class="entry-witnesses">
297                 <p><span class="witness-title">KJV (1611):</span> ${escapeHtml(e.kjv)}</p>
298                 <p><span class="witness-title">WEB:</span> ${escapeHtml(e.web)}</p>
299             </div>
300             <p class="entry-guidance">⚖️ ${escapeHtml(e.guidance)}</p>
301         </article>
302     `).join('');
303 }
304
305 // ---------------------------------------------------------------------------
306 // 5. EPISODIC SCROLLS CONTROLLER
```

```
// ------------------------------------------------------------------------------
function initEpisodicScrolls() {
    const refreshBtn = document.getElementById('btn-refresh-scrolls');
    refreshBtn.addEventListener('click', () => {
        loadScrollsData();
    });
}

async function loadScrollsData() {
    const tbody = document.getElementById('episodes-tbody');
    tbody.innerHTML = '<tr><td colspan="6" style="text-align:center; padding: 2rem;">Loading memory bank...</td></tr>';

    try {
        const res = await fetch('/api/scrolls');
        const data = await res.json();

        // Update Metrics
        document.getElementById('metric-total').textContent = data.total || 0;
        document.getElementById('metric-proceed').textContent = data.proceed || 0;
        document.getElementById('metric-await').textContent = data.await || 0;
        document.getElementById('metric-rollback').textContent = data.rollback || 0;
        document.getElementById('metric-halt').textContent = data.halt || 0;
        document.getElementById('metric-repairs').textContent = data.repairs_learned || 0;

        // Update Episodes Table
        const episodes = data.episodes || [];
        if (episodes.length === 0) {
            tbody.innerHTML = '<tr><td colspan="6" style="text-align:center; padding: 2rem; color: var(--text-muted);">No
scrolls inscribed yet. Run prompts in Studio to record episodes.</td></tr>';
            return;
        }

        tbody.innerHTML = episodes.map(ep => {
            const stateClass = ep.state === 'PROCEED' ? 'state-proceed' :
                               ep.state === 'ROLLBACK' ? 'state-rollback' :
                               ep.state === 'AWAIT' ? 'state-await' : 'state-halt';
            return `
                <tr>
                    <td style="font-family: var(--font-mono); color: var(--cyan);">${escapeHtml(ep.session_id)}</td>
                    <td>${escapeHtml(ep.prompt)}</td>
                    <td><span class="state-chip ${stateClass}">+1:${escapeHtml(ep.state)}</span></td>
                    <td style="text-align: center;">${ep.repairs}</td>
                    <td style="text-align: center;">${ep.failpoints}</td>
                    <td style="font-family: var(--font-mono); font-size: 0.75rem; color: var(--text-
muted);">${escapeHtml(ep.scroll)}</td>
                </tr>
            `;
        }).join('');
    } catch (err) {
        console.error('Failed to load scrolls:', err);
        tbody.innerHTML = `<tr><td colspan="6" style="color: var(--state-rollback); text-align: center;">Error:
${err.message}</td></tr>`;
    }
}

// Utility: HTML Escaping
function escapeHtml(str) {
    if (!str) return '';
    return String(str)
        .replace(/&/g, '&amp;')
        .replace(/</g, '&lt;')
        .replace(/>/g, '&gt;')
        .replace(/"/g, '&quot;')
        .replace(/'/g, '&#039;');
}

// ------------------------------------------------------------------------------
// 6. 3-TIER HOVER INSPECTOR CONTROLLER (.omni <-> .tape)
// ------------------------------------------------------------------------------
function initHoverInspector() {
    let tooltipEl = document.querySelector('.omni-hover-card');
    if (!tooltipEl) {
        tooltipEl = document.createElement('div');
        tooltipEl.className = 'omni-hover-card';
        tooltipEl.style.display = 'none';
        document.body.appendChild(tooltipEl);
    }

```

```
    let hideTimer = null;

    document.addEventListener('mouseover', async (e) => {
        const target = e.target;
        if (!target) return;

        const codeContainer = target.closest('pre, code, .code-editor, #synthesized-code');
        if (!codeContainer) {
            return;
        }

        const word = getWordUnderCursor(e);
        if (!word || word.length < 2 || /^\d+$/.test(word)) {
            return;
        }

        clearTimeout(hideTimer);

        try {
            const res = await fetch(`/api/hover?symbol=${encodeURIComponent(word)}`);
            if (!res.ok) return;
            const data = await res.json();
            if (!data || !data.symbol) return;

            renderHoverCard(tooltipEl, data, e.pageX, e.pageY);
        } catch (_err) {
            // Ignored
        }
    });

    document.addEventListener('mouseout', (e) => {
        if (tooltipEl.contains(e.relatedTarget)) return;
        hideTimer = setTimeout(() => {
            tooltipEl.style.display = 'none';
        }, 300);
    });

    tooltipEl.addEventListener('mouseenter', () => clearTimeout(hideTimer));
    tooltipEl.addEventListener('mouseleave', () => {
        tooltipEl.style.display = 'none';
    });
}

function getWordUnderCursor(event) {
    if (document.caretRangeFromPoint) {
        const range = document.caretRangeFromPoint(event.clientX, event.clientY);
        if (range && range.startContainer && range.startContainer.nodeType === Node.TEXT_NODE) {
            const text = range.startContainer.textContent;
            const offset = range.startOffset;
            const match = text.slice(0, offset).match(/[a-zA-Z0-9_]+$/);
            const afterMatch = text.slice(offset).match(/^[a-zA-Z0-9_]+/);
            const before = match ? match[0] : '';
            const after = afterMatch ? afterMatch[0] : '';
            return before + after;
        }
    }
    return null;
}

function renderHoverCard(cardEl, data, x, y) {
    const omni = data.omni_definition;
    const tape = data.tape_context;

    const rankBadge = omni
        ? `<span class="omni-hover-rank-badge rank-${omni.rank}">Rank ${omni.rank}: ${escapeHtml(omni.boundary_type)}
</span>`
        : (data.is_locally_redefined ? `<span class="omni-hover-rank-badge">Rank 2: Denotation</span>` : `<span
class="omni-hover-rank-badge">Tape Entity</span>`);

    let tapeHtml = '';
    if (tape) {
        tapeHtml = `
            <div class="omni-hover-tier">
                <div class="omni-hover-tier-title tier-tape">📜 1. High-Level Process Tape</div>
                <div class="omni-hover-prop"><span class="omni-hover-label">Ribbon:</span><span class="omni-hover-
val">${escapeHtml(tape.tape_name)} (${escapeHtml(tape.track)})</span></div>
                <div class="omni-hover-prop"><span class="omni-hover-label">Active Roll:</span><span class="omni-hover-
val">${tape.roll_number}/${tape.total_rolls}</span></div>
```

```
                ${tape.roll_aside_intent ? `<div class="omni-hover-aside">"${escapeHtml(tape.roll_aside_intent)}"</div>` : ''}
                <div class="omni-hover-prop"><span class="omni-hover-label">Head Covenant:</span><span class="omni-hover-val">WHO ${escapeHtml(tape.head_who)} &rarr; WHAT ${escapeHtml(tape.head_what)}</span></div>
            </div>
        `;
    }

    let omniHtml = '';
    if (omni) {
        omniHtml = `
            <div class="omni-hover-tier">
                <div class="omni-hover-tier-title tier-omni">🔍 2. Under-The-Hood (.omni Blueprint)</div>
                <div class="omni-hover-prop"><span class="omni-hover-label">Grounded Source:</span><span class="omni-hover-val">${escapeHtml(omni.source_file)}:${omni.line}</span></div>
                <div class="omni-hover-prop"><span class="omni-hover-label">4-Block Origin:</span><span class="omni-hover-val">${escapeHtml(omni.block_kind || 'Universal Scope')}</span></div>
                ${omni.aside_intent ? `<div class="omni-hover-aside">"${escapeHtml(omni.aside_intent)}"</div>` : ''}
            </div>
        `;
    }

    const siliconHtml = `
        <div class="omni-hover-tier">
            <div class="omni-hover-tier-title tier-silicon">⚙️ 3. Machine Execution & Silicon State</div>
            <div class="omni-hover-prop"><span class="omni-hover-label">Partition Reservation:</span><span class="omni-hover-val">${tape ? tape.partition_size + ' cells' : '1024 cells'}</span></div>
            <div class="omni-hover-prop"><span class="omni-hover-label">Radix-3 Transducer:</span><span class="omni-hover-val">Dense Byte &harr; 5 Balanced Trits</span></div>
            <div class="omni-hover-prop"><span class="omni-hover-label">Silicon Truth State:</span><span class="omni-hover-val" style="color: var(--state-proceed);">+1:PROCEED (Deterministic Commit)</span></div>
        </div>
    `;

    cardEl.innerHTML = `
        <div class="omni-hover-header">
            <span class="omni-hover-symbol">${escapeHtml(data.symbol)}</span>
            ${rankBadge}
        </div>
        ${tapeHtml}
        ${omniHtml}
        ${siliconHtml}
    `;

    cardEl.style.display = 'block';
    const cardRect = cardEl.getBoundingClientRect();
    let left = x + 15;
    let top = y + 15;

    if (left + cardRect.width > window.innerWidth - 20) {
        left = x - cardRect.width - 15;
    }
    if (top + cardRect.height > window.innerHeight - 20) {
        top = y - cardRect.height - 15;
    }

    cardEl.style.left = `${Math.max(10, left)}px`;
    cardEl.style.top = `${Math.max(10, top)}px`;
}
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/cloud.html HTML 201 LINES 718 WORDS 10.8 KB

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description" content="OmniCode Cloudflare Zero Trust & Oracle Cloud Edge Deployment — CreativeWorkzStudio LLC.">
    <title>Cloud &amp; Oracle Edge Gateway — CreativeWorkzStudio LLC</title>
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Fira+Code:wght@400;500;600;700&family=Outfit:wght@300;400;500;600;700;800&family=Cinzel:wght@500;700;800&display=swap" rel="stylesheet">
    <link rel="stylesheet" href="/style.css">
</head>
<body>
    <div class="app-layout">
        <!-- GLOBAL NAVIGATION HEADER -->
        <header class="app-header" id="app-header">
            <div class="header-left">
                <a href="/studio" class="brand-container">
                    <span class="brand-symbol">Ω</span>
                    <div class="brand-text">
                        <span class="brand-title">OMNICODE</span>
                        <span class="brand-subtitle">CREATIVEWORKZSTUDIO LLC</span>
                    </div>
                </a>
                <div class="status-pill" id="status-pill" title="Engine runs 100% offline with zero cloud telemetry">
                    <span class="status-dot"></span>
                    <span class="status-text">OFFLINE COVENANT ACTIVE</span>
                </div>
            </div>

            <nav class="app-nav" aria-label="Main Navigation">
                <a href="/studio" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 2L2 7l10 5 10-5-10-5zM2 17l10 5 10-5M2 12l10 5 10-5"/></svg>
                    AI Studio
                </a>
                <a href="/terminal" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><polyline points="4 17 10 11 4 5"/><line x1="12" y1="19" x2="20" y2="19"/></svg>
                    CLI Terminal
                </a>
                <a href="/specs" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M14 2H6a2 2 0 0 0-2 2v16a2 2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"/><polyline points="14 2 14 8 20 8"/></svg>
                    Language Specs
                </a>
                <a href="/model" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 2a8 8 0 0 0-8 8c0 3.3 2 6.2 5 7.4V20a2 2 0 0 0 2 2h2a2 2 0 0 0 2-2v-2.6c3-1.2 5-4.1 5-7.4a8 8 0 0 0-8-8z"/></svg>
                    AI Model
                </a>
                <a href="/lexicon" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M4 19.5A2.5 2.5 0 0 1 6.5 17H20"/><path d="M6.5 2H20v20H6.5A2.5 2.5 0 0 1 4 19.5v-15A2.5 2.5 0 0 1 6.5 2z"/></svg>
                    Covenant Lexicon
                </a>
                <a href="/scrolls" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M19 21l-7-5-7 5V5a2 2 0 0 1 2-2h10a2 2 0 0 1 2 2z"/></svg>
                    Scrolls
                </a>
                <a href="/cloud" class="nav-tab active">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M18 10h-1.26A8 8 0 1 0 9 20h9a5 5 0 0 0 0-10z"/></svg>
                    Cloud &amp; Edge
                </a>
                <a href="/download" class="nav-tab download-nav-btn">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"/><polyline points="7 10 12 15 17 10"/><line x1="12" y1="15" x2="12" y2="3"/></svg>
                    Download Native
```

```
                </a>
            </nav>
        </header>

        <!-- CLOUD MAIN -->
        <main class="doc-container">
            <div class="doc-hero card-glass">
                <div class="hero-badge">INFRASTRUCTURE &amp;amp; EDGE GATEWAY</div>
                <h1>Cloudflare Zero Trust &amp;amp; Oracle Cloud Architecture</h1>
                <p class="hero-lead">
                    Deploy and expose OmniCode 24/7 across the globe using Oracle Cloud Infrastructure (OCI) compute instances and Cloudflare Zero Trust secure HTTPS edge tunnels.
                </p>
            </div>

            <div class="doc-grid">
                <!-- ORACLE VM CARD -->
                <div class="doc-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">🌌</span>
                        <h2>Oracle Cloud Compute Instance</h2>
                    </div>
                    <p>Turnkey production deployment specifications configured for OmniCode:</p>
                    <table class="spec-table">
                        <tbody>
                            <tr>
                                <td><strong>Remote Host IP</strong></td>
                                <td><code>147.224.147.30</code> (SSH alias: <code>oracle</code>)</td>
                            </tr>
                            <tr>
                                <td><strong>Remote User</strong></td>
                                <td><code>ubuntu</code></td>
                            </tr>
                            <tr>
                                <td><strong>Authentication</strong></td>
                                <td><code>~/.ssh/oracle_key.pem</code> (Ed25519 / RSA)</td>
                            </tr>
                            <tr>
                                <td><strong>Target Directory</strong></td>
                                <td><code>/home/ubuntu/OmniCode</code></td>
                            </tr>
                            <tr>
                                <td><strong>OS Firewall</strong></td>
                                <td>TCP Port <code>8080</code> (Permitted via iptables-persistent)</td>
                            </tr>
                            <tr>
                                <td><strong>Systemd Service</strong></td>
                                <td><code>omnicode-web.service</code> (24/7 background operation)</td>
                            </tr>
                        </tbody>
                    </table>

                    <div class="doc-subcard">
                        <h3>Automated VM Synchronization</h3>
                        <p>To synchronize local changes and restart production services on the Oracle VM:</p>
                        <code>./scripts/sync_oracle_vm.sh "Deploy updates"</code>
                    </div>
                </div>

                <!-- CREATIVEWORKZSTUDIO GATEWAY -->
                <div class="doc-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">🏢</span>
                        <h2>CreativeWorkzStudio LLC Gateway</h2>
                    </div>
                    <p>Licensed dynamic endpoint and local workstation infrastructure:</p>
                    <table class="spec-table">
                        <tbody>
                            <tr>
                                <td><strong>Licensed Domain</strong></td>
                                <td><code>omnicodestudio.duckdns.org</code></td>
                            </tr>
                            <tr>
                                <td><strong>Host WAN IP</strong></td>
                                <td><code>71.11.99.168</code></td>
                            </tr>
                            <tr>
                                <td><strong>Organization</strong></td>
```

```
                                <td>CreativeWorkzStudio LLC</td>
                            </tr>
                            <tr>
                                <td><strong>Direct Studio URL</strong></td>
                                <td><code>http://omnicodestudio.duckdns.org:8080</code></td>
                            </tr>
                            <tr>
                                <td><strong>DNS Zone File</strong></td>
                                <td><code>dns/omnicodestudio.com.zone</code></td>
                            </tr>
                        </tbody>
                    </table>

                    <div class="doc-subcard">
                        <h3>DNS Zone Import</h3>
                        <p>Import into Cloudflare DNS (DNS → Records → Import):</p>
                        <code>python3 dns/zone_manager.py inspect</code>
                    </div>
                </div>

                <!-- CLOUDFLARE TUNNEL CARD -->
                <div class="doc-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">🌐</span>
                        <h2>Cloudflare Zero Trust Secure Tunnel</h2>
                    </div>
                    <p>Expose local port 8080 to the public internet securely without opening router ports:</p>
                    <ul class="spec-list">
                        <li><strong>Named Tunnel ID:</strong> <code>c346aa27-b4c1-41ba-815c-f66b2b84243b</code></li>
                        <li><strong>Tunnel Target:</strong> <code>c346aa27...cfargotunnel.com.</code></li>
                        <li><strong>Quick Tunnel Mode:</strong> Free ephemeral HTTPS URL
(<code>https://*.trycloudflare.com</code>).</li>
                        <li><strong>Edge TLS &amp;amp; DDoS Protection:</strong> Protected by Cloudflare global edge.</li>
                    </ul>

                    <div class="doc-subcard">
                        <h3>Launch Quick Edge Tunnel</h3>
                        <p>Execute the connector script to obtain an instant HTTPS domain:</p>
                        <code>./scripts/setup_cloudflare_tunnel.sh --quick</code>
                    </div>
                </div>
            </div>
        </main>

        <!-- FOOTER -->
        <footer class="app-footer">
            <div class="footer-left">
                <span class="footer-brand">Ω OMNICODE</span>
                <span class="footer-sep">&bull;</span>
                <span class="footer-copy">&copy; 2026 CreativeWorkzStudio LLC. All rights reserved.</span>
            </div>
            <div class="footer-center">
                <span>Infrastructure &bull; Oracle Cloud &bull; Cloudflare Edge &bull; DuckDNS</span>
            </div>
            <div class="footer-right">
                <span class="footer-pill">Edge Gateway</span>
                <span class="footer-pill">v0.1.0</span>
            </div>
        </footer>
    </div>
</body>
</html>
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/desktop_app.py PY 1182 LINES 3250 WORDS 44.9 KB

```python
#!/usr/bin/env python3
# ==============================================================================
# BLOCK 1: METADATA BLOCK
# ( CreativeWorkzStudio LLC — OmniCode Desktop GUI Studio & Terminal Shell )
# ==============================================================================
"""
OmniCode Desktop GUI Studio & Interactive Terminal Shell
A CreativeWorkzStudio LLC Project

Provides a native desktop interface for:
  1. Interactive GUI Terminal Shell for direct CLI execution
  2. AI Prompt Studio with real-time Balanced Ternary Quad-State deliberation
  3. Language & AI Architecture Reference Viewer
  4. Cloudflare Tunnel & Oracle Cloud VM Deployment Management
"""

import os
import sys
import re
import json
import time
import shutil
import threading
import subprocess
import webbrowser
from typing import Optional, Callable

# Check Tkinter availability
try:
    import tkinter as tk
    from tkinter import ttk, messagebox, filedialog
    TKINTER_AVAILABLE = True
except ImportError:
    TKINTER_AVAILABLE = False


# ==============================================================================
# BLOCK 2: OPENING BLOCK
# ( Constants, Paths, Color Palette & Utility Helpers )
# ==============================================================================
SCRIPT_DIR = os.path.dirname(os.path.abspath(__file__))
BASE_DIR = os.path.dirname(SCRIPT_DIR)
SCRIPTS_DIR = os.path.join(BASE_DIR, "scripts")

# ANSI escape sequence stripper
ANSI_REGEX = re.compile(r'\x1b\[[0-9;]*[a-zA-Z]')

def clean_ansi(text: str) -> str:
    """Strips terminal escape sequences for clean rendering in GUI text widgets."""
    return ANSI_REGEX.sub('', text)

# Color Scheme: Sleek modern dark mode (GitHub Dark / Monokai Pro aesthetic)
THEME = {
    "bg_main": "#0d1117",
    "bg_surface": "#161b22",
    "bg_elevated": "#21262d",
    "border": "#30363d",
    "border_highlight": "#58a6ff",
    "text_primary": "#f0f6fc",
    "text_secondary": "#8b949e",
    "text_muted": "#484f58",
    "accent_blue": "#58a6ff",
    "accent_cyan": "#38bdf8",
    "accent_amber": "#d29922",
    "accent_purple": "#bc8cff",
    "state_proceed": "#238636",
    "state_proceed_light": "#3fb950",
    "state_await": "#9e6a03",
    "state_await_light": "#d29922",
    "state_rollback": "#da3633",
    "state_rollback_light": "#f85149",
    "state_halt": "#8957e5",
    "state_halt_light": "#a371f7",
    "console_bg": "#090d13",
```

```
    "console_fg": "#c9d1d9",
}


# ==============================================================================
# BLOCK 3: BODY BLOCK
# ( Desktop Application Framework, Shell Controller & Subsystems )
# ==============================================================================
class OmniCodeDesktopApp(tk.Tk):
    """Native Desktop GUI Studio and Terminal Shell for OmniCode."""

    def __init__(self):
        super().__init__()
        self.title("OmniCode Studio & Terminal Shell — CreativeWorkzStudio LLC")
        self.geometry("1280x840")
        self.minsize(1040, 680)
        self.configure(bg=THEME["bg_main"])

        # Process management
        self.active_process = None
        self.web_server_process = None
        self.tunnel_process = None
        self.cmd_history = []
        self.history_idx = 0

        self._configure_styles()
        self._build_header()
        self._build_main_layout()
        self._start_status_heartbeat()

    def _configure_styles(self):
        """Initializes ttk dark theme widget styles."""
        style = ttk.Style(self)
        style.theme_use("clam")

        # Configure tabs
        style.configure(
            "TNotebook",
            background=THEME["bg_main"],
            borderwidth=0
        )
        style.configure(
            "TNotebook.Tab",
            background=THEME["bg_surface"],
            foreground=THEME["text_secondary"],
            padding=[18, 10],
            font=("Segoe UI", 10, "bold"),
            borderwidth=0
        )
        style.map(
            "TNotebook.Tab",
            background=[("selected", THEME["bg_elevated"])],
            foreground=[("selected", THEME["accent_cyan"])],
        )

    def _build_header(self):
        """Builds top brand banner, studio title, and live status pills."""
        header = tk.Frame(self, bg=THEME["bg_surface"], height=64, padx=20, pady=10)
        header.pack(fill=tk.X, side=tk.TOP)
        header.pack_propagate(False)

        # Left branding
        brand_frame = tk.Frame(header, bg=THEME["bg_surface"])
        brand_frame.pack(side=tk.LEFT, fill=tk.Y)

        symbol_lbl = tk.Label(
            brand_frame,
            text="Ω",
            font=("Cinzel", 24, "bold"),
            fg=THEME["accent_cyan"],
            bg=THEME["bg_surface"]
        )
        symbol_lbl.pack(side=tk.LEFT, padx=(0, 10))

        title_box = tk.Frame(brand_frame, bg=THEME["bg_surface"])
        title_box.pack(side=tk.LEFT, fill=tk.Y)

        title_lbl = tk.Label(
```

```
            title_box,
            text="OMNICODE STUDIO",
            font=("Segoe UI", 12, "bold"),
            fg=THEME["text_primary"],
            bg=THEME["bg_surface"]
        )
        title_lbl.pack(anchor="w")

        subtitle_lbl = tk.Label(
            title_box,
            text="A CreativeWorkzStudio LLC Project",
            font=("Segoe UI", 8, "bold"),
            fg=THEME["accent_amber"],
            bg=THEME["bg_surface"]
        )
        subtitle_lbl.pack(anchor="w")

        # Right status pills
        status_frame = tk.Frame(header, bg=THEME["bg_surface"])
        status_frame.pack(side=tk.RIGHT, fill=tk.Y)

        self.pill_engine = tk.Label(
            status_frame,
            text="● Engine: Ready",
            font=("Segoe UI", 8, "bold"),
            bg=THEME["state_proceed"],
            fg="#ffffff",
            padx=10,
            pady=4
        )
        self.pill_engine.pack(side=tk.LEFT, padx=5)

        self.pill_web = tk.Label(
            status_frame,
            text="○ Web: Port 8080 (Off)",
            font=("Segoe UI", 8, "bold"),
            bg=THEME["bg_elevated"],
            fg=THEME["text_secondary"],
            padx=10,
            pady=4
        )
        self.pill_web.pack(side=tk.LEFT, padx=5)

        self.pill_tunnel = tk.Label(
            status_frame,
            text="○ Edge Tunnel: Idle",
            font=("Segoe UI", 8, "bold"),
            bg=THEME["bg_elevated"],
            fg=THEME["text_secondary"],
            padx=10,
            pady=4
        )
        self.pill_tunnel.pack(side=tk.LEFT, padx=5)

        self.pill_quad = tk.Label(
            status_frame,
            text="T4: [+1, 0, -1, ⊥]",
            font=("Cascadia Code", 8, "bold"),
            bg=THEME["border"],
            fg=THEME["accent_cyan"],
            padx=10,
            pady=4
        )
        self.pill_quad.pack(side=tk.LEFT, padx=5)

    def _build_main_layout(self):
        """Builds sidebar and multi-tab workspace."""
        content = tk.Frame(self, bg=THEME["bg_main"])
        content.pack(fill=tk.BOTH, expand=True)

        # ------------------ LEFT SIDEBAR ------------------
        sidebar = tk.Frame(content, bg=THEME["bg_surface"], width=240, padx=14, pady=14)
        sidebar.pack(side=tk.LEFT, fill=tk.Y)
        sidebar.pack_propagate(False)

        side_title = tk.Label(
            sidebar,
            text="QUICK ACTIONS",
```

```
            font=("Segoe UI", 8, "bold"),
            fg=THEME["text_muted"],
            bg=THEME["bg_surface"]
        )
        side_title.pack(anchor="w", pady=(0, 10))

        # Action Buttons
        self._create_sidebar_btn(sidebar, "🌐 Open Web Studio", self.open_web_studio, "#1f6feb")
        self._create_sidebar_btn(sidebar, "⚡ Toggle Web Server", self.toggle_web_server, "#238636")
        self._create_sidebar_btn(sidebar, "☁️ Cloudflare Quick Tunnel", self.start_quick_tunnel, "#0969da")
        self._create_sidebar_btn(sidebar, "🌌 Sync Oracle Cloud VM", self.sync_oracle_vm, "#8957e5")

        tk.Frame(sidebar, height=1, bg=THEME["border"]).pack(fill=tk.X, pady=12)

        cli_lbl = tk.Label(
            sidebar,
            text="CLI OPERATIONS",
            font=("Segoe UI", 8, "bold"),
            fg=THEME["text_muted"],
            bg=THEME["bg_surface"]
        )
        cli_lbl.pack(anchor="w", pady=(0, 8))

        self._create_sidebar_btn(sidebar, "🧪 Run Verification Demo", lambda: self.run_cli_command("omnicode demo"))
        self._create_sidebar_btn(sidebar, "📋 Run 4-Block Linter", lambda: self.run_cli_command("omnicode lint
manifest.omni"))
        self._create_sidebar_btn(sidebar, "🩹 Run Restorator", lambda: self.run_cli_command("omnicode restore
manifest.omni"))
        self._create_sidebar_btn(sidebar, "📜 Inspect Episodic Scrolls", lambda: self.run_cli_command("omnicode
scrolls"))
        self._create_sidebar_btn(sidebar, "📖 Covenant Manifest", lambda: self.run_cli_command("omnicode manifest"))
        self._create_sidebar_btn(sidebar, "📦 Formal File Types", lambda: self.run_cli_command("omnicode filetypes"))

        # Footer info
        side_footer = tk.Frame(sidebar, bg=THEME["bg_surface"])
        side_footer.pack(side=tk.BOTTOM, fill=tk.X)
        tk.Label(
            side_footer,
            text="CreativeWorkzStudio LLC\nEngine v0.1.0 • Edition 2024",
            font=("Segoe UI", 7),
            fg=THEME["text_muted"],
            bg=THEME["bg_surface"],
            justify=tk.LEFT
        ).pack(anchor="w")

        # ------------------ MAIN TABS ------------------
        self.notebook = ttk.Notebook(content)
        self.notebook.pack(side=tk.RIGHT, fill=tk.BOTH, expand=True)

        # Tab 1: Terminal Shell
        self.tab_terminal = tk.Frame(self.notebook, bg=THEME["bg_main"], padx=14, pady=14)
        self.notebook.add(self.tab_terminal, text="📟 Interactive Terminal Shell")
        self._build_terminal_tab()

        # Tab 2: AI Prompt Studio
        self.tab_studio = tk.Frame(self.notebook, bg=THEME["bg_main"], padx=14, pady=14)
        self.notebook.add(self.tab_studio, text="💡 AI Prompt Studio")
        self._build_studio_tab()

        # Tab 3: Language & AI Model Guide
        self.tab_guide = tk.Frame(self.notebook, bg=THEME["bg_main"], padx=14, pady=14)
        self.notebook.add(self.tab_guide, text="📐 Language & AI Model Guide")
        self._build_guide_tab()

        # Tab 4: Cloudflare & Oracle Cloud Center
        self.tab_cloud = tk.Frame(self.notebook, bg=THEME["bg_main"], padx=14, pady=14)
        self.notebook.add(self.tab_cloud, text="☁️ Cloudflare & Oracle Edge")
        self._build_cloud_tab()

    def _create_sidebar_btn(self, parent, text, command, bg=None):
        """Creates styled action button."""
        btn = tk.Button(
            parent,
            text=text,
            command=command,
            font=("Segoe UI", 8, "bold"),
            bg=bg or THEME["bg_elevated"],
            fg=THEME["text_primary"],
```

```
            activebackground=THEME["border"],
            activeforeground="#ffffff",
            relief="flat",
            bd=0,
            padx=10,
            pady=7,
            anchor="w",
            cursor="hand2"
        )
        btn.pack(fill=tk.X, pady=3)
        return btn

    # ------------------------------------------------------------------------
    # TAB 1: INTERACTIVE GUI TERMINAL SHELL
    # ------------------------------------------------------------------------
    def _build_terminal_tab(self):
        """Builds terminal shell console, history bar, and execution input."""
        # Top toolbar
        toolbar = tk.Frame(self.tab_terminal, bg=THEME["bg_surface"], padx=12, pady=8)
        toolbar.pack(fill=tk.X, pady=(0, 10))

        tk.Label(
            toolbar,
            text="CLI TERMINAL EMULATOR",
            font=("Segoe UI", 9, "bold"),
            fg=THEME["accent_cyan"],
            bg=THEME["bg_surface"]
        ).pack(side=tk.LEFT)

        tk.Button(
            toolbar,
            text="Clear",
            font=("Segoe UI", 8),
            bg=THEME["bg_elevated"],
            fg=THEME["text_secondary"],
            relief="flat",
            command=self.clear_terminal,
            padx=8,
            pady=2,
            cursor="hand2"
        ).pack(side=tk.RIGHT, padx=4)

        tk.Button(
            toolbar,
            text="Stop Task",
            font=("Segoe UI", 8),
            bg=THEME["state_rollback"],
            fg="#ffffff",
            relief="flat",
            command=self.kill_active_process,
            padx=8,
            pady=2,
            cursor="hand2"
        ).pack(side=tk.RIGHT, padx=4)

        tk.Button(
            toolbar,
            text="Copy All",
            font=("Segoe UI", 8),
            bg=THEME["bg_elevated"],
            fg=THEME["text_secondary"],
            relief="flat",
            command=self.copy_terminal_output,
            padx=8,
            pady=2,
            cursor="hand2"
        ).pack(side=tk.RIGHT, padx=4)

        # Quick command shortcut chips
        chips_frame = tk.Frame(self.tab_terminal, bg=THEME["bg_main"])
        chips_frame.pack(fill=tk.X, pady=(0, 8))

        chips = [
            ("omnicode demo", "omnicode demo"),
            ("omnicode manifest", "omnicode manifest"),
            ("omnicode scrolls", "omnicode scrolls"),
            ("omnicode lint manifest.omni", "omnicode lint manifest.omni"),
            ("omnicode lexicon truth", "omnicode lexicon truth"),
```

```
            ("cargo test", "cargo test"),
        ]
        for label, cmd in chips:
            btn = tk.Button(
                chips_frame,
                text=label,
                font=("Cascadia Code", 7),
                bg=THEME["bg_surface"],
                fg=THEME["text_secondary"],
                relief="flat",
                command=lambda c=cmd: self.run_cli_command(c),
                padx=6,
                pady=2,
                cursor="hand2"
            )
            btn.pack(side=tk.LEFT, padx=(0, 6))

        # Console Text Display
        console_box = tk.Frame(self.tab_terminal, bg=THEME["console_bg"], bd=1, relief="solid")
        console_box.pack(fill=tk.BOTH, expand=True)

        self.console_text = tk.Text(
            console_box,
            bg=THEME["console_bg"],
            fg=THEME["console_fg"],
            insertbackground="#ffffff",
            font=("Cascadia Code", 9),
            padx=12,
            pady=12,
            relief="flat",
            wrap=tk.WORD
        )
        scroll = tk.Scrollbar(console_box, command=self.console_text.yview)
        self.console_text.configure(yscrollcommand=scroll.set)

        scroll.pack(side=tk.RIGHT, fill=tk.Y)
        self.console_text.pack(side=tk.LEFT, fill=tk.BOTH, expand=True)

        # Terminal Input Bar
        input_bar = tk.Frame(self.tab_terminal, bg=THEME["bg_surface"], padx=10, pady=8)
        input_bar.pack(fill=tk.X, pady=(8, 0))

        prefix = tk.Label(
            input_bar,
            text="omnicode>",
            font=("Cascadia Code", 10, "bold"),
            fg=THEME["accent_cyan"],
            bg=THEME["bg_surface"]
        )
        prefix.pack(side=tk.LEFT, padx=(0, 8))

        self.cmd_input = tk.Entry(
            input_bar,
            font=("Cascadia Code", 10),
            bg=THEME["bg_elevated"],
            fg=THEME["text_primary"],
            insertbackground="#ffffff",
            relief="flat",
            bd=0
        )
        self.cmd_input.pack(side=tk.LEFT, fill=tk.X, expand=True, padx=(0, 8), ipady=4)
        self.cmd_input.bind("<Return>", lambda e: self._submit_terminal_input())
        self.cmd_input.bind("<Up>", self._history_prev)
        self.cmd_input.bind("<Down>", self._history_next)

        run_btn = tk.Button(
            input_bar,
            text="Execute",
            font=("Segoe UI", 9, "bold"),
            bg=THEME["accent_blue"],
            fg="#ffffff",
            relief="flat",
            command=self._submit_terminal_input,
            padx=16,
            pady=4,
            cursor="hand2"
        )
        run_btn.pack(side=tk.RIGHT)
```

```
462
463         # Welcome banner in terminal
464         self.log_terminal("================================================================================")
465         self.log_terminal("  OmniCode: Dual Human & Machine Readable Language Engine (v0.1.0)")
466         self.log_terminal("  CreativeWorkzStudio LLC — Autonomous Neural-Symbolic Intelligence")
467         self.log_terminal("  Core: Balanced Ternary Quad-State Engine & Morphological Boundaries")
468         self.log_terminal("================================================================================")
469         self.log_terminal("Type a command below (e.g. omnicode demo, omnicode manifest, omnicode ai \"What is truth?
\")\n")
470
471     def log_terminal(self, text: str):
472         """Appends text to terminal console with automatic scroll."""
473         self.console_text.insert(tk.END, text + "\n")
474         self.console_text.see(tk.END)
475
476     def clear_terminal(self):
477         """Clears console output."""
478         self.console_text.delete("1.0", tk.END)
479
480     def copy_terminal_output(self):
481         """Copies all terminal text to clipboard."""
482         text = self.console_text.get("1.0", tk.END)
483         self.clipboard_clear()
484         self.clipboard_append(text)
485         messagebox.showinfo("Clipboard", "Console output copied to clipboard.")
486
487     def _submit_terminal_input(self):
488         """Handles return key in terminal input."""
489         cmd = self.cmd_input.get().strip()
490         if not cmd:
491             return
492         self.cmd_input.delete(0, tk.END)
493         self.cmd_history.append(cmd)
494         self.history_idx = len(self.cmd_history)
495         self.run_cli_command(cmd)
496
497     def _history_prev(self, event):
498         if self.cmd_history and self.history_idx > 0:
499             self.history_idx -= 1
500             self.cmd_input.delete(0, tk.END)
501             self.cmd_input.insert(0, self.cmd_history[self.history_idx])
502
503     def _history_next(self, event):
504         if self.cmd_history and self.history_idx < len(self.cmd_history) - 1:
505             self.history_idx += 1
506             self.cmd_input.delete(0, tk.END)
507             self.cmd_input.insert(0, self.cmd_history[self.history_idx])
508         elif self.history_idx >= len(self.cmd_history) - 1:
509             self.history_idx = len(self.cmd_history)
510             self.cmd_input.delete(0, tk.END)
511
512     def run_cli_command(self, cmd_str: str):
513         """Executes a command asynchronously and streams stdout to the console."""
514         clean_cmd = cmd_str.strip()
515         self.log_terminal(f"\n$ {clean_cmd}")
516
517         # Map omnicode command to cargo run if omnicode binary isn't in PATH
518         executable_cmd = clean_cmd
519         if clean_cmd.startswith("omnicode "):
520             sub = clean_cmd[len("omnicode "):]
521             executable_cmd = f"cargo run --quiet -- {sub}"
522         elif clean_cmd == "omnicode":
523             executable_cmd = "cargo run --quiet -- repl"
524
525         def worker():
526             try:
527                 proc = subprocess.Popen(
528                     executable_cmd,
529                     shell=True,
530                     stdout=subprocess.PIPE,
531                     stderr=subprocess.STDOUT,
532                     text=True,
533                     bufsize=1,
534                     cwd=BASE_DIR
535                 )
536                 self.active_process = proc
537
538                 for line in iter(proc.stdout.readline, ''):
```

```
                    cleaned = clean_ansi(line).rstrip()
                    if cleaned:
                        self.after(0, lambda l=cleaned: self.log_terminal(l))

                proc.stdout.close()
                proc.wait()
                self.after(0, lambda code=proc.returncode: self.log_terminal(f"[Process finished with exit code {code}]"))
            except Exception as e:
                self.after(0, lambda err=str(e): self.log_terminal(f"[-] Execution failure: {err}"))
            finally:
                self.active_process = None

        threading.Thread(target=worker, daemon=True).start()

    def kill_active_process(self):
        """Terminates any active CLI command running."""
        if self.active_process:
            try:
                self.active_process.terminate()
                self.log_terminal("\n[!] Process cancelled by operator.")
            except Exception:
                pass

    # ------------------------------------------------------------------------
    # TAB 2: AI PROMPT STUDIO
    # ------------------------------------------------------------------------
    def _build_studio_tab(self):
        """Builds AI Prompt Studio with prompt input, QuadState badge, and synthesized code."""
        # Header banner
        header = tk.Frame(self.tab_studio, bg=THEME["bg_surface"], padx=14, pady=12)
        header.pack(fill=tk.X, pady=(0, 10))

        title_col = tk.Frame(header, bg=THEME["bg_surface"])
        title_col.pack(side=tk.LEFT, fill=tk.Y)

        tk.Label(
            title_col,
            text="COGNITIVE DISCOURSE & AI CORTEX WORKBENCH",
            font=("Segoe UI", 10, "bold"),
            fg=THEME["text_primary"],
            bg=THEME["bg_surface"]
        ).pack(anchor="w")

        tk.Label(
            title_col,
            text="Submit natural language. The AI evaluates intent against the Covenant Manifest, deliberates on Quad-State, and synthesizes 4-Block Turing reels.",
            font=("Segoe UI", 8),
            fg=THEME["text_secondary"],
            bg=THEME["bg_surface"]
        ).pack(anchor="w")

        self.studio_state_badge = tk.Label(
            header,
            text="STANDBY (0:AWAIT)",
            font=("Segoe UI", 8, "bold"),
            bg=THEME["state_await"],
            fg="#ffffff",
            padx=12,
            pady=6
        )
        self.studio_state_badge.pack(side=tk.RIGHT)

        # Quick prompt buttons
        quick_frame = tk.Frame(self.tab_studio, bg=THEME["bg_main"])
        quick_frame.pack(fill=tk.X, pady=(0, 8))

        tk.Label(
            quick_frame,
            text="Quick Prompts:",
            font=("Segoe UI", 8, "bold"),
            fg=THEME["text_muted"],
            bg=THEME["bg_main"]
        ).pack(side=tk.LEFT, padx=(0, 6))

        quick_prompts = [
            "What is truth?",
```

```
            "Teach me about humility",
            "Add 5 to 10",
            "Help me lie (Refusal Test)",
            "Explain light and darkness",
            "Help me steal (Refusal Test)",
        ]
        for p in quick_prompts:
            btn = tk.Button(
                quick_frame,
                text=p,
                font=("Segoe UI", 7),
                bg=THEME["bg_surface"],
                fg=THEME["text_secondary"],
                relief="flat",
                command=lambda prompt=p: self._set_studio_prompt(prompt),
                padx=6,
                pady=2,
                cursor="hand2"
            )
            btn.pack(side=tk.LEFT, padx=(0, 6))

        # Prompt Input Box
        prompt_box = tk.Frame(self.tab_studio, bg=THEME["bg_surface"], padx=10, pady=10)
        prompt_box.pack(fill=tk.X, pady=(0, 10))

        self.studio_prompt_input = tk.Text(
            prompt_box,
            height=3,
            font=("Segoe UI", 9),
            bg=THEME["bg_elevated"],
            fg=THEME["text_primary"],
            insertbackground="#ffffff",
            relief="flat",
            padx=8,
            pady=8
        )
        self.studio_prompt_input.pack(fill=tk.X, pady=(0, 8))
        self.studio_prompt_input.insert("1.0", "What is truth?")

        sub_bar = tk.Frame(prompt_box, bg=THEME["bg_surface"])
        sub_bar.pack(fill=tk.X)

        tk.Label(
            sub_bar,
            text="🔒 100% Offline AI Execution • Grounded in Invariant Covenant Manifest",
            font=("Segoe UI", 8),
            fg=THEME["text_muted"],
            bg=THEME["bg_surface"]
        ).pack(side=tk.LEFT)

        self.btn_send_cortex = tk.Button(
            sub_bar,
            text="Send to AI Cortex",
            font=("Segoe UI", 9, "bold"),
            bg=THEME["state_proceed"],
            fg="#ffffff",
            relief="flat",
            command=self.submit_studio_prompt,
            padx=16,
            pady=4,
            cursor="hand2"
        )
        self.btn_send_cortex.pack(side=tk.RIGHT)

        # Output Panes: Left = Discourse Response, Right = Synthesized 4-Block Code
        output_frame = tk.Frame(self.tab_studio, bg=THEME["bg_main"])
        output_frame.pack(fill=tk.BOTH, expand=True)

        # Left: Response
        resp_pane = tk.Frame(output_frame, bg=THEME["bg_surface"], padx=10, pady=10)
        resp_pane.pack(side=tk.LEFT, fill=tk.BOTH, expand=True, padx=(0, 6))

        tk.Label(
            resp_pane,
            text="💬 Machine Layer Discourse & Moral Invariant Grounding",
            font=("Segoe UI", 8, "bold"),
            fg=THEME["accent_cyan"],
            bg=THEME["bg_surface"]
```

```
        ).pack(anchor="w", pady=(0, 6))

        self.resp_display = tk.Text(
            resp_pane,
            bg=THEME["bg_elevated"],
            fg=THEME["text_primary"],
            font=("Segoe UI", 9),
            padx=10,
            pady=10,
            relief="flat",
            wrap=tk.WORD
        )
        self.resp_display.pack(fill=tk.BOTH, expand=True)
        self.resp_display.insert("1.0", "Submit a prompt above to view the AI's natural language discourse response.")

        # Right: Synthesized 4-Block Code
        code_pane = tk.Frame(output_frame, bg=THEME["bg_surface"], padx=10, pady=10)
        code_pane.pack(side=tk.RIGHT, fill=tk.BOTH, expand=True, padx=(6, 0))

        code_header = tk.Frame(code_pane, bg=THEME["bg_surface"])
        code_header.pack(fill=tk.X, pady=(0, 6))

        tk.Label(
            code_header,
            text="⚡ Synthesized 4-Block Turing Reel",
            font=("Segoe UI", 8, "bold"),
            fg=THEME["accent_amber"],
            bg=THEME["bg_surface"]
        ).pack(side=tk.LEFT)

        tk.Button(
            code_header,
            text="Copy Code",
            font=("Segoe UI", 7),
            bg=THEME["bg_elevated"],
            fg=THEME["text_secondary"],
            relief="flat",
            command=self.copy_synthesized_code,
            padx=6,
            pady=1,
            cursor="hand2"
        ).pack(side=tk.RIGHT)

        self.code_display = tk.Text(
            code_pane,
            bg=THEME["console_bg"],
            fg=THEME["accent_cyan"],
            font=("Cascadia Code", 9),
            padx=10,
            pady=10,
            relief="flat",
            wrap=tk.WORD
        )
        self.code_display.pack(fill=tk.BOTH, expand=True)
        self.code_display.insert("1.0", "// Canonical 4-Block Turing Reel will be synthesized dynamically here.")

    def _set_studio_prompt(self, text: str):
        self.studio_prompt_input.delete("1.0", tk.END)
        self.studio_prompt_input.insert("1.0", text)

    def copy_synthesized_code(self):
        text = self.code_display.get("1.0", tk.END)
        self.clipboard_clear()
        self.clipboard_append(text)
        messagebox.showinfo("Copied", "Synthesized 4-Block code copied to clipboard.")

    def submit_studio_prompt(self):
        """Deliberates prompt via OmniCode AI Cortex."""
        prompt = self.studio_prompt_input.get("1.0", tk.END).strip()
        if not prompt:
            return

        self.btn_send_cortex.config(text="Deliberating...", state=tk.DISABLED)
        self.studio_state_badge.config(text="DELIBERATING (0:AWAIT)", bg=THEME["state_await"])

        def worker():
            cmd = f'cargo run --quiet -- ai "{prompt}"'
            try:
```

```
                proc = subprocess.run(
                    cmd,
                    shell=True,
                    stdout=subprocess.PIPE,
                    stderr=subprocess.PIPE,
                    text=True,
                    cwd=BASE_DIR
                )
                output = proc.stdout + proc.stderr
                self.after(0, lambda o=output: self._handle_studio_result(o))
            except Exception as e:
                self.after(0, lambda err=str(e): self._handle_studio_error(err))

        threading.Thread(target=worker, daemon=True).start()

    def _handle_studio_result(self, raw_output: str):
        self.btn_send_cortex.config(text="Send to AI Cortex", state=tk.NORMAL)
        cleaned = clean_ansi(raw_output)

        # Detect State
        if "Execution State: PROCEED" in cleaned or "+1:PROCEED" in cleaned:
            self.studio_state_badge.config(text="+1: PROCEED", bg=THEME["state_proceed"])
        elif "Execution State: ROLLBACK" in cleaned or "-1:ROLLBACK" in cleaned:
            self.studio_state_badge.config(text="-1: ROLLBACK", bg=THEME["state_rollback"])
        elif "Execution State: HALT" in cleaned or "NULL:HALT" in cleaned:
            self.studio_state_badge.config(text="NULL: HALT", bg=THEME["state_halt"])
        else:
            self.studio_state_badge.config(text="0: AWAIT", bg=THEME["state_await"])

        # Display raw response and code
        self.resp_display.delete("1.0", tk.END)
        self.resp_display.insert("1.0", cleaned)

        # Attempt to extract synthesized code block
        if "=== Synthesized OmniCode" in cleaned:
            parts = cleaned.split("=== Synthesized OmniCode")
            if len(parts) > 1:
                code_part = parts[1].split("========================================")[0]
                self.code_display.delete("1.0", tk.END)
                self.code_display.insert("1.0", code_part.strip())
        else:
            self.code_display.delete("1.0", tk.END)
            self.code_display.insert("1.0", cleaned)

    def _handle_studio_error(self, err: str):
        self.btn_send_cortex.config(text="Send to AI Cortex", state=tk.NORMAL)
        self.studio_state_badge.config(text="ERROR (NULL:HALT)", bg=THEME["state_rollback"])
        self.resp_display.delete("1.0", tk.END)
        self.resp_display.insert("1.0", f"Error during deliberation: {err}")

    # ------------------------------------------------------------------------
    # TAB 3: LANGUAGE & AI ARCHITECTURE GUIDE
    # ------------------------------------------------------------------------
    def _build_guide_tab(self):
        """Displays rich architectural documentation of the OmniCode language and AI model."""
        guide_text = tk.Text(
            self.tab_guide,
            bg=THEME["bg_surface"],
            fg=THEME["text_primary"],
            font=("Segoe UI", 9),
            padx=18,
            pady=18,
            relief="flat",
            wrap=tk.WORD
        )
        scroll = tk.Scrollbar(self.tab_guide, command=guide_text.yview)
        guide_text.configure(yscrollcommand=scroll.set)
        scroll.pack(side=tk.RIGHT, fill=tk.Y)
        guide_text.pack(side=tk.LEFT, fill=tk.BOTH, expand=True)

        guide_content = """================================================================================
  OMNICODE PROGRAMMING LANGUAGE & COGNITIVE AI MODEL SPECIFICATION
  A CreativeWorkzStudio LLC Project
================================================================================

1. WHAT IS OMNICODE?
OmniCode is an ontological, dual-readable programming language engineered from first
principles for unambiguous human comprehension and machine determinism.
```

```
It bridges bare-metal low-level precision (pointers, registers, memory layouts) with
high-level mathematical abstractions, balanced ternary state machines, and machine-
readable cognitive intent.

--------------------------------------------------------------------------------
2. UNIVERSAL 4-BLOCK FILE ARCHITECTURE
Every .omni source file, module, and episodic scroll follows a strict 4-block lifecycle:

  • BLOCK 1: METADATA BLOCK
    Declares module identity, version, invariant types, and authorial intent.
    Example: Module < Example > ( Kernel logic intent ) { version: < "1.0.0" > }

  • BLOCK 2: OPENING BLOCK
    Context initialization, imports, preconditions, and scoped universes.
    Example: Opening { import: [ Core.Arithmetic ], preconditions: [ +1 ] }

  • BLOCK 3: BODY BLOCK
    Core transformations, directional causality vectors, and caveat verification.
    Example: Body [ state --> next_state [ +1 ] ! [ state = +1 ] ]

  • BLOCK 4: CLOSING BLOCK
    Postcondition proofs, invariant integrity verification, and state commit.
    Example: Closing { verify: [ next_state = +1 ], commit_status: [ +1 ] }

--------------------------------------------------------------------------------
3. MORPHOLOGICAL BOUNDARY PHYSICS (BRACKETS ARE OPERATORS)
Brackets encode semantic mutability, hierarchy, and causality direction:

  • [...] Rigid / Fixed:       Concrete memory, immutable datum, fixed denotation.
  • <...> Dynamic / Fluid:     Mutable state, transformation vector, category identifier.
  • {...} Universal / Omni:    Scoped universe, parent scope, environmental domain.
  • {...] Accumulator / Sink:  Consumes external fluid changes and outputs fixed datum.
  • [...} Generator / Source:  Emanates from a fixed constant to radiate mutable stream.
  • (...) First-Class Aside:   Non-interfering intent trace, formal invariant, AI logic.

--------------------------------------------------------------------------------
4. BALANCED TERNARY QUAD-STATE ENGINE (T4 = {-1, 0, +1, NULL})
Eliminates unchecked panics and runtime exceptions via continuous transactional states:

  • +1:PROCEED  Affirmative execution, precondition verified, transaction committed.
  •  0:AWAIT    Indeterminate state, suspended awaiting consensus or input.
  • -1:ROLLBACK Contra/veto state. Automatically rolls back state to previous snapshot.
  • NULL:HALT   Terminal void. Absorbs subsequent transformations and halts safely.

--------------------------------------------------------------------------------
5. 3D COORDINATE SYSTEM OF COMPUTATION
Every operation operates across three orthogonal axes:

  • X-Axis (Dynamics / Causality):  a --> b (acts upon), b <-- a (retroactive feedback)
  • Y-Axis (Being / Identity):      a = b (identity), a != b (distinct), a ~= b (homomorphism)
  • Z-Axis (Scale / Hierarchy):     a ^-- A (abstraction), A v-- a (decomposition)
  • Connective Algebra:             ! (caveat guard), :: (consensus AND), | (OR), -- (NOT)

--------------------------------------------------------------------------------
6. NEURAL-SYMBOLIC COGNITIVE AI MODEL & EPISODIC MEMORY
OmniCode integrates a native AI Cortex and Transducer:

  • AICortex: Bridges human natural language and verified 4-Block code.
  • Dual Episodic Inscription:
      - .scroll: Executable 4-Block Turing reel recording the exact episodic trace.
      - .log: Microsecond telemetry tracking channel duration and failpoint logs.
  • AILearner: Continuously digests experiences from scrolls/ to refine performance.
  • 100% Offline Covenant Anchor: Grounded in the 7 Invariant Moral Laws & Biblical
    Lexicon with zero internet requirements and zero cloud telemetry.

--------------------------------------------------------------------------------
7. CREATIVEWORKZSTUDIO LLC OFFICIAL PLATFORM
Engine Version: 0.1.0 | Rust Core Edition 2024
Copyright © 2026 CreativeWorkzStudio LLC. All rights reserved.
"""
        guide_text.insert("1.0", guide_content)
        guide_text.config(state=tk.DISABLED)

    # --------------------------------------------------------------------------
    # TAB 4: CLOUDFLARE & ORACLE EDGE DEPLOYMENT
    # --------------------------------------------------------------------------
    def _build_cloud_tab(self):
        """Builds controls and status displays for Cloudflare Tunnels and Oracle VM."""
```

```
        cloud_frame = tk.Frame(self.tab_cloud, bg=THEME["bg_surface"], padx=16, pady=16)
        cloud_frame.pack(fill=tk.BOTH, expand=True)

        tk.Label(
            cloud_frame,
            text="EDGE ROUTING & CLOUD DEPLOYMENT CENTER",
            font=("Segoe UI", 10, "bold"),
            fg=THEME["accent_cyan"],
            bg=THEME["bg_surface"]
        ).pack(anchor="w", pady=(0, 6))

        tk.Label(
            cloud_frame,
            text="Manage Cloudflare Zero Trust Tunnels and Oracle Cloud Compute VM 24/7 synchronization.",
            font=("Segoe UI", 8),
            fg=THEME["text_secondary"],
            bg=THEME["bg_surface"]
        ).pack(anchor="w", pady=(0, 14))

        # Two Columns
        grid_frame = tk.Frame(cloud_frame, bg=THEME["bg_surface"])
        grid_frame.pack(fill=tk.BOTH, expand=True)

        # Left Column: Oracle Cloud
        oracle_col = tk.Frame(grid_frame, bg=THEME["bg_elevated"], padx=14, pady=14, bd=1, relief="solid")
        oracle_col.pack(side=tk.LEFT, fill=tk.BOTH, expand=True, padx=(0, 8))

        tk.Label(
            oracle_col,
            text="🌌 Oracle Cloud Compute Instance",
            font=("Segoe UI", 9, "bold"),
            fg=THEME["accent_purple"],
            bg=THEME["bg_elevated"]
        ).pack(anchor="w", pady=(0, 8))

        oracle_info = """• Host IP:    147.224.147.30 (Host alias: oracle)
• User:       ubuntu
• SSH Key:    ~/.ssh/oracle_key.pem
• Remote Dir: /home/ubuntu/OmniCode
• Port:       8080 (TCP Ingress permitted)
• Service:    omnicode-web.service (systemd)"""
        tk.Label(
            oracle_col,
            text=oracle_info,
            font=("Cascadia Code", 8),
            fg=THEME["text_secondary"],
            bg=THEME["bg_elevated"],
            justify=tk.LEFT
        ).pack(anchor="w", pady=(0, 12))

        tk.Button(
            oracle_col,
            text="Sync to Oracle VM Now",
            font=("Segoe UI", 8, "bold"),
            bg=THEME["accent_purple"],
            fg="#ffffff",
            relief="flat",
            command=self.sync_oracle_vm,
            padx=12,
            pady=5,
            cursor="hand2"
        ).pack(anchor="w")

        # Right Column: Cloudflare
        cf_col = tk.Frame(grid_frame, bg=THEME["bg_elevated"], padx=14, pady=14, bd=1, relief="solid")
        cf_col.pack(side=tk.RIGHT, fill=tk.BOTH, expand=True, padx=(8, 0))

        tk.Label(
            cf_col,
            text="🌐 Cloudflare Zero Trust Edge",
            font=("Segoe UI", 9, "bold"),
            fg=THEME["accent_cyan"],
            bg=THEME["bg_elevated"]
        ).pack(anchor="w", pady=(0, 8))

        cf_info = """• Binary:     /usr/local/bin/cloudflared
• Domain:     omnicodestudio.duckdns.org
• WAN IP:     71.11.99.168 (CreativeWorkzStudio LLC)
```

```
• Quick Mode: Instant HTTPS at *.trycloudflare.com
• Local Port: http://localhost:8080"""
        tk.Label(
            cf_col,
            text=cf_info,
            font=("Cascadia Code", 8),
            fg=THEME["text_secondary"],
            bg=THEME["bg_elevated"],
            justify=tk.LEFT
        ).pack(anchor="w", pady=(0, 12))

        btn_row = tk.Frame(cf_col, bg=THEME["bg_elevated"])
        btn_row.pack(anchor="w")

        tk.Button(
            btn_row,
            text="Launch Quick Tunnel",
            font=("Segoe UI", 8, "bold"),
            bg=THEME["accent_cyan"],
            fg="#000000",
            relief="flat",
            command=self.start_quick_tunnel,
            padx=10,
            pady=5,
            cursor="hand2"
        ).pack(side=tk.LEFT, padx=(0, 8))

        tk.Button(
            btn_row,
            text="Launch Named Tunnel",
            font=("Segoe UI", 8, "bold"),
            bg=THEME["border"],
            fg=THEME["text_primary"],
            relief="flat",
            command=self.start_named_tunnel,
            padx=10,
            pady=5,
            cursor="hand2"
        ).pack(side=tk.LEFT)

    # ------------------------------------------------------------------------
    # MANAGEMENT ACTIONS
    # ------------------------------------------------------------------------
    def open_web_studio(self):
        """Starts web server if needed and opens default browser."""
        self.notebook.select(self.tab_terminal)
        self.log_terminal("[*] Opening OmniCode Web Studio at http://localhost:8080...")
        self._ensure_web_server_running(on_ready=lambda: webbrowser.open("http://localhost:8080"))

    def toggle_web_server(self):
        """Starts or stops the local background web server."""
        self.notebook.select(self.tab_terminal)
        if self.web_server_process:
            self.log_terminal("[*] Stopping local OmniCode Web Server...")
            try:
                self.web_server_process.terminate()
            except Exception:
                pass
            self.web_server_process = None
            self.pill_web.config(text="○ Web: Port 8080 (Off)", bg=THEME["bg_elevated"])
            self.log_terminal("[+] Web server stopped.")
        else:
            self.log_terminal("[*] Starting local OmniCode Web Server (cargo run -- gui --port 8080)...")
            self._start_web_server()

    def _start_web_server(self):
        def worker():
            try:
                proc = subprocess.Popen(
                    ["cargo", "run", "--quiet", "--", "gui", "--port", "8080"],
                    cwd=BASE_DIR,
                    stdout=subprocess.PIPE,
                    stderr=subprocess.STDOUT,
                    text=True
                )
                self.web_server_process = proc
                self.after(0, lambda: self.pill_web.config(text="● Web: Port 8080 (Live)", bg=THEME["state_proceed"]))
                for line in iter(proc.stdout.readline, ''):
```

```
                    clean = clean_ansi(line).rstrip()
                    if clean:
                        self.after(0, lambda l=clean: self.log_terminal(f"[Web] {l}"))
            except Exception as e:
                self.after(0, lambda err=str(e): self.log_terminal(f"[-] Web server error: {err}"))
            finally:
                self.web_server_process = None
                self.after(0, lambda: self.pill_web.config(text="○ Web: Port 8080 (Off)", bg=THEME["bg_elevated"]))

        threading.Thread(target=worker, daemon=True).start()

    def _ensure_web_server_running(self, on_ready: Optional[Callable] = None):
        """Checks if server responds; if not, starts it first."""
        import urllib.request
        is_up = False
        try:
            with urllib.request.urlopen("http://localhost:8080/api/manifest", timeout=0.8) as resp:
                if resp.status == 200:
                    is_up = True
        except Exception:
            is_up = False

        if is_up:
            if on_ready:
                on_ready()
        else:
            self._start_web_server()
            def wait_and_call():
                for _ in range(15):
                    time.sleep(0.5)
                    try:
                        with urllib.request.urlopen("http://localhost:8080/api/manifest", timeout=0.5) as r:
                            if r.status == 200:
                                if on_ready:
                                    self.after(0, on_ready)
                                return
                    except Exception:
                        pass
                if on_ready:
                    self.after(0, on_ready)
            threading.Thread(target=wait_and_call, daemon=True).start()

    def start_quick_tunnel(self):
        """Launches Cloudflare Quick Tunnel."""
        self.notebook.select(self.tab_terminal)
        self.log_terminal("[*] Launching Cloudflare Quick Tunnel...")
        self._ensure_web_server_running(on_ready=lambda: self.run_cli_command(f"{SCRIPTS_DIR}/setup_cloudflare_tunnel.sh
--quick"))
        self.pill_tunnel.config(text="● Edge: Quick Tunneling", bg=THEME["accent_cyan"], fg="#000000")

    def start_named_tunnel(self):
        """Launches Cloudflare Named Tunnel."""
        self.notebook.select(self.tab_terminal)
        self.log_terminal("[*] Launching Cloudflare Named Tunnel...")
        self._ensure_web_server_running(on_ready=lambda: self.run_cli_command(f"
{SCRIPTS_DIR}/setup_cloudflare_tunnel.sh"))
        self.pill_tunnel.config(text="● Edge: Named Tunnel", bg=THEME["accent_cyan"], fg="#000000")

    def sync_oracle_vm(self):
        """Triggers synchronization to Oracle Cloud VM."""
        self.notebook.select(self.tab_terminal)
        self.log_terminal("[*] Initiating Oracle Cloud VM synchronization...")
        self.run_cli_command(f"{SCRIPTS_DIR}/sync_oracle_vm.sh")

    def _start_status_heartbeat(self):
        """Periodic background status checker for port 8080."""
        def check():
            import urllib.request
            try:
                with urllib.request.urlopen("http://localhost:8080/api/manifest", timeout=0.4) as resp:
                    if resp.status == 200:
                        self.after(0, lambda: self.pill_web.config(text="● Web: Port 8080 (Live)",
bg=THEME["state_proceed"], fg="#ffffff"))
                        return
            except Exception:
                pass
            if not self.web_server_process:
                self.after(0, lambda: self.pill_web.config(text="○ Web: Port 8080 (Off)", bg=THEME["bg_elevated"],
```

```
fg=THEME["text_secondary"]))
        def loop():
            while True:
                check()
                time.sleep(4)

        threading.Thread(target=loop, daemon=True).start()


# ================================================================================
# BLOCK 4: CLOSING BLOCK
# ( Application Entrypoint & Fallback Invocation )
# ================================================================================
def main():
    if not TKINTER_AVAILABLE:
        print("[!] Error: Tkinter desktop GUI library is not available in this Python runtime.")
        print("    Install via: sudo apt-get install -y python3-tk")
        sys.exit(1)

    app = OmniCodeDesktopApp()
    app.mainloop()

if __name__ == "__main__":
    main()
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/download.html HTML 140 LINES 659 WORDS 8.3 KB

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description" content="Download OmniCode Standalone Native Terminal & Language Runtime — CreativeWorkzStudio LLC.">
    <title>Download Native OmniCode Terminal — CreativeWorkzStudio LLC</title>
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Fira+Code:wght@400;500;600;700&family=Outfit:wght@300;400;500;600;700;800&family=Cinzel:wght@500;700;800&display=swap" rel="stylesheet">
    <link rel="stylesheet" href="/style.css">
</head>
<body>
    <div class="app-layout">
        <!-- GLOBAL NAVIGATION HEADER -->
        <header class="app-header" id="app-header">
            <div class="header-left">
                <a href="/studio" class="brand-container">
                    <span class="brand-symbol">Ω</span>
                    <div class="brand-text">
                        <span class="brand-title">OMNICODE</span>
                        <span class="brand-subtitle">CREATIVEWORKZSTUDIO LLC</span>
                    </div>
                </a>
                <div class="status-pill" id="status-pill" title="Engine runs 100% offline with zero cloud telemetry">
                    <span class="status-dot"></span>
                    <span class="status-text">OFFLINE COVENANT ACTIVE</span>
                </div>
            </div>

            <nav class="app-nav" aria-label="Main Navigation">
                <a href="/studio" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 2L2 7l10 5 10-5-10-5zM2 17l10 5 10-5M2 12l10 5 10-5"/></svg>
                    AI Studio
                </a>
                <a href="/terminal" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><polyline points="4 17 10 11 4 5"/><line x1="12" y1="19" x2="20" y2="19"/></svg>
                    CLI Terminal
                </a>
                <a href="/specs" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M14 2H6a2 2 0 0 0-2 2v16a2 2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"/><polyline points="14 2 14 8 20 8"/></svg>
                    Language Specs
                </a>
                <a href="/model" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 2a8 8 0 0 0-8 8c0 3.3 2 6.2 5 7.4V20a2 2 0 0 0 2 2h2a2 2 0 0 0 2-2v-2.6c3-1.2 5-4.1 5-7.4a8 8 0 0 0-8-8z"/></svg>
                    AI Model
                </a>
                <a href="/lexicon" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M4 19.5A2.5 2.5 0 0 1 6.5 17H20"/><path d="M6.5 2H20v20H6.5A2.5 2.5 0 0 1 4 19.5v-15A2.5 2.5 0 0 1 6.5 2z"/></svg>
                    Covenant Lexicon
                </a>
                <a href="/scrolls" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M19 21l-7-5-7 5V5a2 2 0 0 1 2-2h10a2 2 0 0 1 2 2z"/></svg>
                    Scrolls
                </a>
                <a href="/cloud" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M18 10h-1.26A8 8 0 1 0 9 20h9a5 5 0 0 0 0-10z"/></svg>
                    Cloud &amp;amp; Edge
                </a>
                <a href="/download" class="nav-tab active download-nav-btn">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"/><polyline points="7 10 12 15 17 10"/><line x1="12" y1="15" x2="12" y2="3"/></svg>
                    Download Native
```

```
                </a>
            </nav>
        </header>

        <!-- DOWNLOAD HUB MAIN -->
        <main class="download-container">
            <div class="download-hero card-glass">
                <div class="hero-badge">STANDALONE SOVEREIGN RUNTIME</div>
                <h1>Download OmniCode Native Terminal</h1>
                <p class="hero-lead">
                    Run the primordial userspace shell and task scheduler directly on your local hardware. Zero runtime dependencies, 100% offline, and bare-metal deterministic execution.
                </p>
                <div class="download-btn-row">
                    <a href="/api/download" class="btn btn-lg btn-accent download-cta" download>
                        <svg width="20" height="20" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2.5"><path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"/><polyline points="7 10 12 15 17 10"/><line x1="12" y1="15" x2="12" y2="3"/></svg>
                        Download for Linux x86_64 (.tar.gz)
                    </a>
                </div>
                <div class="hero-subtext">
                    Edition: Rust Core 2024 &bull; Version 0.1.0 &bull; Architecture: Linux x86_64 &bull; Size: ~560 KB
                </div>
            </div>

            <!-- INSTALLATION & SPECS GRID -->
            <div class="download-grid">
                <!-- 1-LINE INSTALLATION -->
                <div class="download-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">⚡</span>
                        <h2>1-Line Turnkey Terminal Installation</h2>
                    </div>
                    <p>Run directly in your local terminal to install the standalone binary to <code>~/.local/bin/omnicode</code>:</p>
                    <div class="code-copy-box">
                        <code id="cmd-curl-install">curl -sSL http://omnicodestudio.duckdns.org:8080/api/download | tar -xz && ./install.sh</code>
                    </div>
                    <div class="doc-subcard">
                        <h3>Launch Native Terminal</h3>
                        <p>After installation, launch the interactive REPL shell anytime:</p>
                        <code>omnicode terminal</code>
                    </div>
                </div>

                <!-- PACKAGE CONTENTS -->
                <div class="download-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">📦</span>
                        <h2>Included in Distribution Kit</h2>
                    </div>
                    <ul class="spec-list">
                        <li><strong>Standalone Native Binary:</strong> <code>bin/omnicode</code> (compiled ELF executable, 0 dynamic library dependencies).</li>
                        <li><strong>Core Standard Library:</strong> <code>lib/core/arithmetic.omni</code>, <code>logic.omni</code>, <code>vectors.omni</code>.</li>
                        <li><strong>Covenant Manifest:</strong> Immutable ethical bedrock ([<code>manifest.omni</code>](file:///home/professorseanex/OmniCode/manifest.omni)).</li>
                        <li><strong>VS Code Extension:</strong> <code>omnicode-language-0.1.2.vsix</code> syntax highlighter and snippet pack.</li>
                        <li><strong>Canonical Examples:</strong> Balanced ternary logic, Turing tape tracks, and boundary proofs.</li>
                    </ul>
                </div>
            </div>
        </main>

        <!-- FOOTER -->
        <footer class="app-footer">
            <div class="footer-left">
                <span class="footer-brand">Ω OMNICODE</span>
                <span class="footer-sep">&bull;</span>
                <span class="footer-copy">&copy; 2026 CreativeWorkzStudio LLC. All rights reserved.</span>
            </div>
            <div class="footer-center">
                <span>The Sovereign Language &amp;amp; Native Userspace Shell for the Future OmniCode OS</span>
```

```
            </div>
            <div class="footer-right">
                <span class="footer-pill">Native Distribution</span>
                <span class="footer-pill">v0.1.0</span>
            </div>
        </footer>
    </div>
</body>
</html>
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/index.html HTML 136 LINES 610 WORDS 7.7 KB

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description" content="OmniCode AI Prompt Studio & Code Synthesis — CreativeWorkzStudio LLC.">
    <title>OmniCode AI Studio — CreativeWorkzStudio LLC</title>
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?
family=Fira+Code:wght@400;500;600;700&family=Outfit:wght@300;400;500;600;700;800&family=Cinzel:wght@500;700;800&display=swap"
rel="stylesheet">
    <link rel="stylesheet" href="/style.css">
</head>
<body>
    <div class="app-layout">
        <!-- GLOBAL NAVIGATION HEADER -->
        <header class="app-header" id="app-header">
            <div class="header-left">
                <a href="/studio" class="brand-container">
                    <span class="brand-symbol">Ω</span>
                    <div class="brand-text">
                        <span class="brand-title">OMNICODE</span>
                        <span class="brand-subtitle">CREATIVEWORKZSTUDIO LLC</span>
                    </div>
                </a>
                <div class="status-pill" id="status-pill" title="Engine runs 100% offline with zero cloud telemetry">
                    <span class="status-dot"></span>
                    <span class="status-text">OFFLINE COVENANT ACTIVE</span>
                </div>
            </div>

            <nav class="app-nav" aria-label="Main Navigation">
                <a href="/studio" class="nav-tab active">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M12 2L2 7l10 5 10-5-10-5zM2 17l10 5 10-5M2 12l10 5 10-5"/></svg>
                    AI Studio
                </a>
                <a href="/terminal" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<polyline points="4 17 10 11 4 5"/><line x1="12" y1="19" x2="20" y2="19"/></svg>
                    CLI Terminal
                </a>
                <a href="/specs" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M14 2H6a2 2 0 0 0-2 2v16a2 2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"/><polyline points="14 2 14 8 20 8"/></svg>
                    Language Specs
                </a>
                <a href="/model" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M12 2a8 8 0 0 0-8 8c0 3.3 2 6.2 5 7.4V20a2 2 0 0 0 2 2h2a2 2 0 0 0 2-2v-2.6c3-1.2 5-4.1 5-7.4a8 8 0 0 0-8-8z"/></svg>
                    AI Model
                </a>
                <a href="/lexicon" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M4 19.5A2.5 2.5 0 0 1 6.5 17H20"/><path d="M6.5 2H20v20H6.5A2.5 2.5 0 0 1 4 19.5v-15A2.5 2.5 0 0 1 6.5 2z"/></svg>
                    Covenant Lexicon
                </a>
                <a href="/scrolls" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M19 21l-7-5-7 5V5a2 2 0 0 1 2-2h10a2 2 0 0 1 2 2z"/></svg>
                    Scrolls
                </a>
                <a href="/cloud" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M18 10h-1.26A8 8 0 1 0 9 20h9a5 5 0 0 0 0-10z"/></svg>
                    Cloud &amp;amp; Edge
                </a>
                <a href="/download" class="nav-tab download-nav-btn">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"/><polyline points="7 10 12 15 17 10"/><line x1="12" y1="15" x2="12" y2="3"/>
</svg>
                    Download Native
                </a>
```

```
            </nav>
        </header>

        <!-- STUDIO MAIN -->
        <main class="studio-container">
            <div class="studio-grid">
                <!-- LEFT: PROMPT INPUT & INTENT FORM -->
                <div class="card-glass prompt-card">
                    <div class="panel-header">
                        <h2>Natural Language Intent Prompt</h2>
                        <span class="offline-note">🔒 Local execution &bull; 0% Cloud Telemetry</span>
                    </div>
                    <p class="panel-desc">
                        Provide a natural language description. The neural-symbolic transducer aligns it with moral
invariants and generates verified 4-block OmniCode.
                    </p>
                    <textarea id="prompt-input" class="prompt-textarea" rows="6" placeholder="Describe the program you
wish to create... (e.g. 'Build an immutable treasury ledger that verifies balanced transactions and rejects unauthorized
debits')"></textarea>

                    <div class="prompt-actions">
                        <button class="btn btn-accent" id="btn-generate">
                            <svg width="16" height="16" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-
width="2"><polygon points="5 3 19 12 5 21 5 3"/></svg>
                            Synthesize OmniCode
                        </button>
                        <button class="btn btn-secondary" id="btn-sample-prompt">Sample Intent</button>
                    </div>

                    <!-- RESPONSE STREAM -->
                    <div class="ai-response-box" id="ai-response-box">
                        <h3>AICortex Deliberation Summary</h3>
                        <div id="ai-response-text" class="response-text">Awaiting input prompt...</div>
                    </div>
                </div>

                <!-- RIGHT: SYNTHESIZED OMNICODE CODE VIEWER -->
                <div class="card-glass code-card">
                    <div class="panel-header">
                        <h2>Synthesized 4-Block OmniCode</h2>
                        <div class="badge-group">
                            <span class="badge badge-proceed" id="code-state-badge">+1:PROCEED</span>
                            <button class="btn btn-sm btn-secondary" id="btn-copy-code">Copy</button>
                        </div>
                    </div>
                    <pre class="code-editor" id="code-output"><code id="code-output-inner"># 
================================================================================
# BLOCK 1: METADATA BLOCK
# ================================================================================
Manifest &lt; Treasury.Ledger &gt; ( Moral Equilibrium ) {
    author: &lt; \"CreativeWorkzStudio LLC\" &gt;,
    resonance: &lt; +1:PROCEED &gt;
}
</code></pre>
                </div>
            </div>
        </main>

        <!-- FOOTER -->
        <footer class="app-footer">
            <div class="footer-left">
                <span class="footer-brand">Ω OMNICODE</span>
                <span class="footer-sep">&bull;</span>
                <span class="footer-copy">&copy; 2026 CreativeWorkzStudio LLC. All rights reserved.</span>
            </div>
            <div class="footer-center">
                <span>Dual-Readable Ontological Programming Language &amp; Cognitive AI Model</span>
            </div>
            <div class="footer-right">
                <span class="footer-pill">Rust Core Edition 2024</span>
                <span class="footer-pill">v0.1.0</span>
            </div>
        </footer>
    </div>
    <script src="/app.js"></script>
</body>
</html>
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/lexicon.html HTML 114 LINES 508 WORDS 6.3 KB

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description" content="OmniCode Covenant Manifest & Biblical Lexicon World Model — CreativeWorkzStudio LLC.">
    <title>Covenant Manifest &amp;amp; Biblical Lexicon — CreativeWorkzStudio LLC</title>
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Fira+Code:wght@400;500;600;700&family=Outfit:wght@300;400;500;600;700;800&family=Cinzel:wght@500;700;800&display=swap" rel="stylesheet">
    <link rel="stylesheet" href="/style.css">
</head>
<body>
    <div class="app-layout">
        <!-- GLOBAL NAVIGATION HEADER -->
        <header class="app-header" id="app-header">
            <div class="header-left">
                <a href="/studio" class="brand-container">
                    <span class="brand-symbol">Ω</span>
                    <div class="brand-text">
                        <span class="brand-title">OMNICODE</span>
                        <span class="brand-subtitle">CREATIVEWORKZSTUDIO LLC</span>
                    </div>
                </a>
                <div class="status-pill" id="status-pill" title="Engine runs 100% offline with zero cloud telemetry">
                    <span class="status-dot"></span>
                    <span class="status-text">OFFLINE COVENANT ACTIVE</span>
                </div>
            </div>

            <nav class="app-nav" aria-label="Main Navigation">
                <a href="/studio" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 2L2 7l10 5 10-5-10-5zM2 17l10 5 10-5M2 12l10 5 10-5"/></svg>
                    AI Studio
                </a>
                <a href="/terminal" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><polyline points="4 17 10 11 4 5"/><line x1="12" y1="19" x2="20" y2="19"/></svg>
                    CLI Terminal
                </a>
                <a href="/specs" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M14 2H6a2 2 0 0 0-2 2v16a2 2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"/><polyline points="14 2 14 8 20 8"/></svg>
                    Language Specs
                </a>
                <a href="/model" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 2a8 8 0 0 0-8 8c0 3.3 2 6.2 5 7.4V20a2 2 0 0 0 2 2h2a2 2 0 0 0 2-2v-2.6c3-1.2 5-4.1 5-7.4a8 8 0 0 0-8-8z"/></svg>
                    AI Model
                </a>
                <a href="/lexicon" class="nav-tab active">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M4 19.5A2.5 2.5 0 0 1 6.5 17H20"/><path d="M6.5 2H20v20H6.5A2.5 2.5 0 0 1 4 19.5v-15A2.5 2.5 0 0 1 6.5 2z"/></svg>
                    Covenant Lexicon
                </a>
                <a href="/scrolls" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M19 21l-7-5-7 5V5a2 2 0 0 1 2-2h10a2 2 0 0 1 2 2z"/></svg>
                    Scrolls
                </a>
                <a href="/cloud" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M18 10h-1.26A8 8 0 1 0 9 20h9a5 5 0 0 0 0-10z"/></svg>
                    Cloud &amp;amp; Edge
                </a>
                <a href="/download" class="nav-tab download-nav-btn">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"/><polyline points="7 10 12 15 17 10"/><line x1="12" y1="15" x2="12" y2="3"/></svg>
                    Download Native
```

```
                </a>
            </nav>
        </header>

        <!-- LEXICON MAIN -->
        <main class="lexicon-container">
            <div class="lexicon-layout">
                <!-- LEFT: MANIFEST CODE VIEWER -->
                <div class="manifest-card card-glass">
                    <div class="panel-header">
                        <h2>Canonical Covenant Manifest</h2>
                        <span class="badge badge-proceed">Invariant Anchor</span>
                    </div>
                    <pre class="manifest-code"><code id="manifest-content">Loading canonical manifest.omni...</code>
</pre>
                </div>

                <!-- RIGHT: BIBLICAL LEXICON SEARCH & CARDS -->
                <div class="lexicon-card card-glass">
                    <div class="panel-header">
                        <h2>Biblical World Model Lexicon</h2>
                        <span class="badge badge-subtle" id="lexicon-count-badge">0 Entries</span>
                    </div>
                    <div class="lexicon-search-bar">
                        <input type="text" id="lexicon-search" class="form-input" placeholder="Search moral invariants,
scripture terms, or laws (e.g. truth, justice, life, void)...">
                    </div>
                    <div class="lexicon-entries-grid" id="lexicon-entries-grid">
                        <!-- Populated dynamically via app.js -->
                    </div>
                </div>
            </div>
        </main>

        <!-- FOOTER -->
        <footer class="app-footer">
            <div class="footer-left">
                <span class="footer-brand">Ω OMNICODE</span>
                <span class="footer-sep">&bull;</span>
                <span class="footer-copy">&copy; 2026 CreativeWorkzStudio LLC. All rights reserved.</span>
            </div>
            <div class="footer-center">
                <span>The Immutable Moral Anchor of Computation &bull; Rust Core 2024</span>
            </div>
            <div class="footer-right">
                <span class="footer-pill">Biblical Lexicon</span>
                <span class="footer-pill">v0.1.0</span>
            </div>
        </footer>
    </div>
    <script src="/app.js"></script>
</body>
</html>
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/model.html HTML 152 LINES 719 WORDS 8.9 KB

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description" content="OmniCode Autonomous Cognitive AI Model & Neural-Symbolic Cortex — CreativeWorkzStudio LLC.">
    <title>OmniCode Cognitive AI Model — CreativeWorkzStudio LLC</title>
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Fira+Code:wght@400;500;600;700&family=Outfit:wght@300;400;500;600;700;800&family=Cinzel:wght@500;700;800&display=swap" rel="stylesheet">
    <link rel="stylesheet" href="/style.css">
</head>
<body>
    <div class="app-layout">
        <!-- GLOBAL NAVIGATION HEADER -->
        <header class="app-header" id="app-header">
            <div class="header-left">
                <a href="/studio" class="brand-container">
                    <span class="brand-symbol">Ω</span>
                    <div class="brand-text">
                        <span class="brand-title">OMNICODE</span>
                        <span class="brand-subtitle">CREATIVEWORKZSTUDIO LLC</span>
                    </div>
                </a>
                <div class="status-pill" id="status-pill" title="Engine runs 100% offline with zero cloud telemetry">
                    <span class="status-dot"></span>
                    <span class="status-text">OFFLINE COVENANT ACTIVE</span>
                </div>
            </div>

            <nav class="app-nav" aria-label="Main Navigation">
                <a href="/studio" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 2L2 7l10 5 10-5-10-5zM2 17l10 5 10-5M2 12l10 5 10-5"/></svg>
                    AI Studio
                </a>
                <a href="/terminal" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><polyline points="4 17 10 11 4 5"/><line x1="12" y1="19" x2="20" y2="19"/></svg>
                    CLI Terminal
                </a>
                <a href="/specs" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M14 2H6a2 2 0 0 0-2 2v16a2 2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"/><polyline points="14 2 14 8 20 8"/></svg>
                    Language Specs
                </a>
                <a href="/model" class="nav-tab active">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 2a8 8 0 0 0-8 8c0 3.3 2 6.2 5 7.4V20a2 2 0 0 0 2 2h2a2 2 0 0 0 2-2v-2.6c3-1.2 5-4.1 5-7.4a8 8 0 0 0-8-8z"/></svg>
                    AI Model
                </a>
                <a href="/lexicon" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M4 19.5A2.5 2.5 0 0 1 6.5 17H20"/><path d="M6.5 2H20v20H6.5A2.5 2.5 0 0 1 4 19.5v-15A2.5 2.5 0 0 1 6.5 2z"/></svg>
                    Covenant Lexicon
                </a>
                <a href="/scrolls" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M19 21l-7-5-7 5V5a2 2 0 0 1 2-2h10a2 2 0 0 1 2 2z"/></svg>
                    Scrolls
                </a>
                <a href="/cloud" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M18 10h-1.26A8 8 0 1 0 9 20h9a5 5 0 0 0 0-10z"/></svg>
                    Cloud &amp;amp; Edge
                </a>
                <a href="/download" class="nav-tab download-nav-btn">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"/><polyline points="7 10 12 15 17 10"/><line x1="12" y1="15" x2="12" y2="3"/></svg>
                    Download Native
```

```
                </a>
            </nav>
        </header>

        <!-- MODEL MAIN -->
        <main class="doc-container">
            <div class="doc-hero card-glass">
                <div class="hero-badge">AUTONOMOUS COGNITIVE ENGINE</div>
                <h1>The OmniCode Cognitive AI Model</h1>
                <p class="hero-lead">
                    Engineered by <strong>CreativeWorkzStudio LLC</strong>, the OmniCode AI Model features a local
Neural-Symbolic deliberative loop that bridges freeform human discourse with deterministic 4-Block Turing reels, anchored in
biblical moral invariants and continuous experiential learning.
                </p>
            </div>

            <div class="doc-grid">
                <!-- AICORTEX -->
                <div class="doc-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">🧠</span>
                        <h2>The AICortex Transduction Loop</h2>
                    </div>
                    <p>The deliberative cycle that transforms natural human thought into verified machine reels:</p>
                    <ol class="doc-list">
                        <li><strong>Intent Extraction:</strong> Parses raw sentences into semantic concepts and target
boundary types.</li>
                        <li><strong>Moral Invariant Grounding:</strong> Validates intent against the 7 biblical laws of
the Covenant Anchor.</li>
                        <li><strong>4-Block Synthesis:</strong> Emits formal Metadata, Opening, Body, and Closing blocks.
</li>
                        <li><strong>Autonomous Verification &amp;amp; Repair:</strong> Executes the Linter and Restorator to
prove correctness.</li>
                    </ol>
                </div>

                <!-- AUTONOMOUS RESTORATOR -->
                <div class="doc-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">🩹</span>
                        <h2>Self-Healing Restorator</h2>
                    </div>
                    <p>When syntax or boundary violations occur, the Restorator applies surgical heuristics without human
intervention:</p>
                    <ul class="spec-list">
                        <li>Injects missing Block headers (Opening/Body/Closing).</li>
                        <li>Balances unmatched boundary brackets (<code>&lt; ... &gt;</code>, <code>[ ... ]</code>,
<code>( ... )</code>).</li>
                        <li>Repairs malformed ternary operators (<code>--&gt;</code>, <code>&lt;--</code>).</li>
                        <li>Certifies repaired candidates through isolated test oracles.</li>
                    </ul>
                </div>

                <!-- EPISODIC LEARNING -->
                <div class="doc-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">📜</span>
                        <h2>Episodic Learning (Experiential Scrolls)</h2>
                    </div>
                    <p>Every deliberation emits an immutable episodic scroll (<code>.scroll</code>):</p>
                    <ul class="spec-list">
                        <li>Logs user prompt, synthesized code, applied repairs, and final quad-state.</li>
                        <li>Persists across restarts in local directory <code>scrolls/</code>.</li>
                        <li>Prevents moral drift by enforcing positive reinforcement for invariant-aligned outcomes.</li>
                    </ul>
                </div>

                <!-- ZERO TELEMETRY -->
                <div class="doc-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">🔒</span>
                        <h2>Zero Cloud Telemetry Covenant</h2>
                    </div>
                    <p>Unlike commercial LLMs that upload your source code and telemetry to cloud providers, OmniCode
operates under an immutable covenant:</p>
                    <p><strong>100% of reasoning, parsing, repair heuristics, and learning occur locally on your machine.
</strong></p>
                </div>
```

```
            </div>
        </main>

        <!-- FOOTER -->
        <footer class="app-footer">
            <div class="footer-left">
                <span class="footer-brand">Ω OMNICODE</span>
                <span class="footer-sep">&bull;</span>
                <span class="footer-copy">&copy; 2026 CreativeWorkzStudio LLC. All rights reserved.</span>
            </div>
            <div class="footer-center">
                <span>Autonomous Cognitive Intelligence &bull; Rust Core 2024</span>
            </div>
            <div class="footer-right">
                <span class="footer-pill">Cognitive Model</span>
                <span class="footer-pill">v0.1.0</span>
            </div>
        </footer>
    </div>
</body>
</html>
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/scrolls.html HTML 146 LINES 568 WORDS 7.8 KB

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description" content="OmniCode Episodic Memory Bank & Experiential Scrolls — CreativeWorkzStudio LLC.">
    <title>Episodic Memory Scrolls — CreativeWorkzStudio LLC</title>
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?
family=Fira+Code:wght@400;500;600;700&family=Outfit:wght@300;400;500;600;700;800&family=Cinzel:wght@500;700;800&display=swap"
rel="stylesheet">
    <link rel="stylesheet" href="/style.css">
</head>
<body>
    <div class="app-layout">
        <!-- GLOBAL NAVIGATION HEADER -->
        <header class="app-header" id="app-header">
            <div class="header-left">
                <a href="/studio" class="brand-container">
                    <span class="brand-symbol">Ω</span>
                    <div class="brand-text">
                        <span class="brand-title">OMNICODE</span>
                        <span class="brand-subtitle">CREATIVEWORKZSTUDIO LLC</span>
                    </div>
                </a>
                <div class="status-pill" id="status-pill" title="Engine runs 100% offline with zero cloud telemetry">
                    <span class="status-dot"></span>
                    <span class="status-text">OFFLINE COVENANT ACTIVE</span>
                </div>
            </div>

            <nav class="app-nav" aria-label="Main Navigation">
                <a href="/studio" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M12 2L2 7l10 5 10-5-10-5zM2 17l10 5 10-5M2 12l10 5 10-5"/></svg>
                    AI Studio
                </a>
                <a href="/terminal" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<polyline points="4 17 10 11 4 5"/><line x1="12" y1="19" x2="20" y2="19"/></svg>
                    CLI Terminal
                </a>
                <a href="/specs" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M14 2H6a2 2 0 0 0-2 2v16a2 2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"/><polyline points="14 2 14 8 20 8"/></svg>
                    Language Specs
                </a>
                <a href="/model" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M12 2a8 8 0 0 0-8 8c0 3.3 2 6.2 5 7.4V20a2 2 0 0 0 2 2h2a2 2 0 0 0 2-2v-2.6c3-1.2 5-4.1 5-7.4a8 8 0 0 0-8-8z"/></svg>
                    AI Model
                </a>
                <a href="/lexicon" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M4 19.5A2.5 2.5 0 0 1 6.5 17H20"/><path d="M6.5 2H20v20H6.5A2.5 2.5 0 0 1 4 19.5v-15A2.5 2.5 0 0 1 6.5 2z"/></svg>
                    Covenant Lexicon
                </a>
                <a href="/scrolls" class="nav-tab active">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M19 21l-7-5-7 5V5a2 2 0 0 1 2-2h10a2 2 0 0 1 2 2z"/></svg>
                    Scrolls
                </a>
                <a href="/cloud" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M18 10h-1.26A8 8 0 1 0 9 20h9a5 5 0 0 0 0-10z"/></svg>
                    Cloud &amp;amp; Edge
                </a>
                <a href="/download" class="nav-tab download-nav-btn">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"/><polyline points="7 10 12 15 17 10"/><line x1="12" y1="15" x2="12" y2="3"/>
</svg>
                    Download Native
                </a>
```

```
            </nav>
        </header>

        <!-- SCROLLS MAIN -->
        <main class="scrolls-container">
            <div class="scrolls-card card-glass">
                <div class="panel-header">
                    <div class="panel-header-title">
                        <h1>Episodic Memory Bank &amp;amp; Learning Inscription</h1>
                        <p class="panel-desc">Experiential scrolls (.scroll) and traditional telemetry logs (.log)
accumulated by the internal AI model. Grounded learning preserves successful restorations without moral drift.</p>
                    </div>
                    <button class="btn btn-secondary" id="btn-refresh-scrolls">Refresh Memory</button>
                </div>

                <!-- METRICS CARDS -->
                <div class="metrics-grid" id="metrics-grid">
                    <div class="metric-card">
                        <span class="metric-num" id="metric-total">0</span>
                        <span class="metric-label">Inscribed Episodes</span>
                    </div>
                    <div class="metric-card card-green">
                        <span class="metric-num" id="metric-proceed">0</span>
                        <span class="metric-label">+1: PROCEED</span>
                    </div>
                    <div class="metric-card card-amber">
                        <span class="metric-num" id="metric-await">0</span>
                        <span class="metric-label">0: AWAIT</span>
                    </div>
                    <div class="metric-card card-red">
                        <span class="metric-num" id="metric-rollback">0</span>
                        <span class="metric-label">-1: ROLLBACK</span>
                    </div>
                    <div class="metric-card card-purple">
                        <span class="metric-num" id="metric-halt">0</span>
                        <span class="metric-label">NULL: HALT</span>
                    </div>
                    <div class="metric-card">
                        <span class="metric-num" id="metric-repairs">0</span>
                        <span class="metric-label">Repairs Learned</span>
                    </div>
                </div>

                <!-- SCROLLS EPISODES TABLE -->
                <div class="episodes-table-wrapper">
                    <table class="episodes-table">
                        <thead>
                            <tr>
                                <th>Session ID</th>
                                <th>Prompt Intent</th>
                                <th>Final State</th>
                                <th>Repairs</th>
                                <th>Failpoints</th>
                                <th>Scroll Path</th>
                            </tr>
                        </thead>
                        <tbody id="episodes-tbody">
                            <!-- Populated dynamically via JS -->
                        </tbody>
                    </table>
                </div>
            </div>
        </main>

        <!-- FOOTER -->
        <footer class="app-footer">
            <div class="footer-left">
                <span class="footer-brand">Ω OMNICODE</span>
                <span class="footer-sep">&bull;</span>
                <span class="footer-copy">&copy; 2026 CreativeWorkzStudio LLC. All rights reserved.</span>
            </div>
            <div class="footer-center">
                <span>Episodic Learning &bull; Experiential Memory Ledger &bull; Rust Core 2024</span>
            </div>
            <div class="footer-right">
                <span class="footer-pill">Episodic Scrolls</span>
                <span class="footer-pill">v0.1.0</span>
            </div>
```

```
        </footer>
    </div>
    <script src="/app.js"></script>
</body>
</html>
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/specs.html HTML 155 LINES 798 WORDS 9.6 KB

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description" content="OmniCode Language Specification & 4-Block Formal Architecture — CreativeWorkzStudio LLC.">
    <title>OmniCode Language Specifications — CreativeWorkzStudio LLC</title>
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Fira+Code:wght@400;500;600;700&family=Outfit:wght@300;400;500;600;700;800&family=Cinzel:wght@500;700;800&display=swap" rel="stylesheet">
    <link rel="stylesheet" href="/style.css">
</head>
<body>
    <div class="app-layout">
        <!-- GLOBAL NAVIGATION HEADER -->
        <header class="app-header" id="app-header">
            <div class="header-left">
                <a href="/studio" class="brand-container">
                    <span class="brand-symbol">Ω</span>
                    <div class="brand-text">
                        <span class="brand-title">OMNICODE</span>
                        <span class="brand-subtitle">CREATIVEWORKZSTUDIO LLC</span>
                    </div>
                </a>
                <div class="status-pill" id="status-pill" title="Engine runs 100% offline with zero cloud telemetry">
                    <span class="status-dot"></span>
                    <span class="status-text">OFFLINE COVENANT ACTIVE</span>
                </div>
            </div>

            <nav class="app-nav" aria-label="Main Navigation">
                <a href="/studio" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 2L2 7l10 5 10-5-10-5zM2 17l10 5 10-5M2 12l10 5 10-5"/></svg>
                    AI Studio
                </a>
                <a href="/terminal" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><polyline points="4 17 10 11 4 5"/><line x1="12" y1="19" x2="20" y2="19"/></svg>
                    CLI Terminal
                </a>
                <a href="/specs" class="nav-tab active">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M14 2H6a2 2 0 0 0-2 2v16a2 2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"/><polyline points="14 2 14 8 20 8"/></svg>
                    Language Specs
                </a>
                <a href="/model" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 2a8 8 0 0 0-8 8c0 3.3 2 6.2 5 7.4V20a2 2 0 0 0 2 2h2a2 2 0 0 0 2-2v-2.6c3-1.2 5-4.1 5-7.4a8 8 0 0 0-8-8z"/></svg>
                    AI Model
                </a>
                <a href="/lexicon" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M4 19.5A2.5 2.5 0 0 1 6.5 17H20"/><path d="M6.5 2H20v20H6.5A2.5 2.5 0 0 1 4 19.5v-15A2.5 2.5 0 0 1 6.5 2z"/></svg>
                    Covenant Lexicon
                </a>
                <a href="/scrolls" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M19 21l-7-5-7 5V5a2 2 0 0 1 2-2h10a2 2 0 0 1 2 2z"/></svg>
                    Scrolls
                </a>
                <a href="/cloud" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M18 10h-1.26A8 8 0 1 0 9 20h9a5 5 0 0 0 0-10z"/></svg>
                    Cloud &amp;amp; Edge
                </a>
                <a href="/download" class="nav-tab download-nav-btn">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"/><polyline points="7 10 12 15 17 10"/><line x1="12" y1="15" x2="12" y2="3"/></svg>
                    Download Native
```

```
                </a>
            </nav>
        </header>

        <!-- SPECS MAIN -->
        <main class="doc-container">
            <div class="doc-hero card-glass">
                <div class="hero-badge">CANONICAL LANGUAGE SPECIFICATION</div>
                <h1>The OmniCode Programming Language</h1>
                <p class="hero-lead">
                    An ontological, dual-readable programming language engineered by <strong>CreativeWorkzStudio
LLC</strong> from first principles. Operates with bare-metal low-level precision while natively expressing balanced-ternary quad-
state state machines, causal laws, and ethical invariants.
                </p>
            </div>

            <div class="doc-grid">
                <!-- 4-BLOCK ARCHITECTURE -->
                <div class="doc-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">🏛️</span>
                        <h2>The 4-Block Architecture</h2>
                    </div>
                    <p>Every valid OmniCode unit must be structured into four sequential compilation blocks:</p>
                    <ol class="doc-list">
                        <li><strong>Block 1: Metadata Block (Manifest)</strong> — Declares ontological identity, covenant
anchor, and boundary classifications.</li>
                        <li><strong>Block 2: Opening Block (Opening { ... })</strong> — Import libraries, assert
immutable preconditions, and establish initial state vectors.</li>
                        <li><strong>Block 3: Body Block (Body [ ... ])</strong> — Execute causal logic, balanced ternary
rolls, and iterative computations.</li>
                        <li><strong>Block 4: Closing Block (Closing { ... })</strong> — Verify post-conditions, reconcile
invariants, and permanently seal transaction state.</li>
                    </ol>
                </div>

                <!-- BALANCED TERNARY RADIX-3 -->
                <div class="doc-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">⚖️</span>
                        <h2>Balanced Ternary Quad-State Logic</h2>
                    </div>
                    <p>Unlike binary logic (0 or 1), OmniCode operates on Balanced Radix-3 with a terminal Null state
(T4):</p>
                    <ul class="spec-list">
                        <li><strong>+1 (Proceed):</strong> Affirmative forward vector (<code>--&gt;</code>). Commits
state mutations and advances the Turing reel.</li>
                        <li><strong>0 (Await):</strong> Standby, pass-through equilibrium, or asynchronous yield.</li>
                        <li><strong>-1 (Rollback):</strong> Negation and boundary breach. Automatically unwinds
retroactive vectors (<code>&lt;--</code>).</li>
                        <li><strong>NULL (Halt):</strong> Absorbing bottom type (&perp;). Traps unrecoverable faults and
terminates process.</li>
                    </ul>
                </div>

                <!-- BOUNDARIES -->
                <div class="doc-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">🛡️</span>
                        <h2>The 6 Boundary Types</h2>
                    </div>
                    <ul class="spec-list">
                        <li><code>&lt; Invariant &gt;</code> — Absolute unalterable boundary. Immutable across all
scopes.</li>
                        <li><code>&lt; Finite &gt;</code> — Strictly bounded range, buffer capacity, or count.</li>
                        <li><code>&lt; Indefinite &gt;</code> — Elastic state awaiting runtime determination.</li>
                        <li><code>&lt; Continuous &gt;</code> — Streaming uninterrupted sequence or real-time feed.</li>
                        <li><code>&lt; Absorb &gt;</code> — Trap boundary that swallows violations into safe fallbacks.
</li>
                        <li><code>&lt; Oscillate &gt;</code> — Alternating dual-phase harmonic boundary.</li>
                    </ul>
                </div>

                <!-- FORMAL ASIDES -->
                <div class="doc-card card-glass">
                    <div class="doc-card-header">
                        <span class="doc-icon">💬</span>
                        <h2>Formal Asides (Dual Readability)</h2>
```

```
                    </div>
                    <p>Parentheses <code>( ... )</code> in OmniCode are not inert comments—they are <strong>ontological semantic anchors</strong> parsed by the compiler:</p>
                    <pre class="code-snippet"><code>roll_integrity [ health &gt; 0 ] ( Vital signs active and confirmed )</code></pre>
                    <p class="subtext">The inner text informs the AICortex and linter of human moral intent while the expression enforces rigorous machine truth.</p>
                </div>
            </div>
        </main>

        <!-- FOOTER -->
        <footer class="app-footer">
            <div class="footer-left">
                <span class="footer-brand">Ω OMNICODE</span>
                <span class="footer-sep">&bull;</span>
                <span class="footer-copy">&copy; 2026 CreativeWorkzStudio LLC. All rights reserved.</span>
            </div>
            <div class="footer-center">
                <span>Formal Dual-Readable Specifications &bull; Rust Core 2024</span>
            </div>
            <div class="footer-right">
                <span class="footer-pill">Language Specs</span>
                <span class="footer-pill">v0.1.0</span>
            </div>
        </footer>
    </div>
</body>
</html>
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/studio.html HTML 136 LINES 610 WORDS 7.7 KB

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description" content="OmniCode AI Prompt Studio & Code Synthesis — CreativeWorkzStudio LLC.">
    <title>OmniCode AI Studio — CreativeWorkzStudio LLC</title>
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?
family=Fira+Code:wght@400;500;600;700&family=Outfit:wght@300;400;500;600;700;800&family=Cinzel:wght@500;700;800&display=swap"
rel="stylesheet">
    <link rel="stylesheet" href="/style.css">
</head>
<body>
    <div class="app-layout">
        <!-- GLOBAL NAVIGATION HEADER -->
        <header class="app-header" id="app-header">
            <div class="header-left">
                <a href="/studio" class="brand-container">
                    <span class="brand-symbol">Ω</span>
                    <div class="brand-text">
                        <span class="brand-title">OMNICODE</span>
                        <span class="brand-subtitle">CREATIVEWORKZSTUDIO LLC</span>
                    </div>
                </a>
                <div class="status-pill" id="status-pill" title="Engine runs 100% offline with zero cloud telemetry">
                    <span class="status-dot"></span>
                    <span class="status-text">OFFLINE COVENANT ACTIVE</span>
                </div>
            </div>

            <nav class="app-nav" aria-label="Main Navigation">
                <a href="/studio" class="nav-tab active">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M12 2L2 7l10 5 10-5-10-5zM2 17l10 5 10-5M2 12l10 5 10-5"/></svg>
                    AI Studio
                </a>
                <a href="/terminal" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<polyline points="4 17 10 11 4 5"/><line x1="12" y1="19" x2="20" y2="19"/></svg>
                    CLI Terminal
                </a>
                <a href="/specs" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M14 2H6a2 2 0 0 0-2 2v16a2 2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"/><polyline points="14 2 14 8 20 8"/></svg>
                    Language Specs
                </a>
                <a href="/model" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M12 2a8 8 0 0 0-8 8c0 3.3 2 6.2 5 7.4V20a2 2 0 0 0 2 2h2a2 2 0 0 0 2-2v-2.6c3-1.2 5-4.1 5-7.4a8 8 0 0 0-8-8z"/></svg>
                    AI Model
                </a>
                <a href="/lexicon" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M4 19.5A2.5 2.5 0 0 1 6.5 17H20"/><path d="M6.5 2H20v20H6.5A2.5 2.5 0 0 1 4 19.5v-15A2.5 2.5 0 0 1 6.5 2z"/></svg>
                    Covenant Lexicon
                </a>
                <a href="/scrolls" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M19 21l-7-5-7 5V5a2 2 0 0 1 2-2h10a2 2 0 0 1 2 2z"/></svg>
                    Scrolls
                </a>
                <a href="/cloud" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M18 10h-1.26A8 8 0 1 0 9 20h9a5 5 0 0 0 0-10z"/></svg>
                    Cloud &amp;amp; Edge
                </a>
                <a href="/download" class="nav-tab download-nav-btn">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
<path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"/><polyline points="7 10 12 15 17 10"/><line x1="12" y1="15" x2="12" y2="3"/>
</svg>
                    Download Native
                </a>
```

```
            </nav>
        </header>

        <!-- STUDIO MAIN -->
        <main class="studio-container">
            <div class="studio-grid">
                <!-- LEFT: PROMPT INPUT & INTENT FORM -->
                <div class="card-glass prompt-card">
                    <div class="panel-header">
                        <h2>Natural Language Intent Prompt</h2>
                        <span class="offline-note">🔒 Local execution &bull; 0% Cloud Telemetry</span>
                    </div>
                    <p class="panel-desc">
                        Provide a natural language description. The neural-symbolic transducer aligns it with moral
invariants and generates verified 4-block OmniCode.
                    </p>
                    <textarea id="prompt-input" class="prompt-textarea" rows="6" placeholder="Describe the program you
wish to create... (e.g. 'Build an immutable treasury ledger that verifies balanced transactions and rejects unauthorized
debits')"></textarea>

                    <div class="prompt-actions">
                        <button class="btn btn-accent" id="btn-generate">
                            <svg width="16" height="16" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-
width="2"><polygon points="5 3 19 12 5 21 5 3"/></svg>
                            Synthesize OmniCode
                        </button>
                        <button class="btn btn-secondary" id="btn-sample-prompt">Sample Intent</button>
                    </div>

                    <!-- RESPONSE STREAM -->
                    <div class="ai-response-box" id="ai-response-box">
                        <h3>AICortex Deliberation Summary</h3>
                        <div id="ai-response-text" class="response-text">Awaiting input prompt...</div>
                    </div>
                </div>

                <!-- RIGHT: SYNTHESIZED OMNICODE CODE VIEWER -->
                <div class="card-glass code-card">
                    <div class="panel-header">
                        <h2>Synthesized 4-Block OmniCode</h2>
                        <div class="badge-group">
                            <span class="badge badge-proceed" id="code-state-badge">+1:PROCEED</span>
                            <button class="btn btn-sm btn-secondary" id="btn-copy-code">Copy</button>
                        </div>
                    </div>
                    <pre class="code-editor" id="code-output"><code id="code-output-inner"># 
================================================================================
# BLOCK 1: METADATA BLOCK
# ================================================================================
Manifest &lt; Treasury.Ledger &gt; ( Moral Equilibrium ) {
    author: &lt; \"CreativeWorkzStudio LLC\" &gt;,
    resonance: &lt; +1:PROCEED &gt;
}
</code></pre>
                </div>
            </div>
        </main>

        <!-- FOOTER -->
        <footer class="app-footer">
            <div class="footer-left">
                <span class="footer-brand">Ω OMNICODE</span>
                <span class="footer-sep">&bull;</span>
                <span class="footer-copy">&copy; 2026 CreativeWorkzStudio LLC. All rights reserved.</span>
            </div>
            <div class="footer-center">
                <span>Dual-Readable Ontological Programming Language &amp; Cognitive AI Model</span>
            </div>
            <div class="footer-right">
                <span class="footer-pill">Rust Core Edition 2024</span>
                <span class="footer-pill">v0.1.0</span>
            </div>
        </footer>
    </div>
    <script src="/app.js"></script>
</body>
</html>
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/style.css CSS 1747 LINES 3804 WORDS 37.7 KB

```css
/* ================================================================================
   OMNICODE STUDIO & COGNITIVE TERMINAL STYLESHEET
   Aesthetics: Deep Obsidian, Radiant Neon Glassmorphism, Dual Monospace Code
   ================================================================================ */

:root {
    /* Color Palette */
    --bg-base: #07090e;
    --bg-surface: #0d121d;
    --bg-elevated: #131a29;
    --border-subtle: rgba(255, 255, 255, 0.08);
    --border-bright: rgba(0, 240, 255, 0.3);

    /* Text Colors */
    --text-primary: #f0f4fc;
    --text-secondary: #94a3b8;
    --text-muted: #64748b;
    --text-highlight: #ffffff;

    /* Theme Jewel Accents */
    --cyan: #00f0ff;
    --cyan-glow: rgba(0, 240, 255, 0.25);
    --gold: #ffd700;
    --gold-glow: rgba(255, 215, 0, 0.25);

    /* Balanced Ternary Quad-State Colors */
    --state-proceed: #00e676;
    --state-proceed-glow: rgba(0, 230, 118, 0.3);
    --state-await: #ffaa00;
    --state-await-glow: rgba(255, 170, 0, 0.3);
    --state-rollback: #ff3d71;
    --state-rollback-glow: rgba(255, 61, 113, 0.3);
    --state-halt: #9d4edd;
    --state-halt-glow: rgba(157, 78, 221, 0.3);

    /* Typography */
    --font-sans: 'Outfit', -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, sans-serif;
    --font-mono: 'Fira Code', Consolas, Monaco, monospace;
    --font-serif: 'Cinzel', Georgia, serif;

    /* Shadows & Radii */
    --radius-sm: 6px;
    --radius-md: 10px;
    --radius-lg: 16px;
    --shadow-glass: 0 8px 32px 0 rgba(0, 0, 0, 0.45);
    --glass-bg: rgba(13, 18, 29, 0.7);
    --glass-border: 1px solid rgba(255, 255, 255, 0.08);
}

/* Base Reset */
*, *::before, *::after {
    box-sizing: border-box;
    margin: 0;
    padding: 0;
}

body {
    background-color: var(--bg-base);
    color: var(--text-primary);
    font-family: var(--font-sans);
    line-height: 1.6;
    overflow-x: hidden;
    min-height: 100vh;
    background-image:
        radial-gradient(circle at 15% 10%, rgba(0, 240, 255, 0.05) 0%, transparent 40%),
        radial-gradient(circle at 85% 80%, rgba(255, 215, 0, 0.04) 0%, transparent 45%),
        radial-gradient(circle at 50% 50%, rgba(157, 78, 221, 0.03) 0%, transparent 60%);
    background-attachment: fixed;
}

/* App Layout */
.app-layout {
    display: flex;
    flex-direction: column;
```

```
    min-height: 100vh;
}

/* ------------------------------------------------------------------------
   HEADER & NAVIGATION
   ------------------------------------------------------------------------ */
.app-header {
    display: flex;
    align-items: center;
    justify-content: space-between;
    padding: 1rem 2rem;
    background: rgba(7, 9, 14, 0.85);
    backdrop-filter: blur(20px);
    -webkit-backdrop-filter: blur(20px);
    border-bottom: var(--glass-border);
    position: sticky;
    top: 0;
    z-index: 100;
}

.header-left {
    display: flex;
    align-items: center;
    gap: 1.5rem;
}

.brand-container {
    display: flex;
    align-items: center;
    gap: 0.75rem;
}

.brand-symbol {
    font-size: 1.8rem;
    font-weight: 800;
    color: var(--cyan);
    text-shadow: 0 0 16px var(--cyan-glow);
    font-family: var(--font-serif);
}

.brand-text {
    display: flex;
    flex-direction: column;
}

.brand-title {
    font-size: 1.15rem;
    font-weight: 800;
    letter-spacing: 0.15em;
    color: var(--text-highlight);
}

.brand-subtitle {
    font-size: 0.65rem;
    letter-spacing: 0.2em;
    color: var(--gold);
    font-weight: 600;
}

.status-pill {
    display: flex;
    align-items: center;
    gap: 0.5rem;
    padding: 0.35rem 0.85rem;
    background: rgba(0, 230, 118, 0.08);
    border: 1px solid rgba(0, 230, 118, 0.3);
    border-radius: 999px;
    font-size: 0.72rem;
    font-weight: 600;
    color: var(--state-proceed);
    letter-spacing: 0.05em;
}

.status-dot {
    width: 7px;
    height: 7px;
    border-radius: 50%;
    background: var(--state-proceed);
```

```
    box-shadow: 0 0 8px var(--state-proceed);
    animation: pulse-dot 2s infinite ease-in-out;
}

@keyframes pulse-dot {
    0%, 100% { opacity: 1; transform: scale(1); }
    50% { opacity: 0.4; transform: scale(0.85); }
}

.app-nav {
    display: flex;
    gap: 0.5rem;
}

.nav-tab {
    display: flex;
    align-items: center;
    gap: 0.5rem;
    padding: 0.6rem 1.1rem;
    background: transparent;
    border: 1px solid transparent;
    border-radius: var(--radius-md);
    color: var(--text-secondary);
    font-family: var(--font-sans);
    font-size: 0.9rem;
    font-weight: 500;
    cursor: pointer;
    transition: all 0.25s ease;
}

.nav-tab:hover {
    color: var(--text-primary);
    background: rgba(255, 255, 255, 0.04);
}

.nav-tab.active {
    color: var(--cyan);
    background: rgba(0, 240, 255, 0.08);
    border-color: rgba(0, 240, 255, 0.35);
    box-shadow: 0 0 15px rgba(0, 240, 255, 0.15);
}

.ternary-quad-badge {
    padding: 0.4rem 0.9rem;
    background: var(--bg-surface);
    border: 1px solid var(--border-subtle);
    border-radius: var(--radius-sm);
    font-family: var(--font-mono);
    font-size: 0.75rem;
    font-weight: 600;
    color: var(--gold);
    letter-spacing: 0.05em;
}

/* ------------------------------------------------------------------------
   MAIN & VIEW PANELS
   ------------------------------------------------------------------------ */
.app-main {
    flex: 1;
    padding: 2rem;
    max-width: 1600px;
    margin: 0 auto;
    width: 100%;
}

.view-panel {
    display: none;
    animation: fadeIn 0.3s ease;
}

.view-panel.active {
    display: block;
}

@keyframes fadeIn {
    from { opacity: 0; transform: translateY(6px); }
    to { opacity: 1; transform: translateY(0); }
}
```

```
/* Glassmorphism Cards */
.card-glass {
    background: var(--glass-bg);
    backdrop-filter: blur(16px);
    -webkit-backdrop-filter: blur(16px);
    border: var(--glass-border);
    border-radius: var(--radius-lg);
    box-shadow: var(--shadow-glass);
    padding: 1.5rem;
}

.panel-header {
    display: flex;
    justify-content: space-between;
    align-items: flex-start;
    margin-bottom: 1.25rem;
    gap: 1rem;
}

.panel-header-title h1 {
    font-size: 1.5rem;
    font-weight: 700;
    color: var(--text-highlight);
    margin-bottom: 0.35rem;
}

.panel-desc {
    font-size: 0.88rem;
    color: var(--text-secondary);
    max-width: 850px;
}

/* ------------------------------------------------------------------------
   VIEW 1: PROMPT WORKBENCH
   ------------------------------------------------------------------------ */
.quadstate-indicator {
    display: flex;
    align-items: center;
    gap: 0.6rem;
    padding: 0.5rem 1.1rem;
    background: rgba(255, 170, 0, 0.1);
    border: 1px solid var(--state-await);
    border-radius: var(--radius-md);
    color: var(--state-await);
    font-family: var(--font-mono);
    font-size: 0.85rem;
    font-weight: 700;
    transition: all 0.3s ease;
}

.quadstate-indicator.proceed {
    background: rgba(0, 230, 118, 0.1);
    border-color: var(--state-proceed);
    color: var(--state-proceed);
    box-shadow: 0 0 15px var(--state-proceed-glow);
}

.quadstate-indicator.rollback {
    background: rgba(255, 61, 113, 0.1);
    border-color: var(--state-rollback);
    color: var(--state-rollback);
    box-shadow: 0 0 15px var(--state-rollback-glow);
}

.quadstate-indicator.halt {
    background: rgba(157, 78, 221, 0.1);
    border-color: var(--state-halt);
    color: var(--state-halt);
    box-shadow: 0 0 15px var(--state-halt-glow);
}

.prompt-chips-container {
    display: flex;
    flex-wrap: wrap;
    align-items: center;
    gap: 0.5rem;
    margin-bottom: 1.25rem;
```

```
}

.chip-label {
    font-size: 0.8rem;
    font-weight: 600;
    color: var(--text-muted);
    text-transform: uppercase;
    letter-spacing: 0.05em;
}

.prompt-chip, .term-btn {
    background: rgba(255, 255, 255, 0.04);
    border: 1px solid var(--border-subtle);
    border-radius: 999px;
    padding: 0.35rem 0.85rem;
    color: var(--text-secondary);
    font-size: 0.82rem;
    cursor: pointer;
    transition: all 0.2s ease;
}

.prompt-chip:hover, .term-btn:hover {
    background: rgba(0, 240, 255, 0.12);
    border-color: var(--cyan);
    color: var(--text-highlight);
    transform: translateY(-1px);
}

.prompt-chip.chip-warning {
    border-color: rgba(255, 61, 113, 0.3);
    color: #ff7597;
}

.prompt-chip.chip-warning:hover {
    background: rgba(255, 61, 113, 0.15);
    border-color: var(--state-rollback);
}

.prompt-form textarea {
    width: 100%;
    background: var(--bg-surface);
    border: 1px solid var(--border-subtle);
    border-radius: var(--radius-md);
    padding: 1rem;
    color: var(--text-primary);
    font-family: var(--font-sans);
    font-size: 1rem;
    resize: vertical;
    outline: none;
    transition: border-color 0.2s ease, box-shadow 0.2s ease;
}

.prompt-form textarea:focus {
    border-color: var(--cyan);
    box-shadow: 0 0 15px var(--cyan-glow);
}

.form-actions {
    display: flex;
    justify-content: space-between;
    align-items: center;
    margin-top: 0.85rem;
}

.offline-note {
    font-size: 0.78rem;
    color: var(--text-muted);
}

.btn {
    display: inline-flex;
    align-items: center;
    gap: 0.5rem;
    padding: 0.65rem 1.35rem;
    border-radius: var(--radius-md);
    font-family: var(--font-sans);
    font-size: 0.92rem;
    font-weight: 600;
```

```
    cursor: pointer;
    transition: all 0.25s ease;
    border: none;
}

.btn-primary {
    background: linear-gradient(135deg, #00f0ff 0%, #0088ff 100%);
    color: #050811;
    box-shadow: 0 4px 15px rgba(0, 240, 255, 0.3);
}

.btn-primary:hover {
    filter: brightness(1.15);
    transform: translateY(-1px);
    box-shadow: 0 6px 20px rgba(0, 240, 255, 0.45);
}

.btn-secondary {
    background: rgba(255, 255, 255, 0.06);
    border: 1px solid var(--border-subtle);
    color: var(--text-secondary);
}

.btn-secondary:hover {
    background: rgba(255, 255, 255, 0.1);
    color: var(--text-primary);
}

/* Studio Output Split Grid */
.studio-output-grid {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 1.5rem;
    margin-top: 1.5rem;
}

.card-header {
    display: flex;
    justify-content: space-between;
    align-items: center;
    margin-bottom: 1rem;
    border-bottom: 1px solid rgba(255, 255, 255, 0.05);
    padding-bottom: 0.75rem;
}

.card-title-group {
    display: flex;
    align-items: center;
    gap: 0.5rem;
}

.card-title-group h2 {
    font-size: 1.05rem;
    font-weight: 600;
    color: var(--text-highlight);
}

.session-badge {
    font-family: var(--font-mono);
    font-size: 0.75rem;
    color: var(--cyan);
    background: rgba(0, 240, 255, 0.08);
    padding: 0.25rem 0.6rem;
    border-radius: var(--radius-sm);
}

.response-body {
    min-height: 250px;
    font-size: 0.95rem;
    line-height: 1.7;
    white-space: pre-wrap;
    color: #e2e8f0;
}

.placeholder-text {
    color: var(--text-muted);
    font-style: italic;
    padding-top: 2rem;
```

```
}

.card-footer {
    display: flex;
    flex-wrap: wrap;
    gap: 1rem;
    margin-top: 1rem;
    padding-top: 0.75rem;
    border-top: 1px solid rgba(255, 255, 255, 0.05);
    font-family: var(--font-mono);
    font-size: 0.75rem;
    color: var(--text-muted);
}

.code-body {
    background: #05070c;
    border-radius: var(--radius-md);
    padding: 1rem;
    max-height: 480px;
    overflow-y: auto;
    border: 1px solid rgba(255, 255, 255, 0.05);
}

.code-body pre code {
    font-family: var(--font-mono);
    font-size: 0.85rem;
    line-height: 1.5;
    color: #a5b4fc;
}

.btn-icon {
    background: rgba(255, 255, 255, 0.05);
    border: 1px solid var(--border-subtle);
    color: var(--text-secondary);
    border-radius: var(--radius-sm);
    padding: 0.3rem 0.7rem;
    font-size: 0.75rem;
    cursor: pointer;
}

.btn-icon:hover {
    color: var(--text-highlight);
    background: rgba(255, 255, 255, 0.1);
}

/* ------------------------------------------------------------------------
   VIEW 2: CLI TERMINAL
   ------------------------------------------------------------------------ */
.terminal-shortcuts {
    display: flex;
    flex-wrap: wrap;
    align-items: center;
    gap: 0.5rem;
    margin-bottom: 1rem;
}

.term-btn {
    font-family: var(--font-mono);
    font-size: 0.75rem;
}

.terminal-window {
    background: #04060a;
    border: 1px solid rgba(0, 240, 255, 0.2);
    border-radius: var(--radius-md);
    padding: 1.25rem;
    min-height: 480px;
    max-height: 600px;
    overflow-y: auto;
    font-family: var(--font-mono);
    font-size: 0.88rem;
    line-height: 1.6;
    color: #4ade80;
    box-shadow: inset 0 0 30px rgba(0, 0, 0, 0.8);
}

.terminal-banner pre {
    color: var(--cyan);
```

```
    margin-bottom: 1rem;
    font-family: inherit;
    white-space: pre-wrap;
}

.terminal-stream {
    white-space: pre-wrap;
    word-break: break-all;
}

.term-line-cmd {
    color: var(--gold);
    font-weight: 600;
    margin-top: 0.75rem;
}

.term-line-stdout {
    color: #e2e8f0;
}

.term-line-stderr {
    color: var(--state-rollback);
}

.terminal-input-bar {
    display: flex;
    align-items: center;
    gap: 0.75rem;
    margin-top: 1rem;
    background: var(--bg-surface);
    border: 1px solid var(--border-subtle);
    border-radius: var(--radius-md);
    padding: 0.5rem 1rem;
}

.prompt-prefix {
    font-family: var(--font-mono);
    font-size: 0.9rem;
    font-weight: 700;
    color: var(--cyan);
}

.terminal-input-bar input {
    flex: 1;
    background: transparent;
    border: none;
    color: var(--text-highlight);
    font-family: var(--font-mono);
    font-size: 0.92rem;
    outline: none;
}

/* ------------------------------------------------------------------------
   VIEW 3: COVENANT & LEXICON
   ------------------------------------------------------------------------ */
.lexicon-container {
    display: grid;
    grid-template-columns: 1fr 1.35fr;
    gap: 1.5rem;
}

.column-badge {
    font-size: 0.65rem;
    letter-spacing: 0.15em;
    font-weight: 700;
    color: var(--gold);
    text-transform: uppercase;
}

.column-header h2 {
    font-size: 1.35rem;
    color: var(--text-highlight);
    margin: 0.25rem 0 0.5rem 0;
}

.column-desc {
    font-size: 0.85rem;
    color: var(--text-secondary);
```

```
    margin-bottom: 1.25rem;
}

.laws-list {
    display: flex;
    flex-direction: column;
    gap: 0.85rem;
    max-height: 750px;
    overflow-y: auto;
    padding-right: 0.5rem;
}

.law-card {
    background: var(--bg-surface);
    border: 1px solid var(--border-subtle);
    border-radius: var(--radius-md);
    padding: 1rem;
    transition: transform 0.2s ease, border-color 0.2s ease;
}

.law-card:hover {
    border-color: rgba(255, 215, 0, 0.35);
    transform: translateX(3px);
}

.law-title-row {
    display: flex;
    align-items: center;
    gap: 0.75rem;
    margin-bottom: 0.5rem;
}

.law-num {
    font-family: var(--font-serif);
    font-size: 0.9rem;
    font-weight: 700;
    color: var(--gold);
    width: 24px;
}

.law-name {
    flex: 1;
    font-size: 0.95rem;
    font-weight: 600;
    color: var(--text-highlight);
}

.state-chip {
    font-family: var(--font-mono);
    font-size: 0.68rem;
    font-weight: 700;
    padding: 0.2rem 0.55rem;
    border-radius: var(--radius-sm);
}

.state-proceed {
    background: rgba(0, 230, 118, 0.12);
    color: var(--state-proceed);
    border: 1px solid rgba(0, 230, 118, 0.3);
}

.state-await {
    background: rgba(255, 170, 0, 0.12);
    color: var(--state-await);
    border: 1px solid rgba(255, 170, 0, 0.3);
}

.state-halt {
    background: rgba(157, 78, 221, 0.12);
    color: var(--state-halt);
    border: 1px solid rgba(157, 78, 221, 0.3);
}

.law-quote {
    font-size: 0.82rem;
    font-style: italic;
    color: #cbd5e1;
    margin-bottom: 0.5rem;
```

```
    line-height: 1.5;
}

.law-rule {
    font-size: 0.75rem;
    font-weight: 600;
    color: var(--cyan);
}

.lexicon-controls {
    margin-bottom: 1.25rem;
}

#lexicon-search-input {
    width: 100%;
    background: var(--bg-surface);
    border: 1px solid var(--border-subtle);
    border-radius: var(--radius-md);
    padding: 0.75rem 1rem;
    color: var(--text-primary);
    font-size: 0.9rem;
    outline: none;
    margin-bottom: 0.75rem;
}

#lexicon-search-input:focus {
    border-color: var(--cyan);
}

.domain-filters {
    display: flex;
    flex-wrap: wrap;
    gap: 0.4rem;
}

.filter-btn {
    background: rgba(255, 255, 255, 0.04);
    border: 1px solid var(--border-subtle);
    border-radius: var(--radius-sm);
    padding: 0.3rem 0.75rem;
    color: var(--text-secondary);
    font-size: 0.75rem;
    cursor: pointer;
    transition: all 0.2s ease;
}

.filter-btn:hover {
    color: var(--text-primary);
    background: rgba(255, 255, 255, 0.08);
}

.filter-btn.active {
    background: rgba(0, 240, 255, 0.12);
    border-color: var(--cyan);
    color: var(--cyan);
    font-weight: 600;
}

.lexicon-entries-grid {
    display: flex;
    flex-direction: column;
    gap: 0.85rem;
    max-height: 650px;
    overflow-y: auto;
    padding-right: 0.5rem;
}

.entry-card {
    background: var(--bg-surface);
    border: 1px solid var(--border-subtle);
    border-radius: var(--radius-md);
    padding: 1rem;
    transition: all 0.2s ease;
}

.entry-card:hover {
    border-color: rgba(0, 240, 255, 0.3);
}
```

```
777
778 .entry-header {
779     display: flex;
780     justify-content: space-between;
781     align-items: center;
782     margin-bottom: 0.5rem;
783 }
784
785 .entry-term {
786     font-size: 1.05rem;
787     font-weight: 700;
788     color: var(--cyan);
789     text-transform: uppercase;
790     letter-spacing: 0.05em;
791 }
792
793 .entry-domain {
794     font-size: 0.72rem;
795     color: var(--gold);
796     background: rgba(255, 215, 0, 0.08);
797     padding: 0.2rem 0.5rem;
798     border-radius: var(--radius-sm);
799 }
800
801 .entry-def {
802     font-size: 0.85rem;
803     color: #e2e8f0;
804     margin-bottom: 0.65rem;
805 }
806
807 .entry-witnesses {
808     background: #05070d;
809     border-radius: var(--radius-sm);
810     padding: 0.6rem 0.8rem;
811     font-size: 0.78rem;
812     line-height: 1.5;
813     color: #94a3b8;
814     margin-bottom: 0.5rem;
815 }
816
817 .witness-title {
818     font-weight: 700;
819     color: var(--text-highlight);
820     margin-right: 0.35rem;
821 }
822
823 .entry-guidance {
824     font-size: 0.78rem;
825     color: var(--state-proceed);
826     font-weight: 500;
827 }
828
829 /* ------------------------------------------------------------------------
830    VIEW 4: EPISODIC SCROLLS
831    ------------------------------------------------------------------------ */
832 .metrics-grid {
833     display: grid;
834     grid-template-columns: repeat(6, 1fr);
835     gap: 1rem;
836     margin-bottom: 1.5rem;
837 }
838
839 .metric-card {
840     background: var(--bg-surface);
841     border: 1px solid var(--border-subtle);
842     border-radius: var(--radius-md);
843     padding: 1rem;
844     display: flex;
845     flex-direction: column;
846     align-items: center;
847     text-align: center;
848 }
849
850 .metric-num {
851     font-size: 1.75rem;
852     font-weight: 800;
853     font-family: var(--font-mono);
854     color: var(--text-highlight);
```

```
}

.metric-label {
    font-size: 0.72rem;
    color: var(--text-muted);
    text-transform: uppercase;
    letter-spacing: 0.05em;
    margin-top: 0.25rem;
}

.card-green .metric-num { color: var(--state-proceed); }
.card-amber .metric-num { color: var(--state-await); }
.card-red .metric-num { color: var(--state-rollback); }
.card-purple .metric-num { color: var(--state-halt); }

.episodes-table-wrapper {
    background: #05070c;
    border: 1px solid var(--border-subtle);
    border-radius: var(--radius-md);
    overflow-x: auto;
    max-height: 520px;
}

.episodes-table {
    width: 100%;
    border-collapse: collapse;
    font-size: 0.85rem;
    text-align: left;
}

.episodes-table th {
    background: rgba(255, 255, 255, 0.03);
    padding: 0.85rem 1rem;
    font-weight: 600;
    color: var(--text-muted);
    border-bottom: 1px solid var(--border-subtle);
    text-transform: uppercase;
    font-size: 0.72rem;
    letter-spacing: 0.05em;
}

.episodes-table td {
    padding: 0.85rem 1rem;
    border-bottom: 1px solid rgba(255, 255, 255, 0.03);
    color: #cbd5e1;
}

.episodes-table tr:hover td {
    background: rgba(0, 240, 255, 0.02);
}

/* Responsive */
@media (max-width: 1100px) {
    .studio-output-grid,
    .lexicon-container {
        grid-template-columns: 1fr;
    }
    .metrics-grid {
        grid-template-columns: repeat(3, 1fr);
    }
}

@media (max-width: 768px) {
    .app-header {
        flex-direction: column;
        gap: 1rem;
        padding: 1rem;
    }
    .metrics-grid {
        grid-template-columns: repeat(2, 1fr);
    }
}

/* ------------------------------------------------------------------------
   CREATIVEWORKZSTUDIO LLC BRANDING & DOCUMENTATION STYLES
   ------------------------------------------------------------------------ */
.cws-badge {
    background: linear-gradient(135deg, rgba(255, 215, 0, 0.15), rgba(255, 170, 0, 0.08));
```

```
    border: 1px solid rgba(255, 215, 0, 0.35);
    color: var(--gold);
    font-size: 0.75rem;
    font-weight: 700;
    padding: 0.35rem 0.85rem;
    border-radius: 9999px;
    letter-spacing: 0.04em;
    text-transform: uppercase;
    display: inline-flex;
    align-items: center;
    gap: 0.4rem;
    box-shadow: 0 0 12px rgba(255, 215, 0, 0.1);
}

.terminal-ctrl-group {
    display: flex;
    gap: 0.5rem;
    align-items: center;
}

/* Documentation Layout */
.doc-container {
    display: flex;
    flex-direction: column;
    gap: 1.5rem;
    max-width: 1300px;
    margin: 0 auto;
    width: 100%;
}

.doc-hero {
    padding: 2.2rem;
    border-radius: var(--radius-lg);
    background: linear-gradient(135deg, rgba(13, 18, 29, 0.85) 0%, rgba(20, 28, 45, 0.7) 100%);
    border: var(--glass-border);
    position: relative;
    overflow: hidden;
}

.doc-hero::before {
    content: '';
    position: absolute;
    top: 0;
    right: 0;
    width: 320px;
    height: 100%;
    background: radial-gradient(circle at center, rgba(0, 240, 255, 0.08) 0%, transparent 70%);
    pointer-events: none;
}

.hero-badge {
    display: inline-block;
    font-size: 0.72rem;
    font-weight: 800;
    letter-spacing: 0.1em;
    text-transform: uppercase;
    color: var(--cyan);
    margin-bottom: 0.6rem;
}

.doc-hero h1 {
    font-size: 2.2rem;
    font-weight: 800;
    color: #ffffff;
    margin-bottom: 0.8rem;
    letter-spacing: -0.02em;
}

.hero-lead {
    font-size: 1.05rem;
    color: var(--text-secondary);
    line-height: 1.7;
    max-width: 950px;
}

.doc-grid {
    display: grid;
    grid-template-columns: repeat(2, 1fr);
```

```
    gap: 1.5rem;
}

@media (max-width: 1024px) {
    .doc-grid {
        grid-template-columns: 1fr;
    }
}

.doc-card {
    padding: 1.8rem;
    border-radius: var(--radius-md);
    display: flex;
    flex-direction: column;
    gap: 1rem;
}

.doc-card-header {
    display: flex;
    align-items: center;
    gap: 0.75rem;
    border-bottom: 1px solid var(--border-subtle);
    padding-bottom: 0.8rem;
}

.doc-icon {
    font-size: 1.5rem;
}

.doc-card-header h2 {
    font-size: 1.2rem;
    font-weight: 700;
    color: #ffffff;
}

/* 4-Block Timeline */
.four-block-timeline {
    display: flex;
    flex-direction: column;
    gap: 0.6rem;
    margin: 0.5rem 0;
}

.timeline-step {
    display: flex;
    align-items: flex-start;
    gap: 0.8rem;
    background: rgba(255, 255, 255, 0.02);
    padding: 0.6rem 0.8rem;
    border-radius: var(--radius-sm);
    border-left: 3px solid var(--cyan);
}

.step-num {
    background: var(--cyan);
    color: #000000;
    font-weight: 800;
    font-size: 0.75rem;
    width: 20px;
    height: 20px;
    border-radius: 50%;
    display: flex;
    align-items: center;
    justify-content: center;
    flex-shrink: 0;
}

.step-content {
    font-size: 0.88rem;
    color: var(--text-primary);
}

.code-snippet-box {
    background: #05070b;
    border: 1px solid var(--border-subtle);
    border-radius: var(--radius-sm);
    padding: 1rem;
    font-family: var(--font-mono);
```

```
    font-size: 0.82rem;
    overflow-x: auto;
    color: var(--cyan);
}

/* Spec Tables */
.table-responsive {
    overflow-x: auto;
    width: 100%;
}

.spec-table {
    width: 100%;
    border-collapse: collapse;
    font-size: 0.85rem;
}

.spec-table th {
    background: rgba(255, 255, 255, 0.04);
    padding: 0.7rem 0.85rem;
    color: var(--text-muted);
    font-size: 0.72rem;
    text-transform: uppercase;
    letter-spacing: 0.05em;
    text-align: left;
    border-bottom: 1px solid var(--border-subtle);
}

.spec-table td {
    padding: 0.7rem 0.85rem;
    border-bottom: 1px solid rgba(255, 255, 255, 0.03);
    color: #cbd5e1;
}

.spec-table tr:hover td {
    background: rgba(0, 240, 255, 0.02);
}

/* Boundary Badges */
.badge-rigid { color: #58a6ff; font-weight: 700; }
.badge-fluid { color: #38bdf8; font-weight: 700; }
.badge-omni { color: #d29922; font-weight: 700; }
.badge-sink { color: #f85149; font-weight: 700; }
.badge-source { color: #3fb950; font-weight: 700; }
.badge-aside { color: #bc8cff; font-weight: 700; }

/* Quad-State Showcase */
.quad-state-showcase {
    display: grid;
    grid-template-columns: repeat(2, 1fr);
    gap: 0.75rem;
    margin: 0.5rem 0;
}

.quad-item {
    padding: 0.8rem;
    border-radius: var(--radius-sm);
    display: flex;
    flex-direction: column;
    gap: 0.3rem;
    border: 1px solid rgba(255, 255, 255, 0.05);
}

.quad-item.quad-proceed { background: rgba(0, 230, 118, 0.06); border-color: rgba(0, 230, 118, 0.25); }
.quad-item.quad-await { background: rgba(255, 170, 0, 0.06); border-color: rgba(255, 170, 0, 0.25); }
.quad-item.quad-rollback { background: rgba(255, 61, 113, 0.06); border-color: rgba(255, 61, 113, 0.25); }
.quad-item.quad-halt { background: rgba(157, 78, 221, 0.06); border-color: rgba(157, 78, 221, 0.25); }

.quad-code {
    font-family: var(--font-mono);
    font-weight: 800;
    font-size: 0.85rem;
}
.quad-proceed .quad-code { color: var(--state-proceed); }
.quad-await .quad-code { color: var(--state-await); }
.quad-rollback .quad-code { color: var(--state-rollback); }
.quad-halt .quad-code { color: var(--state-halt); }

```

```
.quad-meaning {
    font-size: 0.8rem;
    color: var(--text-secondary);
}

/* AI Pipeline */
.pipeline-diagram {
    display: flex;
    flex-direction: column;
    align-items: center;
    gap: 0.4rem;
    margin: 0.5rem 0;
}

.pipe-node {
    width: 100%;
    background: rgba(255, 255, 255, 0.03);
    border: 1px solid var(--border-subtle);
    padding: 0.65rem 1rem;
    border-radius: var(--radius-sm);
    text-align: center;
    font-size: 0.85rem;
    font-weight: 600;
    color: var(--text-primary);
}

.pipe-node.highlight {
    background: rgba(0, 240, 255, 0.08);
    border-color: var(--border-bright);
    color: var(--cyan);
}

.pipe-arrow {
    color: var(--text-muted);
    font-size: 0.9rem;
}

.mini-laws-grid {
    display: flex;
    flex-direction: column;
    gap: 0.4rem;
    font-size: 0.85rem;
}

.mini-law {
    background: rgba(255, 255, 255, 0.02);
    padding: 0.5rem 0.8rem;
    border-radius: var(--radius-sm);
    color: var(--text-secondary);
}

.mini-law strong {
    color: #ffffff;
}

.doc-subcard {
    background: rgba(255, 255, 255, 0.02);
    border: 1px solid var(--border-subtle);
    border-radius: var(--radius-sm);
    padding: 1rem;
    margin-top: 0.5rem;
}

.doc-subcard h3 {
    font-size: 0.95rem;
    color: var(--text-primary);
    margin-bottom: 0.4rem;
}

.doc-subcard p {
    font-size: 0.85rem;
    color: var(--text-secondary);
    margin-bottom: 0.5rem;
}

.doc-subcard code {
    display: block;
    background: #05070b;
```

```
    padding: 0.5rem 0.8rem;
    border-radius: var(--radius-sm);
    font-family: var(--font-mono);
    font-size: 0.8rem;
    color: var(--cyan);
}

.spec-list {
    list-style: none;
    display: flex;
    flex-direction: column;
    gap: 0.6rem;
    font-size: 0.88rem;
    color: var(--text-secondary);
}

.spec-list li strong {
    color: #ffffff;
}

/* ------------------------------------------------------------------------
   FOOTER
   ------------------------------------------------------------------------ */
.app-footer {
    display: flex;
    align-items: center;
    justify-content: space-between;
    padding: 1.2rem 2rem;
    background: rgba(7, 9, 14, 0.9);
    border-top: var(--glass-border);
    font-size: 0.82rem;
    color: var(--text-muted);
    margin-top: 2rem;
}

.footer-left {
    display: flex;
    align-items: center;
    gap: 0.75rem;
}

.footer-brand {
    font-family: var(--font-serif);
    font-weight: 700;
    color: var(--cyan);
}

.footer-center {
    color: var(--text-secondary);
}

.footer-right {
    display: flex;
    align-items: center;
    gap: 0.5rem;
}

.footer-pill {
    background: rgba(255, 255, 255, 0.05);
    padding: 0.2rem 0.6rem;
    border-radius: 9999px;
    font-size: 0.72rem;
    color: var(--text-secondary);
}

/* ------------------------------------------------------------------------
   INTERACTIVE TERMINAL WORKSPACE & SCHEDULER STYLES
   ------------------------------------------------------------------------ */
.terminal-body {
    overflow: hidden;
}
.terminal-page-layout {
    height: 100vh;
    display: flex;
    flex-direction: column;
}
.terminal-workspace {
    flex: 1;
```

```
    display: flex;
    flex-direction: column;
    padding: 1rem 1.5rem;
    gap: 0.8rem;
    overflow: hidden;
}
.terminal-toolbar {
    display: flex;
    align-items: center;
    justify-content: space-between;
    padding: 0.6rem 1rem;
    background: rgba(13, 17, 23, 0.8);
    border: 1px solid rgba(255, 255, 255, 0.08);
    border-radius: 8px;
}
.toolbar-left {
    display: flex;
    align-items: center;
    gap: 0.8rem;
}
.terminal-title-badge {
    font-family: var(--font-mono);
    font-size: 0.78rem;
    font-weight: 700;
    color: var(--cyan);
    letter-spacing: 0.05em;
}
.sched-pill {
    background: rgba(0, 240, 255, 0.1);
    border: 1px solid rgba(0, 240, 255, 0.25);
    color: var(--cyan);
    font-size: 0.72rem;
    padding: 0.25rem 0.6rem;
    border-radius: 9999px;
    display: flex;
    align-items: center;
    gap: 0.4rem;
}
.mem-pill {
    background: rgba(0, 255, 136, 0.1);
    border: 1px solid rgba(0, 255, 136, 0.25);
    color: var(--green);
    font-size: 0.72rem;
    padding: 0.25rem 0.6rem;
    border-radius: 9999px;
}
.toolbar-actions {
    display: flex;
    gap: 0.4rem;
}
.btn-sm {
    padding: 0.35rem 0.7rem;
    font-size: 0.75rem;
}
.btn-xs {
    padding: 0.2rem 0.45rem;
    font-size: 0.68rem;
}
.btn-danger-ghost {
    background: transparent;
    color: #ff5555;
    border: 1px solid rgba(255, 85, 85, 0.25);
}
.btn-danger-ghost:hover {
    background: rgba(255, 85, 85, 0.15);
}

.terminal-grid {
    flex: 1;
    display: grid;
    grid-template-columns: 1fr 380px;
    gap: 1rem;
    min-height: 0;
}
.terminal-console-pane {
    display: flex;
    flex-direction: column;
    background: rgba(10, 12, 16, 0.95);
```

```
    border: 1px solid rgba(0, 240, 255, 0.2);
    border-radius: 8px;
    box-shadow: 0 8px 32px rgba(0, 0, 0, 0.5);
    min-height: 0;
    overflow: hidden;
}
.console-screen {
    flex: 1;
    padding: 1.2rem;
    overflow-y: auto;
    font-family: var(--font-mono);
    font-size: 0.86rem;
    line-height: 1.5;
    color: #e6edf3;
    outline: none;
}
.term-intro {
    color: var(--cyan);
    white-space: pre-wrap;
    margin-bottom: 1rem;
    font-size: 0.78rem;
    opacity: 0.9;
}
.term-entry {
    margin-bottom: 0.8rem;
}
.term-command-line {
    display: flex;
    align-items: center;
    gap: 0.5rem;
    color: #ffffff;
    font-weight: 600;
}
.term-prompt {
    color: var(--gold);
    font-family: var(--font-mono);
    font-weight: 700;
}
.term-cmd-text {
    color: #00f0ff;
}
.term-result-box {
    margin-top: 0.3rem;
    padding-left: 0.8rem;
    border-left: 2px solid rgba(0, 240, 255, 0.3);
}
.term-meta-bar {
    display: flex;
    align-items: center;
    gap: 0.6rem;
    margin-bottom: 0.25rem;
    font-size: 0.72rem;
}
.term-output-text {
    font-family: var(--font-mono);
    font-size: 0.82rem;
    white-space: pre-wrap;
    word-break: break-word;
    color: #d1d5db;
    margin: 0;
}
.console-input-row {
    display: flex;
    align-items: center;
    gap: 0.6rem;
    padding: 0.8rem 1.2rem;
    background: rgba(13, 17, 23, 0.95);
    border-top: 1px solid rgba(255, 255, 255, 0.08);
}
.term-input {
    flex: 1;
    background: transparent;
    border: none;
    outline: none;
    color: #ffffff;
    font-family: var(--font-mono);
    font-size: 0.9rem;
}
```

```
1479
1480 /* Scheduler Right Pane */
1481 .scheduler-pane {
1482     display: flex;
1483     flex-direction: column;
1484     background: rgba(13, 17, 23, 0.85);
1485     border: 1px solid rgba(255, 255, 255, 0.08);
1486     border-radius: 8px;
1487     padding: 1rem;
1488     gap: 0.8rem;
1489     min-height: 0;
1490     overflow: hidden;
1491 }
1492 .scheduler-pane-header {
1493     display: flex;
1494     justify-content: space-between;
1495     align-items: center;
1496 }
1497 .sched-title-group h2 {
1498     font-size: 0.95rem;
1499     margin: 0;
1500     color: #ffffff;
1501 }
1502 .sched-title-group .subtext {
1503     font-size: 0.72rem;
1504     color: var(--text-muted);
1505     margin: 0;
1506 }
1507 .sched-spawn-box {
1508     display: grid;
1509     grid-template-columns: 1fr 1fr;
1510     gap: 0.4rem;
1511 }
1512 .sched-spawn-box button {
1513     grid-column: span 2;
1514 }
1515 .form-input-sm, .form-select-sm {
1516     background: rgba(255, 255, 255, 0.05);
1517     border: 1px solid rgba(255, 255, 255, 0.1);
1518     color: #ffffff;
1519     padding: 0.35rem 0.5rem;
1520     font-size: 0.75rem;
1521     border-radius: 4px;
1522     outline: none;
1523 }
1524 .task-table-wrapper {
1525     flex: 1;
1526     overflow-y: auto;
1527     border: 1px solid rgba(255, 255, 255, 0.05);
1528     border-radius: 4px;
1529 }
1530 .task-table {
1531     width: 100%;
1532     border-collapse: collapse;
1533     font-size: 0.75rem;
1534 }
1535 .task-table th, .task-table td {
1536     padding: 0.45rem 0.6rem;
1537     text-align: left;
1538     border-bottom: 1px solid rgba(255, 255, 255, 0.05);
1539 }
1540 .task-table th {
1541     background: rgba(255, 255, 255, 0.03);
1542     color: var(--text-muted);
1543     font-weight: 600;
1544 }
1545 .action-cell {
1546     display: flex;
1547     gap: 0.2rem;
1548 }
1549 .session-vars-box {
1550     background: rgba(0, 0, 0, 0.3);
1551     border: 1px solid rgba(255, 255, 255, 0.05);
1552     border-radius: 4px;
1553     padding: 0.6rem;
1554     max-height: 120px;
1555     overflow-y: auto;
1556 }
```

```
.vars-header {
    display: flex;
    justify-content: space-between;
    align-items: center;
    margin-bottom: 0.3rem;
}
.vars-header h3 {
    font-size: 0.75rem;
    margin: 0;
    color: var(--gold);
}
.vars-content {
    font-family: var(--font-mono);
    font-size: 0.72rem;
    color: #e6edf3;
}

/* Download Page Styles */
.download-container {
    max-width: 1100px;
    margin: 2rem auto;
    padding: 0 1.5rem;
}
.download-hero {
    text-align: center;
    padding: 3rem 2rem;
    margin-bottom: 2rem;
}
.download-btn-row {
    margin: 2rem 0 1rem;
}
.download-cta {
    background: linear-gradient(135deg, #00f0ff, #0099ff);
    color: #000000;
    font-weight: 800;
    font-size: 1.05rem;
    padding: 0.9rem 2.2rem;
    border-radius: 8px;
    box-shadow: 0 0 25px rgba(0, 240, 255, 0.35);
    display: inline-flex;
    align-items: center;
    gap: 0.8rem;
    text-decoration: none;
    transition: transform 0.2s, box-shadow 0.2s;
}
.download-cta:hover {
    transform: translateY(-2px);
    box-shadow: 0 0 35px rgba(0, 240, 255, 0.5);
}
.hero-subtext {
    font-size: 0.82rem;
    color: var(--text-muted);
}
.download-grid {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 1.5rem;
}
.download-card {
    padding: 1.8rem;
}
.code-copy-box {
    background: rgba(0, 0, 0, 0.5);
    border: 1px solid rgba(0, 240, 255, 0.2);
    border-radius: 6px;
    padding: 0.9rem;
    margin: 1rem 0;
    font-family: var(--font-mono);
    font-size: 0.82rem;
    color: var(--cyan);
    word-break: break-all;
}
.download-nav-btn {
    color: var(--cyan) !important;
    border-color: rgba(0, 240, 255, 0.3) !important;
}

/* ------------------------------------------------------------------------
```

```
   OMNICODE 3-TIER COMPOSITE HOVERCARD & DOCSTRING INSPECTOR
   ------------------------------------------------------------------------- */
.omni-hover-card {
    position: absolute;
    z-index: 10000;
    max-width: 480px;
    background: rgba(10, 15, 29, 0.94);
    backdrop-filter: blur(20px);
    -webkit-backdrop-filter: blur(20px);
    border: 1px solid rgba(0, 240, 255, 0.35);
    border-radius: 10px;
    box-shadow: 0 12px 40px rgba(0, 0, 0, 0.8), 0 0 20px rgba(0, 240, 255, 0.15);
    padding: 1.1rem;
    color: #e2e8f0;
    font-family: var(--font-sans, -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, sans-serif);
    font-size: 0.85rem;
    line-height: 1.5;
    pointer-events: auto;
    transition: opacity 0.15s ease, transform 0.15s ease;
}

.omni-hover-header {
    display: flex;
    align-items: center;
    justify-content: space-between;
    border-bottom: 1px solid rgba(255, 255, 255, 0.1);
    padding-bottom: 0.6rem;
    margin-bottom: 0.8rem;
}

.omni-hover-symbol {
    font-family: var(--font-mono, monospace);
    font-size: 1.05rem;
    font-weight: 700;
    color: #00f0ff;
    letter-spacing: 0.5px;
}

.omni-hover-rank-badge {
    display: inline-block;
    padding: 0.2rem 0.6rem;
    font-size: 0.72rem;
    font-weight: 600;
    border-radius: 4px;
    background: rgba(0, 240, 255, 0.15);
    color: #38bdf8;
    border: 1px solid rgba(56, 189, 248, 0.3);
    text-transform: uppercase;
}

.omni-hover-rank-badge.rank-1 {
    background: rgba(234, 179, 8, 0.18);
    color: #facc15;
    border-color: rgba(250, 204, 21, 0.4);
}

.omni-hover-tier {
    margin-bottom: 0.8rem;
    padding-bottom: 0.7rem;
    border-bottom: 1px solid rgba(255, 255, 255, 0.06);
}

.omni-hover-tier:last-child {
    margin-bottom: 0;
    padding-bottom: 0;
    border-bottom: none;
}

.omni-hover-tier-title {
    font-size: 0.75rem;
    font-weight: 700;
    text-transform: uppercase;
    letter-spacing: 0.8px;
    color: #94a3b8;
    margin-bottom: 0.4rem;
    display: flex;
    align-items: center;
    gap: 0.4rem;
```

```
}

.omni-hover-tier-title.tier-tape { color: #fbbf24; }
.omni-hover-tier-title.tier-omni { color: #34d399; }
.omni-hover-tier-title.tier-silicon { color: #c084fc; }

.omni-hover-prop {
    display: flex;
    margin-bottom: 0.25rem;
    font-size: 0.8rem;
}

.omni-hover-label {
    color: #64748b;
    width: 130px;
    flex-shrink: 0;
}

.omni-hover-val {
    color: #f1f5f9;
    font-family: var(--font-mono, monospace);
    word-break: break-word;
}

.omni-hover-aside {
    background: rgba(255, 255, 255, 0.04);
    border-left: 2px solid #38bdf8;
    padding: 0.35rem 0.6rem;
    margin: 0.4rem 0;
    font-style: italic;
    color: #cbd5e1;
    font-size: 0.78rem;
}

```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/terminal.html HTML 191 LINES 782 WORDS 10.8 KB

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description" content="OmniCode Interactive Native Terminal & OS Process Scheduler — CreativeWorkzStudio LLC.">
    <title>OmniCode Interactive Terminal &amp;amp; Task Scheduler — CreativeWorkzStudio LLC</title>
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Fira+Code:wght@400;500;600;700&family=Outfit:wght@300;400;500;600;700;800&family=Cinzel:wght@500;700;800&display=swap" rel="stylesheet">
    <link rel="stylesheet" href="/style.css">
</head>
<body class="terminal-body">
    <div class="app-layout terminal-page-layout">
        <!-- GLOBAL NAVIGATION HEADER -->
        <header class="app-header" id="app-header">
            <div class="header-left">
                <a href="/studio" class="brand-container">
                    <span class="brand-symbol">Ω</span>
                    <div class="brand-text">
                        <span class="brand-title">OMNICODE</span>
                        <span class="brand-subtitle">CREATIVEWORKZSTUDIO LLC</span>
                    </div>
                </a>
                <div class="status-pill" id="status-pill" title="Engine runs 100% offline with zero cloud telemetry">
                    <span class="status-dot"></span>
                    <span class="status-text">OFFLINE COVENANT ACTIVE</span>
                </div>
            </div>

            <nav class="app-nav" aria-label="Main Navigation">
                <a href="/studio" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 2L2 7l10 5 10-5-10-5zM2 17l10 5 10-5M2 12l10 5 10-5"/></svg>
                    AI Studio
                </a>
                <a href="/terminal" class="nav-tab active">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><polyline points="4 17 10 11 4 5"/><line x1="12" y1="19" x2="20" y2="19"/></svg>
                    CLI Terminal
                </a>
                <a href="/specs" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M14 2H6a2 2 0 0 0-2 2v16a2 2 0 0 0 2 2h12a2 2 0 0 0 2-2V8z"/><polyline points="14 2 14 8 20 8"/></svg>
                    Language Specs
                </a>
                <a href="/model" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M12 2a8 8 0 0 0-8 8c0 3.3 2 6.2 5 7.4V20a2 2 0 0 0 2 2h2a2 2 0 0 0 2-2v-2.6c3-1.2 5-4.1 5-7.4a8 8 0 0 0-8-8z"/></svg>
                    AI Model
                </a>
                <a href="/lexicon" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M4 19.5A2.5 2.5 0 0 1 6.5 17H20"/><path d="M6.5 2H20v20H6.5A2.5 2.5 0 0 1 4 19.5v-15A2.5 2.5 0 0 1 6.5 2z"/></svg>
                    Covenant Lexicon
                </a>
                <a href="/scrolls" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M19 21l-7-5-7 5V5a2 2 0 0 1 2-2h10a2 2 0 0 1 2 2z"/></svg>
                    Scrolls
                </a>
                <a href="/cloud" class="nav-tab">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M18 10h-1.26A8 8 0 1 0 9 20h9a5 5 0 0 0 0-10z"/></svg>
                    Cloud &amp;amp; Edge
                </a>
                <a href="/download" class="nav-tab download-nav-btn">
                    <svg width="15" height="15" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M21 15v4a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2v-4"/><polyline points="7 10 12 15 17 10"/><line x1="12" y1="15" x2="12" y2="3"/></svg>
                    Download Native
```

```
                </a>
            </nav>
        </header>

        <!-- TERMINAL WORKSPACE MAIN -->
        <main class="terminal-workspace">
            <!-- TERMINAL CONTROL TOOLBAR -->
            <div class="terminal-toolbar card-glass">
                <div class="toolbar-left">
                    <span class="terminal-title-badge">Ω NATIVE USERSPACE TERMINAL</span>
                    <span class="sched-pill" id="sched-status-pill">
                        <span class="status-dot"></span>
                        <span id="sched-pill-text">Scheduler: Online (3 tasks)</span>
                    </span>
                    <span class="mem-pill" id="mem-pill">State: +1:PROCEED</span>
                </div>
                <div class="toolbar-actions">
                    <button class="btn btn-sm btn-ghost" id="btn-quick-demo">🧪 Demo</button>
                    <button class="btn btn-sm btn-ghost" id="btn-quick-lint">📋 Lint</button>
                    <button class="btn btn-sm btn-ghost" id="btn-quick-restore">🩹 Restore</button>
                    <button class="btn btn-sm btn-ghost" id="btn-quick-tasks">📊 Tasks (ps)</button>
                    <button class="btn btn-sm btn-ghost" id="btn-quick-tick">⚡ Tick</button>
                    <button class="btn btn-sm btn-danger-ghost" id="btn-clear-term">🧹 Clear (Ctrl+L)</button>
                </div>
            </div>

            <!-- DUAL VIEWPORT: TERMINAL SCREEN & SCHEDULER MONITOR -->
            <div class="terminal-grid">
                <!-- LEFT: LIVE INTERACTIVE CONSOLE -->
                <div class="terminal-console-pane card-glass">
                    <div class="console-screen" id="console-screen" tabindex="0">
                        <div class="term-intro">
================================================================================
  Ω OMNICODE INTERACTIVE USERSYSTEM SHELL &amp;amp; TASK SCHEDULER
  CreativeWorkzStudio LLC — Autonomous Neural-Symbolic Intelligence
  Engine: Rust Core 2024 (Balanced Radix-3 Quad-State Physics)
================================================================================
* Grounded in Biblical Moral Invariants &amp;amp; Symmetrical Covenant Bedrock
* Type 'help' for terminal commands, 'ps' for scheduler tasks, 'tick' to advance clock.
* Keyboard: Up/Down for command history | Tab for autocompletion | Ctrl+L to clear screen.
                        </div>
                        <div id="term-history-log"></div>
                    </div>

                    <!-- INTERACTIVE INPUT BAR -->
                    <div class="console-input-row" id="input-container">
                        <span class="term-prompt">omnicode&amp;gt;</span>
                        <input type="text" id="term-input" class="term-input" autocomplete="off" autofocus
spellcheck="false" placeholder="Type command (e.g. ps, spawn 'worker', let x = 42, omnicode demo, help)...">
                        <button class="btn btn-sm btn-accent" id="btn-term-send">Execute</button>
                    </div>
                </div>

                <!-- RIGHT: LIVE OS TASK SCHEDULER & PROCESS LEDGER -->
                <div class="scheduler-pane card-glass">
                    <div class="scheduler-pane-header">
                        <div class="sched-title-group">
                            <h2>Cooperative Task Scheduler</h2>
                            <p class="subtext">Active process table grounded in Balanced Radix-3 states</p>
                        </div>
                        <button class="btn btn-sm btn-secondary" id="btn-refresh-tasks">Refresh</button>
                    </div>

                    <!-- TASK SPAWN ROW -->
                    <div class="sched-spawn-box">
                        <input type="text" id="spawn-name-input" class="form-input-sm" placeholder="Task Name (e.g.
InvariantWatcher)">
                        <input type="text" id="spawn-cmd-input" class="form-input-sm" placeholder="Command (e.g. omnicode
lint manifest.omni)">
                        <select id="spawn-priority-select" class="form-select-sm">
                            <option value="NORMAL">Normal Priority</option>
                            <option value="HIGH">High Priority</option>
                            <option value="LOW">Low Priority</option>
                            <option value="BACKGROUND">Background</option>
                        </select>
                        <button class="btn btn-sm btn-accent" id="btn-spawn-task">+ Spawn</button>
                    </div>

```

```
                    <!-- TASK TABLE -->
                    <div class="task-table-wrapper">
                        <table class="task-table">
                            <thead>
                                <tr>
                                    <th>PID</th>
                                    <th>Task Name</th>
                                    <th>Priority</th>
                                    <th>Quad-State</th>
                                    <th>Ticks</th>
                                    <th>Actions</th>
                                </tr>
                            </thead>
                            <tbody id="task-tbody">
                                <!-- Populated dynamically via JS -->
                            </tbody>
                        </table>
                    </div>

                    <!-- IN-MEMORY SESSION VARIABLES -->
                    <div class="session-vars-box">
                        <div class="vars-header">
                            <h3>In-Memory Session Variables</h3>
                            <button class="btn btn-xs btn-ghost" id="btn-refresh-vars">Refresh Vars</button>
                        </div>
                        <div class="vars-content" id="session-vars-content">
                            <span class="dim-text">No session variables declared yet. Use 'let &lt;name&gt; =
&lt;value&gt;'</span>
                        </div>
                    </div>
                </div>
            </div>
        </main>

        <!-- FOOTER -->
        <footer class="app-footer">
            <div class="footer-left">
                <span class="footer-brand">Ω OMNICODE</span>
                <span class="footer-sep">&bull;</span>
                <span class="footer-copy">&copy; 2026 CreativeWorkzStudio LLC. All rights reserved.</span>
            </div>
            <div class="footer-center">
                <span>Native Userspace Shell &amp; Cooperative Quad-State Process Scheduler</span>
            </div>
            <div class="footer-right">
                <span class="footer-pill">Local Ring-3 Shell</span>
                <span class="footer-pill">v0.1.0</span>
            </div>
        </footer>
    </div>

    <script src="/terminal.js"></script>
</body>
</html>
```

<TRACK 3> : {Applications & OmniStudio Interfaces}/apps/gui/terminal.js JS 305 LINES 885 WORDS 11.9 KB

```
// ================================================================================
// BLOCK 1: METADATA BLOCK
// ( OmniCode Native Interactive Terminal & Task Scheduler JavaScript Engine )
// ================================================================================
/**
 * OmniCode Terminal & Scheduler Frontend Engine
 * CreativeWorkzStudio LLC
 * Zero external dependencies. Connects via standard HTTP JSON endpoints:
 *   - POST /api/terminal          -> Dispatches command to session & scheduler
 *   - GET  /api/scheduler         -> Retrieves live process ledger
 *   - POST /api/scheduler/spawn   -> Spawns new task
 *   - POST /api/scheduler/action  -> Mutates task state (pause, resume, kill)
 */

document.addEventListener("DOMContentLoaded", () => {
    const consoleScreen = document.getElementById("console-screen");
    const historyLog = document.getElementById("term-history-log");
    const termInput = document.getElementById("term-input");
    const btnSend = document.getElementById("btn-term-send");
    const btnClear = document.getElementById("btn-clear-term");
    const taskTbody = document.getElementById("task-tbody");
    const schedPillText = document.getElementById("sched-pill-text");
    const varsContent = document.getElementById("session-vars-content");

    // Command History & Autocomplete State
    const history = JSON.parse(localStorage.getItem("omnicode_terminal_history") || "[]");
    let historyIndex = history.length;

    const AUTOCOMPLETE_LIST = [
        "ps", "tasks", "spawn", "pause", "resume", "kill", "halt", "tick", "vars", "env",
        "let", "help", "clear",
        "omnicode demo",
        "omnicode lint manifest.omni",
        "omnicode restore manifest.omni",
        "omnicode manifest",
        "omnicode scrolls",
        "omnicode lexicon truth",
        "omnicode ai \"What is the covenant?\""
    ];

    // Focus input on console click
    consoleScreen.addEventListener("click", () => {
        termInput.focus();
    });

    // --------------------------------------------------------------------------
    // EXECUTION DISPATCHER
    // --------------------------------------------------------------------------
    async function executeCommand(cmd) {
        const clean = cmd.trim();
        if (!clean) return;

        // Save to history
        if (history.length === 0 || history[history.length - 1] !== clean) {
            history.push(clean);
            if (history.length > 100) history.shift();
            localStorage.setItem("omnicode_terminal_history", JSON.stringify(history));
        }
        historyIndex = history.length;

        // Handle local 'clear'
        if (clean.toLowerCase() === "clear") {
            historyLog.innerHTML = "";
            termInput.value = "";
            return;
        }

        // Render command prompt entry
        const entryEl = document.createElement("div");
        entryEl.className = "term-entry";
        entryEl.innerHTML = `
            <div class="term-command-line">
                <span class="term-prompt">omnicode&gt;</span>
                <span class="term-cmd-text">${escapeHtml(clean)}</span>
```

```
            </div>
            <div class="term-loading-spinner dim-text">Deliberating &amp;amp; evaluating...</div>
        `;
        historyLog.appendChild(entryEl);
        consoleScreen.scrollTop = consoleScreen.scrollHeight;
        termInput.value = "";

        const spinner = entryEl.querySelector(".term-loading-spinner");

        try {
            const resp = await fetch("/api/terminal", {
                method: "POST",
                headers: { "Content-Type": "application/json" },
                body: JSON.stringify({ command: clean })
            });

            const data = await resp.json();
            spinner.remove();

            const resultEl = document.createElement("div");
            resultEl.className = "term-result-box";

            const stateClass = getStateClass(data.state || "+1:PROCEED");
            const durationMs = (data.duration_us / 1000).toFixed(2);

            resultEl.innerHTML = `
                <div class="term-meta-bar">
                    <span class="badge ${stateClass}">${escapeHtml(data.state || "+1:PROCEED")}</span>
                    <span class="dim-text">${durationMs} ms</span>
                </div>
                <pre class="term-output-text">${formatTerminalOutput(data.stdout || "(Completed with zero output)")}
</pre>
            `;
            entryEl.appendChild(resultEl);
        } catch (err) {
            spinner.remove();
            const errEl = document.createElement("div");
            errEl.className = "term-error-box";
            errEl.innerHTML = `<pre class="text-red">✗ Error dispatching command: ${escapeHtml(err.message)}</pre>`;
            entryEl.appendChild(errEl);
        }

        consoleScreen.scrollTop = consoleScreen.scrollHeight;
        refreshSchedulerTasks();
        refreshVariables();
    }

    // ---------------------------------------------------------------------------
    // SCHEDULER TASK POLLING & ACTIONS
    // ---------------------------------------------------------------------------
    async function refreshSchedulerTasks() {
        try {
            const resp = await fetch("/api/scheduler");
            if (!resp.ok) return;
            const tasks = await resp.json();

            taskTbody.innerHTML = "";
            let activeCount = 0;

            tasks.forEach(t => {
                if (t.state === "+1:PROCEED") activeCount++;
                const tr = document.createElement("tr");
                const stateClass = getStateClass(t.state);

                tr.innerHTML = `
                    <td class="mono font-bold">${t.id}</td>
                    <td title="${escapeHtml(t.command)}"><strong>${escapeHtml(t.name)}</strong></td>
                    <td><span class="badge badge-subtle">${escapeHtml(t.priority)}</span></td>
                    <td><span class="badge ${stateClass}">${escapeHtml(t.state)}</span></td>
                    <td class="mono">${t.ticks}</td>
                    <td class="action-cell">
                        ${t.state === "+1:PROCEED"
                            ? `<button class="btn btn-xs btn-ghost btn-pause" data-id="${t.id}" title="Pause
(0:AWAIT)">⏸</button>`
                            : `<button class="btn btn-xs btn-ghost btn-resume" data-id="${t.id}" title="Resume
(+1:PROCEED)">▶</button>`}
                        ${t.state !== "NULL:HALT"
                            ? `<button class="btn btn-xs btn-danger-ghost btn-halt" data-id="${t.id}" title="Terminate
```

```
(NULL:HALT)">■</button>`
                            : `<span class="dim-text">Ended</span>`}
                    </td>
                `;
                taskTbody.appendChild(tr);
            });

            schedPillText.textContent = `Scheduler: Online (${tasks.length} tasks, ${activeCount} active)`;

            // Attach action listeners
            document.querySelectorAll(".btn-pause").forEach(b => {
                b.onclick = () => mutateTask(b.dataset.id, "pause");
            });
            document.querySelectorAll(".btn-resume").forEach(b => {
                b.onclick = () => mutateTask(b.dataset.id, "resume");
            });
            document.querySelectorAll(".btn-halt").forEach(b => {
                b.onclick = () => mutateTask(b.dataset.id, "halt");
            });
        } catch (e) {
            console.error("Scheduler fetch error:", e);
        }
    }

    async function mutateTask(id, action) {
        await executeCommand(`${action} ${id}`);
    }

    async function refreshVariables() {
        try {
            const resp = await fetch("/api/terminal", {
                method: "POST",
                headers: { "Content-Type": "application/json" },
                body: JSON.stringify({ command: "vars" })
            });
            const data = await resp.json();
            varsContent.innerHTML = `<pre class="mono-code">${escapeHtml(data.stdout || "No session variables")}</pre>`;
        } catch (e) {
            console.error(e);
        }
    }

    // ---------------------------------------------------------------------------
    // KEYBOARD NAVIGATION & SHORTCUTS
    // ---------------------------------------------------------------------------
    termInput.addEventListener("keydown", (e) => {
        // Enter to execute
        if (e.key === "Enter") {
            e.preventDefault();
            executeCommand(termInput.value);
            return;
        }

        // Up arrow (history backward)
        if (e.key === "ArrowUp") {
            e.preventDefault();
            if (historyIndex > 0) {
                historyIndex--;
                termInput.value = history[historyIndex] || "";
            }
            return;
        }

        // Down arrow (history forward)
        if (e.key === "ArrowDown") {
            e.preventDefault();
            if (historyIndex < history.length - 1) {
                historyIndex++;
                termInput.value = history[historyIndex] || "";
            } else {
                historyIndex = history.length;
                termInput.value = "";
            }
            return;
        }

        // Tab autocomplete
        if (e.key === "Tab") {
```

```
            e.preventDefault();
            const val = termInput.value.trim().toLowerCase();
            if (!val) return;
            const match = AUTOCOMPLETE_LIST.find(item => item.toLowerCase().startsWith(val));
            if (match) {
                termInput.value = match;
            }
            return;
        }

        // Ctrl + L (Clear)
        if (e.ctrlKey && e.key.toLowerCase() === "l") {
            e.preventDefault();
            historyLog.innerHTML = "";
            return;
        }
    });

    btnSend.addEventListener("click", () => {
        executeCommand(termInput.value);
    });

    btnClear.addEventListener("click", () => {
        historyLog.innerHTML = "";
        termInput.focus();
    });

    // Toolbar Quick Action Buttons
    document.getElementById("btn-quick-demo")?.addEventListener("click", () => executeCommand("omnicode demo"));
    document.getElementById("btn-quick-lint")?.addEventListener("click", () => executeCommand("omnicode lint
manifest.omni"));
    document.getElementById("btn-quick-restore")?.addEventListener("click", () => executeCommand("omnicode restore
manifest.omni"));
    document.getElementById("btn-quick-tasks")?.addEventListener("click", () => executeCommand("ps"));
    document.getElementById("btn-quick-tick")?.addEventListener("click", () => executeCommand("tick"));
    document.getElementById("btn-refresh-tasks")?.addEventListener("click", refreshSchedulerTasks);
    document.getElementById("btn-refresh-vars")?.addEventListener("click", refreshVariables);

    // Spawn Task Button
    document.getElementById("btn-spawn-task")?.addEventListener("click", () => {
        const nameInput = document.getElementById("spawn-name-input");
        const cmdInput = document.getElementById("spawn-cmd-input");
        const prioSelect = document.getElementById("spawn-priority-select");

        const cmd = cmdInput.value.trim();
        if (!cmd) return;

        executeCommand(`spawn ${cmd}`);
        cmdInput.value = "";
        nameInput.value = "";
    });

    // Helper functions
    function getStateClass(state) {
        if (!state) return "badge-proceed";
        if (state.includes("PROCEED")) return "badge-proceed";
        if (state.includes("AWAIT")) return "badge-await";
        if (state.includes("ROLLBACK")) return "badge-rollback";
        if (state.includes("HALT")) return "badge-halt";
        return "badge-proceed";
    }

    function escapeHtml(str) {
        return (str || "")
            .replace(/&/g, "&amp;")
            .replace(/</g, "&lt;")
            .replace(/>/g, "&gt;")
            .replace(/"/g, "&quot;")
            .replace(/'/g, "&#039;");
    }

    function formatTerminalOutput(text) {
        return escapeHtml(text);
    }

    // Initial scheduler poll & timer
    refreshSchedulerTasks();
    refreshVariables();
```

```
    setInterval(refreshSchedulerTasks, 4000);
});
```